A PRICE ON THE SACRED: WHEN RELIGION MOBILIZES AND RESTRAINS[*]

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October 14, 2026

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Abstract

How does religious observance affect political violence and protests? I argue that heightened observance raises the return to political actions that an actor's doctrine can frame as worship, while raising the opportunity cost of actions that compete with worship for time. I test this using Ramadan, which rotates through the solar year. In predominantly Muslim states, armed conflict rises by about 4% during Ramadan, driven by religious insurgencies. Hourly military data from Afghanistan show that fasting changes when insurgents fight: close combat rises nearly 9% and concentrates after meals. A text analysis of a decade of weekly insurgent newsletters shows that their messaging turns substantially more devotional during the month, tying worship to combat. On the civilian side, protests fall by nearly 7%, and Muslims surveyed during Ramadan pray more and report less political interest and willingness to protest, while non-Muslims do not. State-scripted religious messaging turns more devotional but no more political. The same shock mobilizes religious militants, for whom combat is worship, and restrains citizens, for whom protest competes with prayer.

Keywords: Religion, Political violence, Insurgency, Protests, Ramadan
JEL classification: Z12; D74; D71; D23; Z13; C55

[*]I am grateful to Omer Ali, Sascha O. Becker, Eli Berman, Michael Callen, Michael Colaresi, Oeindrila Dube, Claire Duquennois, Andreas Ferrara, Rania Gihleb, Osea Giuntella, David Huffman, Mohammad Isaqzadeh, Muhammad Yasir Khan, Timur Kuran, Janet Lewis, Guillermo Lezama, Laura Liu, Lester Lusher, Eduardo Montero, Jared Rubin, Gabriel Tourek, Richard Van Weelden, Randall Walsh, and Austin Wright for their helpful comments and feedback. I also thank participants at ASREC 2025, ESOC Annual Meeting 2025, University of Chicago Dev Tea seminar, and IRES Workshop 2026 for thorough feedback. This document is an output from the research initiative 'Reducing Conflict and Improving Performance in the Economy' (ReCIPE), a programme funded by the Foreign, Commonwealth & Development Office (FCDO). The views expressed are not necessarily those of FCDO. All errors remain my own.

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1 Introduction

Religion can strengthen cooperation by sustaining social ties and shared norms (Ibn Khaldun, 1377; Greif, 1994), but it can also make political objectives sacred and resistant to material compromise (Atran et al., 2007; Bénabou and Tirole, 2011). A recurring tension is that the same cultural tradition can intensify the violence of some actors while drawing others out of politics (e.g., Swidler, 1986; Appleby, 2000).

This tension confounds both policymakers and scholars. Policymakers often treat sacred time as an instrument of restraint. For instance, in March 2024, the United Nations Security Council twice set Ramadan as the term of a ceasefire, once in Sudan and once in Gaza, and the Secretary-General urged the Sudanese parties to honour the values of the month by stopping the fighting.[1] The premise is that a month devoted to fasting, prayer, and self-restraint lowers the appetite for fighting, and the existing evidence, drawn largely from data on terrorism, points the same way (Reese et al., 2017). If the premise fails, the month is not a window for de-escalation but a season of heightened risk.

This paper tests that premise directly. To measure armed conflict, I use event data from the Uppsala Conflict Data Program (UCDP) covering 107 countries over 1989–2022, together with declassified U.S. military records from Afghanistan whose time stamps are typically precise to the minute. To measure protest, I use Armed Conflict Location & Event Data (ACLED) records of peaceful demonstrations in 48 African countries over 1997–2023. For individual behavior, I use World Values Survey (WVS) interviews with roughly 40,000 Muslims in 23 countries, placed on the Hijri calendar by interview date. To observe how actors frame the month, I assemble a decade of the Islamic State's weekly newsletter and more than 1,000 Friday sermons scripted by the state religious ministries of Turkey and Egypt. A large language model classifies who initiated each conflict event and scores the religious and political content of each text.

Contrary to the premise, armed conflict rises during the month in predominantly Muslim countries,[2] driven by religious insurgencies rather than governments or secular-nationalist groups, while peaceful protest falls. This is not what researchers expected: in a forecast survey I collected from 61 researchers through the Social Science Prediction Platform (DellaVigna et al., 2019), three in four predicted a decline in armed conflict during Ramadan, and the median forecaster expected no change in protest (Figure A.1).

Why would a month of fasting and prayer bring more fighting and less protest? I argue religious observance reprices political action according to its doctrinal compatibility with worship. When an actor already holds a reading that renders an activity an act of worship, greater

[1] S/RES/2724 called for an immediate cessation of hostilities in Sudan during Ramadan (United Nations Security Council, 2024a), and S/RES/2728 demanded an immediate ceasefire in Gaza for the month (United Nations Security Council, 2024b). Neither resolution states what the Council expected violence to do absent the demand, but the Secretary-General made the premise explicit, telling the Council that a Ramadan cessation of hostilities "can help stem the suffering and usher the way to sustainable peace" (Guterres, 2024). Ramadan has anchored many earlier truces as well, including India's suspension of offensive operations in Jammu and Kashmir for the month in 2018 (Yasir and Gettleman, 2018) and the UN-brokered truce in Yemen, which took effect on the first day of Ramadan in 2022 (Guterres, 2022).

[2] I classify a country as predominantly Muslim if Muslims are at least 75% of its population.

observance raises its return because the same effort produces both political and religious value. When no such reading is available, observance instead raises its opportunity cost by increasing the value of time devoted to worship. The direction of the response thus depends entirely on how the political activity enters the actor's objective function, and the mechanism operates on the intensive margin of actors already in place.

This logic builds on the insight that a single tradition can support opposed readings of the same political act, as worship or as a distraction from it. What sets the direction of the response is not the tradition but the actor's own doctrinal reading, which fixes whether intensified observance acts as a subsidy for political effort or a tax on time.[3] Periods of intense ritual practice should therefore amplify the commitments these readings establish. For religious militant organizations, heightened observance reaffirms violent struggle as an act of devotion, while for actors holding a quietist reading, the same rituals deepen their restraint.

I formalize this in a relative-price framework showing how a common shock to the value of religious time generates opposite comparative statics depending on the doctrinal weight of the political act (Becker, 1965; Azzi and Ehrenberg, 1975), and I test these predictions empirically using Ramadan, whose competing readings let the two forces operate in opposite directions. Religious militant organizations invoke early Islamic victories to cast Ramadan as a season of conquest, so for an armed group that can frame combat as worship, the return to violence rises.[4] Classical teaching, by contrast, emphasizes the lunar month as a season of self-restraint and reconciliation, likening the fast to a shield and enjoining the faster to refrain even from quarreling. For ordinary citizens, who largely retain this reading, intensified fasting and prayer raise the shadow price of secular time, crowding out political activity such as protest that lacks doctrinal compatibility.

Observed by nearly two billion Muslims worldwide (Pew Research Center, 2017), Ramadan generates a recurring increase in religious observance through daytime fasting, intensified prayer, and communal worship (Lugo et al., 2012). Since the lunar calendar is about eleven days shorter than the Gregorian calendar, Ramadan rotates through the solar year, making its timing determined by astronomy rather than political or economic conditions (Campante and Yanagizawa-Drott, 2015). I exploit this 33-year drift with a balanced panel using country-by-Hijri-year and Gregorian-month fixed effects, isolating within-country, within-year variation net of common seasonal patterns. I sharpen this with a novel matched-date design that improves upon the literature's standard month-level or fasting-hour variation.[5] Specifically, the

[3] The pattern recurs across traditions. The *Bhagavad Gita*, in which Krishna instructs Arjuna to fight as a sacred duty, was read by some as a warrant for armed struggle and by Gandhi as an allegory for nonviolent resistance (Robinson, 2014). Religious Zionists have read the return to the Land of Israel as divinely sanctioned state-building, while anti-Zionist Haredim have drawn on the same textual tradition to oppose Jewish sovereignty before the Messiah (Ravitzky, 1996). Black churches have cast protest as an act of faith through Exodus, while other Christian clergy have invoked biblical injunctions to obey governing authorities as grounds for restraint (Chappell, 2005).

[4] The foundational precedents are the Battle of Badr (624 AD) and the Conquest of Mecca (630 AD), both occurring during Ramadan. Badr, fought on the seventeenth of the month against a far larger Meccan force, is remembered as a divinely aided triumph and is referenced in the Quran. The Conquest brought Mecca and its sanctuary under the control of the early Muslim community.

[5] Campante and Yanagizawa-Drott (2015) identify Ramadan effects from the interaction of latitude with the month's Gregorian timing, which generates differences in daily fasting hours across countries and years. I find that fasting duration is unrelated to either conflict or protest (Table A.15), consistent with a mechanism that operates

final days of Ramadan in one year occupy the same Gregorian dates as the days just after Ramadan in the following year, once the month has receded. Matching these days holds the solar calendar fixed while switching the treatment on and off.[6] I also leverage within-day variation from hourly conflict records, which provides the first hour-by-hour measurement of the fast's behavioral effects, to test how daytime fasting affects the distribution of attacks.

In the UCDP panel, armed conflict rises by 3.8% during Ramadan in the 31 predominantly Muslim countries in the sample. Restricting attention to the 14 countries averaging more than one rebel-government incident per month nearly doubles the Ramadan coefficient, to a 7.0% increase, implying approximately 26 additional armed clashes across the subsample during a typical Ramadan. The result replicates in independent data: ACLED yields effects nearly twice as large, and province-level military records from Afghanistan show an 8.7% monthly increase in insurgent direct fire. An event study confirms this escalation represents excess violence rather than temporal substitution, as adjacent months show no compensating decline. Using a large language model to classify the initiator in more than 100,000 conflict reports, I show that the escalation originates primarily from rebel offensives rather than government operations.[7]

The response splits along doctrine. I classify each insurgency from its founding charter, stated objectives, and the reported role of religious doctrine (Brosché and Sundberg, 2024; Crenshaw and Robinson, 2025), treating ideology as a predetermined constraint rather than reading it backward from wartime conduct. Doctrine in this sense is an organizational identity, not a position that can be revised over the horizon of a fasting month. The Ramadan increase is concentrated among religious insurgencies. Secular-nationalist organizations show no detectable response. The two sets operate in the same observant societies and face the same governments. What separates them is whether their stated objectives make combat an act of worship whose reward Ramadan multiplies.

If doctrine is the mechanism, it should leave a distinct footprint in how these groups communicate and whom they target. I score a decade of the self-proclaimed Islamic State's Arabic-language weekly newsletter for religious framing with an LLM. Its public messaging turns sharply more devotional during Ramadan, while its operational battle reporting, a within-issue placebo, does not.[8] Because harming non-combatants violates mainstream Islamic jurisprudence and invites the public backlash that prior work documents around sacred dates (Hassner, 2017; Reese et al., 2017), a doctrinally driven escalation should stop at the civilian

through the value of religious observance rather than through the physiological or economic burden of a longer fast.

[6] This comparison nets out any confound tied to a specific calendar date, such as weather or a fixed-date holiday, that month-level fixed effects cannot, and it isolates the month's most demanding slice, when worship and the cumulative toll of fasting both peak. I exclude the holiday of Eid al-Fitr from the control window.

[7] The attribution uses the GABRIEL text-analysis framework (Asirvatham et al., 2026), which classifies each event from its own source description as government-initiated, rebel-initiated, or neutral via log-probability extraction at temperature zero. The resulting patterns are consistent with the SIGACTS military records, which separately log friendly and enemy actions.

[8] The Islamic State's prolific media output (Zelin, 2015), specifically the consistently dated weekly *al-Naba*, provides a rare, systematic corpus for within-year comparisons. While prior work quantifies jihadist media at scale (Vergani and Bliuc, 2015), its devotional intensity across the liturgical calendar remains unexplored. Acknowledging limits to external validity, I treat this as a single-group mechanism probe, plausibly an upper bound on the reframing a typical religious insurgency would exhibit, and rely on the conflict panels for cross-group contrasts.

boundary. Based on UCDP and Global Terrorism Database records, civilian targeting does not rise during Ramadan for the general run of religious groups. The Islamic State is the sole exception. Its violence too is directed primarily against the state, but its takfiri doctrine strips whole categories of people of protected status, including Muslims it judges apostate (Bunzel, 2015).

The timing of violence provides a further test. The mechanism requires fighters to fast while operating. While Islamic jurisprudence explicitly exempts combatants, fighters frequently observe the fast to stack its spiritual merit alongside that of combat, a culture reinforced by insurgent codes of conduct (Hegghammer, 2017). As long as enough fighters fast to reduce average daytime capacity, the physical toll of abstinence should push labor-intensive combat into the hours following a meal. Using hourly military records from Afghanistan (Shaver and Wright, 2017), I find that the Ramadan increase in direct-fire attacks concentrates exactly in these two windows, after the pre-dawn suhoor and after iftar, and vanishes during the midday fast. Alternative accounts based on darkness, defender inattention, or publicity cannot explain why high-exertion combat peaks specifically in both post-meal windows.

Civilian protest provides the discriminating contrast. In ACLED data from 1997–2023, I find that demonstrations in predominantly Muslim countries decline by 6.9% during Ramadan. The military records of protest in Afghanistan also reveal a relative drop in Ramadan. To recover the micro-foundations of this decline, I exploit the timing of WVS interviews around Ramadan. Across 23 Muslim-majority countries between 2001 and 2022, tens of thousands of Muslims interviewed around the holy month report increased daily prayer alongside declines in political interest and willingness to demonstrate. Non-Muslim respondents interviewed in the same country-months show no comparable changes.

A remaining possibility, which the survey footprint alone cannot rule out, is that the state engineers the decline from the pulpit by using the obligatory Friday sermon to preach political quietism. In the centrally scripted sermons of Turkey and Egypt, however, devotional content surges during Ramadan while political content does not move, so the state amplifies the devotional pull of the month rather than loading the pulpit with political messaging.

A natural objection is that Ramadan is a bundled treatment, changing working hours, sleep, markets, and the operational tempo of the state along with observance. The ideology gradient addresses this. Every such change falls equally on religious and secular-nationalist insurgencies fighting the same governments, so any channel running through the bundle rather than through doctrine predicts escalation in both, and the secular-nationalist nulls reject it. A weakened-state account fails for a further reason, since state forces raise rather than lower their own operational tempo during the month. Section 8 addresses the remaining accounts: the propaganda value of attacks during the month, Ramadan almsgiving as militant finance (Limodio, 2022), displacement across months, fasting intensity, and media coverage, which cannot explain identical patterns in declassified military records.

The central contribution of this paper is to show that religion's seemingly contradictory political consequences, militancy and restraint, are two responses to the same economic force: religious observance changes the relative price of political action according to whether doctrine

makes that action part of worship. Prior work identifies important channels through which religion shapes economic and political behavior. Costly ritual sustains commitment and diverts effort from secular activity (Iannaccone, 1992; Berman and Laitin, 2008; Berman, 2011; Aimone et al., 2013), and longer Ramadan fasts lower output growth in a pattern consistent with club-good screening (Campante and Yanagizawa-Drott, 2015). Supernatural beliefs can reduce the perceived risk of death in combat and generate group-level benefits (Nunn and Sánchez de la Sierra, 2017). My framework adds a margin these channels do not address: whether the political action itself carries religious value. Once this doctrinal margin is recognized, a common increase in religious observance can generate opposite political responses among actors facing the same environment. Where ritual bulletproofing emboldens fighters by lowering the perceived risk of death, Ramadan raises the religious return to combat itself for militants whose doctrine sanctifies it, a return that martyrdom does not end but crowns in the hereafter. Unlike Ramadan's output effects, which grow with the length of the daily fast, the conflict and protest effects do not, pointing to the value of worship rather than the strictness of the fast. Religious observance thus acts as a subsidy for political action that doctrine renders sacred and as a tax on action that competes with worship for scarce time.

While a large literature studies the causal determinants of armed conflict (Miguel et al., 2004; Blattman and Miguel, 2010; Besley and Persson, 2011; Dube and Vargas, 2013; Nunn and Qian, 2014; McGuirk and Nunn, 2025), comparatively little work isolates the role of religious observance itself (Iyer, 2016).[9] Existing studies of Islamic countries largely emphasize restraint, documenting fewer incidents of terrorist attacks around holidays or long run effects of fasting on terrorism (Reese et al., 2017; Hodler et al., 2024). I show instead that greater observance can increase political violence, but only for religious organizations. I document this mechanism both in rhetoric, through a text analysis of insurgent messaging, and in behavior, leveraging within-day conflict variation unavailable in prior work. Furthermore, while terrorism-based measures primarily capture indiscriminate attacks, the Ramadan escalation I identify occurs through active armed conflict – a far more prevalent form of violence.

The paper also contributes to the strands of literature on collective action (Campante and Chor, 2012; Madestam et al., 2013; Enikolopov et al., 2020) and how religious practice shapes behavior (Clingingsmith et al., 2009; Campante and Yanagizawa-Drott, 2015; Becker and Pascali, 2019; Bazzi et al., 2020; Montero and Yang, 2022). Existing work emphasizes the mobilizing role of religion, whether through the motivation that devotion supplies (Hoffman and Jamal, 2014) or the coordination that communal assembly provides (Chaney, 2013; Kikuta, 2024). By contrast, I find the rising value of devotional time increases the opportunity cost of activities that lack religious compatibility, reducing civilian protest despite the coordination benefits generated by collective religious practice. I document this demobilization across aggregate protest behavior and individual political attitudes, while using a novel corpus of state-scripted sermons to show that the state amplifies the month's devotional pull without a corresponding surge in top-down political directives.

[9] A parallel literature examines the reverse relationship, asking how conflict and adversity shape religious commitment (Zussman, 2014; Henrich et al., 2019; Bentzen, 2019; Dube et al., 2022).

2 Background

Ramadan is the ninth month of the Hijri calendar, a purely lunar calendar of twelve months of 29 or 30 days that causes Ramadan to occur 10 to 12 days earlier each year relative to the Gregorian calendar. It is the month in which the Qur'an was first revealed, and fasting from dawn to sunset is one of the five pillars of Islam. The fast is obligatory for all Muslims who have reached puberty, with exemptions for those who are acutely or chronically ill, traveling, elderly, breastfeeding, diabetic, or pregnant. Observance requires abstaining from food and drink.

Ramadan transforms the daily lives of those who observe it. Adherents wake before dawn to eat (*suhur*) and break the fast at sunset (*iftar*), often in the company of family and co-religionists, and mosques host communal iftar meals throughout the month. Muslims pray five times daily and gather for Friday congregational prayers year-round, with heightened emphasis during Ramadan, and both men and women frequently attend additional night prayers (*tarawih*) after iftar. Predominantly Muslim countries have the highest rates of daily and weekly worship of any religious population (Pew Research Center, 2018).

The theological basis for this intensification is explicit in the prophetic traditions.[10] One canonical tradition holds that whoever fasts Ramadan in sincere faith, seeking God's reward, is forgiven his past sins (Sahih Bukhari, Book 2, Hadith 38), establishing the month as a period of uniquely heightened religious receptivity. A second (Sahih Muslim, Book 13, Hadith 212) likens the fast to a shield and instructs the faster to refrain from obscene language and quarreling. Together these traditions give ordinary Muslims strong incentives to redirect time and energy from secular activities toward worship, prayer, and self-restraint during the month.

While the Hijri calendar governs religious observance, virtually all Muslim-majority countries use the Gregorian calendar for civil, commercial, and governmental functions (Aydin, 2017). Work schedules, school terms, agricultural cycles, and military operations follow Gregorian months and seasons; even Saudi Arabia, which historically used the Hijri calendar for official purposes, adopted the Gregorian system for administrative functions in 2016. Ramadan's annual rotation therefore causes the fast to overlap with different seasonal and institutional conditions each year.

The Doctrinal Framing of Warfare

Islam designates four sacred months during which warfare is prohibited except in self-defense.[11] Ramadan is not among them. Foundational military victories in early Islam occurred during the month, including the Battle of Badr, the first major engagement between the early Muslims and the Meccan Quraysh, and the Conquest of Mecca. This precedent carried into the modern era: the 1973 Arab-Israeli War is remembered in the Arab world as the Ramadan War, and Iran named its 1982 offensive against Iraq Operation Ramadan to invoke the memory of early Islamic victories.

[10] All hadith references follow the standard numbering of the Sahih al-Bukhari and Sahih Muslim collections, accessed via `https://sunnah.com`.

[11] Muharram, Rajab, Dhul Qa'dah, and Dhu al-Hijjah (Al-Dawoody, 2011).

Both state forces and insurgent groups continue to draw on this narrative for religious and strategic legitimacy. Classical Islamic teaching emphasizes the "Greater Jihad," the internal struggle for self-restraint, while modern radical Islamist organizations prioritize the "Lesser Jihad" of offensive warfare, framing Ramadan as a temporal window in which the spiritual rewards of martyrdom and military success are multiplied. The Afghan Taliban have rejected calls for seasonal ceasefires as an "ignorance of religion", characterizing combat during the fast as obligatory worship (*ibadat*) whose merit is multiplied seventyfold (Taylor and Harooni, 2013). Spokesmen for the Islamic State have called Ramadan the "month of conquest and jihad", drawing an explicit parallel to the victories of the Prophet (Bin Ali, 2017; Amarasingam and Winter, 2017), and treat the hardship of the fast as ascetic preparation that increases the spiritual merit (*thawab*) of the fighter.

The doctrinal raw material for this appropriation is available in the canonical collections (Cook, 2015; Hegghammer, 2017). One tradition on guarding the frontier (*ribat*) promises a reward equal to a month of fasting and night prayer to whoever guards the frontier for a day and a night in the cause of God.[12] The hadith is conventionally read as honoring defensive frontier service. Militants invert the emphasis: issues of the Islamic State's weekly *al-Naba* published during Ramadan 1438 (June 2017) describe Ramadan as a month of "ribat, jihad, giving, and martyrdom" and argue that a single night of frontline guard duty surpasses the spiritual merit of worship on Laylat al-Qadr, the most sacred night of the year (Zelin, 2026). Equating modern offensive operations with classical defensive ribat is the interpretive move that recasts combat as the month's highest devotion.

Insurgent fighters frequently maintain the fast while conducting operations. Hegghammer (2017) documents fasting among jihadist fighters, including before suicide attacks, and the Taliban's code of conduct requires fighters to uphold core religious obligations (Munir, 2011). This compliance is the premise of the within-day analysis: if enough fighters fast for average daytime capacity to fall, the physical cost of dawn-to-sunset abstinence should shift labor-intensive combat into the hours after a meal.

3 Conceptual Framework

Sacred time raises the return to devotion. Its effect on political activity depends on whether the activity itself earns religious merit and on the worship it displaces. Building on the time-allocation approach of Becker (1965) and Azzi and Ehrenberg (1975), I model both margins within a common budget. Political effort rises when its additional religious return exceeds the increase in its time opportunity cost. The framework connects escalation by religious insurgents and declining civilian participation through this threshold. Proofs and extensions appear in Appendix A.

[12] Sunan an-Nasa'i, Book 25, Hadith 83, available at https://sunnah.com/nasai:3167.

3.1 Activities and religious production

An actor chooses political effort a, ordinary worship r, and other activities ℓ, subject to

$$\tau a + r + \ell = T, \qquad 0 \le a \le \bar{a}, \quad r, \ell \ge 0. \tag{1}$$

The parameter $\tau \in [0,1]$ measures the discretionary time displaced by political effort. Voluntary participation drawn entirely from the actor's own time has $\tau = 1$. An organizational deployment can instead replace another task within a fixed duty schedule, approaching $\tau = 0$. Deployment effort and the value of displaced tasks enter the net political return $B(a)$; Appendix A.1 derives this constraint from a full time budget.

Let $M(R) > 0$, with $M'(R) > 0$, denote the perceived religious reward multiplier associated with sacred salience R. For a stable religious valuation $\beta \ge 0$, write $m(R) = \beta M(R)$. Ordinary worship produces $f(r)$ units of religious merit, and political effort produces θa.[13] The actor solves

$$\max_{a,r,\ell} \; B(a) + u(\ell) + m(R)[f(r) + \theta a] \quad \text{subject to (1)}. \tag{2}$$

Assume $B'' < 0$, $u' > 0$, $u'' < 0$, $f' > 0$, and $f'' < 0$. Other activities can include leisure, household activity, or work. The main comparative statics concern $\beta > 0$ and an interior optimum; the appendix treats the nonreligious benchmark separately.

The coefficient θ is the action's *direct doctrinal credit*: positive for merit, zero for neutrality, and negative for a violation. A demonstration with $\theta = 0$ can become more costly during Ramadan because it displaces worship. The time budget generates that cost separately from doctrine. Founding documents, prior rulings, and recruitment material provide evidence on the ordering of θ across actor–action pairs; organizational arrangements inform τ. These interpretations require evidence independent of the Ramadan outcomes being explained. Religious valuation β concerns devotion, whereas θ concerns the credit attached to a particular act.[14]

3.2 The price of political effort

Let λ be the multiplier on the time constraint. At an interior optimum,

$$u'(\ell) = m f'(r) = \lambda, \qquad B'(a) + m\theta = \tau\lambda. \tag{3}$$

The effective marginal price of political effort is

$$p(a, m; \theta, \tau) = \tau\lambda(a, m) - m\theta, \qquad B'(a) = p(a, m; \theta, \tau). \tag{4}$$

It combines the shadow value of displaced time with the religious credit earned by the act. Holding political effort fixed and allowing worship and other activities to adjust, the shadow

[13] Utility is linear in religious output $Q = f(r) + \theta a$. Appendix A.8 replaces mQ with a concave $V(mQ)$, retaining the same Q, and shows that the threshold retains its direction when relative curvature is below one.

[14] Doctrine is a belief about religious reward. Treating M as perceived productivity holds preferences fixed, but the available outcomes cannot distinguish this interpretation from a changing utility weight on religious activity.

value of time rises with religious rewards:

$$\psi(a,m) \equiv \left.\frac{\partial \lambda}{\partial m}\right|_a = \frac{-u''(\ell)}{-u''(\ell) - mf''(r)} f'(r) > 0. \tag{5}$$

Strict concavity then gives

$$\frac{\mathrm{d}a^*}{\mathrm{d}R} = \frac{m'(R)}{\mathcal{D}}[\theta - \tau\psi(a^*,m)], \qquad \mathcal{D} > 0, \tag{6}$$

where Appendix A.2 derives the curvature term $\mathcal{D}$.

Proposition 1 (Religious credit and the opportunity cost of time)**.** *For $\beta > 0$ at an interior optimum, political effort rises with sacred salience if and only if $\theta > \tau\psi(a^*,m)$, and falls when the inequality is reversed. Ordinary worship rises whenever $\theta \leq 0$ and $\tau > 0$.*

Two benchmarks clarify the threshold. When $\tau = 0$, $\mathrm{d}a^*/\mathrm{d}m = \theta/E$, where $E = -B'' > 0$: meritorious action rises, neutral action is unchanged, and prohibited action falls. This recovers the reduced-form compatibility model as the case in which political effort replaces other duty tasks. When $\theta = 0$ and $\tau = 1$, $\mathrm{d}a^*/\mathrm{d}m = -\psi/\mathcal{D} < 0$, and worship rises. This is devotional displacement. With both margins active, even a meritorious action can decline if the worship opportunities it displaces become sufficiently valuable. As other activities become less able to absorb the adjustment, ψ approaches $f'(r)$ and the threshold compares the merit produced by political effort with that of the worship forgone.

3.3 Individuals and organizations

For observant civilians whose demonstrations receive no direct credit, $\theta = 0$ and $\tau > 0$ imply less political effort and more worship. The empirical test pairs declining participation with increased religious practice and compares individuals exposed to the Ramadan reward increase with those whose religious returns are unchanged. A common increase in assembly costs instead lowers participation in both groups.[15]

For an armed organization, leaders choose deployments while members bear their time and effort costs. Appendix A.4 gives an implementation in which leaders compensate members for participation. Religious credit lowers the compensation required; displaced devotional opportunities raise it. Under an adjustable transfer and a binding participation constraint, the leader's problem has the form of (2). A leader who internalizes the same member costs provides another interpretation.

This distinction gives the secular comparison a concrete benchmark. With $\theta_n = 0$, the direct response is $-m_n'\tau_n\psi_n/\mathcal{D}_n$. It approaches zero when deployments replace other duty tasks and is negative when they displace valued discretionary time. Professionalized duty scheduling thus offers a specific explanation for a secular null, conditional on evidence about

[15] A cost shock can also raise worship by releasing time from protest. The comparison therefore uses differential exposure to the religious shock as well as the joint movement of outcomes. It assumes comparable exposure to assembly costs and accounts for differences in response curvature; Appendix A.3 gives the exact responses. Surveyed interest and willingness are proxies for political engagement.

organizational arrangements. Religious insurgents with positive combat credit escalate when $\theta_r > \tau_r \psi_r$. Estimating the secular response and the religious–secular difference jointly assesses these configurations.[16]

The mechanism locates the direct increase in insurgent-initiated attacks. Under a neutral-credit, fixed-duty benchmark for state operations, government activity has no direct religious response, while strategic reactions can still raise its level. The attribution test therefore asks whether the religious–secular gap is concentrated in rebel initiation rather than government initiation. Similar government responses across opponent types would support this localization.[17]

3.4 Targets, timing, and communication

Doctrinal credit varies across targets. Under a shared time budget and separable concave returns, nonpositive civilian credit and nonnegative state-directed credit imply locally non-increasing civilian-directed effort. Conditional on total effort, sacred salience shifts allocation toward the target carrying greater credit (Appendix A.5). These results predict restraint toward doctrinally excluded targets alongside escalation on credited margins that cross the threshold. A doctrine crediting attacks on a civilian category changes the prediction for that category, subject to the same shadow-cost condition. The empirical comparison uses independently established target classifications and distinguishes civilians in event data from legitimate targets under an actor's doctrine.

Timing supplies a complementary test of the costs borne by participants. When fighters fast and operations can be rescheduled, rising exertion costs favor periods of recovery after meals. Holding a tactic's total output fixed, the shift is weaker for events requiring little exertion at execution, under comparable scheduling curvature. A profile based on time since nourishment accommodates both early-morning and post-iftar activity. Appendix A.6 derives these predictions and the additional conditions for attenuation against a night-capable adversary or invariance of monthly totals to fasting duration.

Finally, communication provides evidence on the proposed religious return. Statements linking political action to spiritual merit and sacred time support the interpretation of θ and M. Comparing insurgent messages with centrally scripted sermons helps assess whether this language extends beyond a general seasonal devotional turn.[18]

Together, these comparisons assess where escalation originates, which actions receive religious credit, and how participants allocate their time. Each relies on a classification supported by evidence independent of the estimated Ramadan responses: prior doctrine and founding material for θ, organizational arrangements for τ, and calendar exposure for m'. The threshold

[16] A near-zero coefficient restricts the scaled net response, not τ_n alone. Common shocks can offset displacement, and heterogeneous curvature affects group contrasts. Remark 2 states the additional restrictions needed to interpret the contrast as a difference in religious incentives.

[17] The state benchmark is an auxiliary assumption about the operations studied, rather than a claim about soldiers' religious beliefs. Mapping effort into event counts requires a monotone relationship under sufficiently stable production and reporting conditions.

[18] Communication is supporting mechanism evidence: the baseline model contains no communication choice. Generic references to prayer or fasting alone do not measure the religious credit attached to political action.

therefore acquires empirical content from the joint pattern across these decisions rather than from any single coefficient.

4 Data

Cross-country conflict and protest outcomes come from UCDP and ACLED. Hourly and province-level Afghan outcomes come from declassified military records. Individual behavior comes from the WVS. Two text corpora, a militant weekly and two centrally scripted state sermon systems, measure how actors themselves frame the month. Muslim population shares and daily fasting durations enter as sample restrictions and interaction terms.

Armed conflict

The primary source of armed conflict data is the UCDP Georeferenced Event Dataset (GED), covering 1989 to 2022. An event involves the use of armed force by the government of a state against one or more organized opposition groups, with at least one direct battle death (Sundberg and Melander, 2013). Although each event requires only a single battle death, inclusion is set at the dyad-year level: a government-opposition dyad enters the sample only if it produces at least 25 battle-related deaths within a calendar year, UCDP's standard threshold for an active conflict. GED therefore disaggregates into individual events precisely the conflicts that constitute UCDP's country-year armed conflict data rather than recording all low-intensity violence.

I restrict attention to state-based conflict, which accounts for more than 70% of all UCDP events. Each event records the identity of the rebel group involved, 139 distinct organizations across the sample, together with a source-article field containing a brief narrative description. Both are essential to the ideology and aggressor coding introduced in Section 6.1. The panel covers 107 countries, essentially all Muslim-majority states and most of the rest of the world (Figure A.2), and spans one complete rotation of Ramadan through the solar year.

Two categories of violence fall outside this outcome. Non-state violence has no government counterparty and therefore no analogue to the rebel-state margin studied here, so I exclude it throughout. One-sided violence and terrorism databases are a different case: they are unsuited to measuring rebel-state combat, which is where the escalation operates, but they are the appropriate instruments for civilian targeting, and I return to both in Section 7.2. This distinction also locates the paper relative to the terrorism literature. Databases built around a terrorism definition, such as the GTD, restrict coverage to civilian-directed attacks outside conventional warfare and frequently leave the perpetrator unattributed. They under-record the battlefield activity in which the escalation channel operates while over-representing the targets where backlash deters operations (Reese et al., 2017). The divergence between my conflict results and that literature therefore reflects a difference in targets and tactics.

Within-day attack timing

Hourly data on insurgent violence come from the U.S. Department of Defense Significant Activities (SIGACTS) dataset for Operation Enduring Freedom, covering more than 200,000 insurgent attacks in Afghanistan from 2002 through early 2015 (Condra et al., 2018; Shaver and Wright, 2017). Records were collected jointly by Afghan security forces and NATO's International Security Assistance Force under standardized military reporting protocols. Because they bypass media reporting entirely, they are not subject to the biases documented in media-based incident compilation (Weidmann, 2016). Each incident is geo-referenced using military grid coordinates and carries a time stamp, typically precise to the minute.

The data distinguish three attack types that differ in the physical exertion they demand at the moment of execution. Direct fire (DF) involves small arms and rocket-propelled grenades in close combat and requires sustained presence and effort. Indirect fire (IDF) involves mortars and rockets launched from prepared positions and requires less continuous exertion once the firing point is established. Improvised explosive devices (IEDs) are pre-positioned and triggered remotely or by command wire, requiring minimal real-time effort.

Two cleaning steps follow Condra et al. (2018) and address known reporting artifacts in the time stamps. First, I drop incidents reported at exactly midnight (00:00:00). A disproportionate mass of observations carries this exact stamp, which Condra et al. (2018) read as a default assigned when the actual time was unrecorded; observations at any other minute of the midnight hour are retained. Second, the minute-level distribution is skewed toward round numbers, with events more frequently listed on the 30th or 45th minute than on the 31st or 43rd, reflecting reporter rounding rather than attack timing. I therefore aggregate to the hour for the hour-of-day profile and to half-day shifts for the within-day reallocation tests.

Civilian protest

Protest outcomes come from the Armed Conflict Location and Event Data (ACLED) project, covering 48 African countries from 1997 to 2023 (Raleigh et al., 2023). I restrict the outcome to *peaceful protests*, gatherings in which demonstrators neither engage in violence or property destruction nor are met with force or intervention. The vast majority of recorded demonstrations meet this definition, and the sub-event type isolates non-violent civilian political behavior as the conceptual counterpart to the violence outcomes examined elsewhere. Peaceful protests are 88% of all recorded demonstrations in the predominantly Muslim sample, with 9% met by intervention and 3% by excessive force. I also use ACLED's politically violent events, restricted to state-rebel incidents to approximate the UCDP definition, as a cross-dataset check on the conflict results.

The African restriction is a coverage constraint rather than a design choice: ACLED began with Africa and offers no comparable pre-2016 coverage elsewhere, so this is the only region over which protest can be observed across a substantial portion of a lunar cycle. The setting is nonetheless well suited to the question. Africa contains more than half a billion Muslims and at least 17 Muslim-majority countries, nearly a third of the global Muslim population

(Kettani, 2010), and Pew Research Center (2018) reports that it registers the highest shares of respondents describing religion as very important and the highest daily prayer rates of any region, around 80%. Sample countries appear in Figure A.3.

I supplement this with the SIGACTS demonstration variable for Afghan provinces, which provides consistent coverage including remote districts where media reporting is sparse. The SIGACTS category is broader than ACLED's peaceful-protest definition, since military reporters did not consistently code participant identity, so I treat it as within-country corroboration outside of Africa rather than a second cross-country estimate.

Religious messaging corpora

To observe how actors frame the month in their own words, I assemble two text corpora. The first is 533 issues of the Islamic State's Arabic-language weekly *al-Naba*, published between December 2015 and April 2026, of which 46 fall in Ramadan, scraped from Zelin (2026) and segmented into editorials, feature essays, and operational attack reports. The newsletter is a numbered weekly issued almost without interruption for a decade, which is the structure a within-Hijri-year comparison requires and which no other armed group's output provides.

The second comprises the centrally drafted Friday sermon scripts of two state religious ministries: Turkey's *Diyanet*, 610 weekly scripts from March 2014 to February 2026, of which 50 fall in Ramadan, and Egypt's *Wizārat al-Awqāf*, 460 scripts from July 2016, when the ministry moved to verbatim distribution, to March 2026, of which 35 fall in Ramadan. In both systems the ministry drafts one script and distributes it for verbatim delivery nationwide, so each observation is the state's chosen message for a given week rather than a sample of what local preachers said. Appendix B documents the scraping, segmentation, attribute definitions, and scoring procedure for both corpora.

Individual survey outcomes

To recover the individual-level behavior underlying the protest decline, I use waves 4 through 7 of the WVS (Haerpfer et al., 2022), restricted to the 41 737 Muslim respondents across 23 Muslim-majority countries with exact interview dates, fielded between 2001 and 2022.[19] Exact dates allow each respondent to be matched to the Hijri calendar and classified by whether the interview fell during Ramadan.

I construct three binary outcomes. Prayer frequency equals one for respondents reporting prayer several times a day or once a day.[20] Political interest equals one for respondents who are "very" or "somewhat" interested in politics. Protest willingness equals one for respondents

[19] Countries enter at a 50% Muslim population share, a lower bar than the 75% threshold used in the aggregate panels. The aggregate panels require a high share because the outcome is a country-level count and the treatment's bite depends on how much of the population observes the month. Here the sample is conditioned on the respondent being Muslim, so exposure is established at the individual level and the country threshold serves only to ensure that Ramadan is socially prevalent. Of the countries meeting it, 23 have waves recording exact interview dates, which the Hijri matching requires.

[20] The underlying item is WVS `F028B`, "How often do you pray." The threshold isolates intense daily observance in an already devout sample: 75.1% pray at least once a day, 83.1% at least several times a week, and 93.7% at least occasionally.

who "have done" or "might do" peaceful demonstrations, and zero for those who "would never" participate.

Muslim population share

I merge the conflict and protest panels with country-level Muslim population shares from the World Religion Project (Maoz and Henderson, 2013) and Hackett et al. (2025). The WRP reports adherent counts and total population at five-year intervals from 1985 to 2010, which I interpolate to annual frequency; for 2010 to 2023 I interpolate the Pew Research Center series, which is reported at ten-year intervals. A country is classified as predominantly Muslim if its average share over the sample period is at least 75%. This yields 31 countries in the UCDP conflict panel and 14 in the ACLED protest panel.

Fasting duration

To test whether the intensity of the fast rather than the occurrence of the month drives the results, I construct daily fasting duration for each country-year as the interval between civil twilight and sunset at the capital city, web-scraped from the U.S. Naval Observatory's Astronomical Applications Department.[21] Duration varies across countries and years through two channels: Ramadan's rotation between long summer and short winter days, and latitude, which amplifies seasonal daylight variation. Equatorial countries face nearly constant fasting durations while countries farther from the equator face swings of up to seven hours across cycles. Because the variation is determined entirely by geography and astronomy, it is orthogonal to conflict conditions.

All outcome panels are balanced, with hours, days, or months recording no incidents coded as zero. Gregorian dates are converted to the Hijri calendar using the HijriDate Python package and cross-checked against independent sources.[22] Summary statistics appear in Table A.1 and Table A.2.

5 Identification Strategy

The baseline specification is a balanced country-by-Hijri-month panel, estimated on the thirty-one countries with a Muslim population share above 75%:

$$\log(y_{cmt} + 1) = \beta_1 \text{Ramadan}_{mt} + \gamma_g + \theta_{ct} + \epsilon_{cmt} \tag{7}$$

where y_{cmt} is the number of state-based armed conflicts or peaceful protests in country c, Hijri month m, and Hijri year t; Ramadan_{mt} is an indicator equal to one for the ninth Hijri month; γ_g are Gregorian month fixed effects; and θ_{ct} are country-by-Hijri-year fixed effects. Standard errors are clustered at the country level in the cross-country specifications and at the

[21] Civil twilight occurs when the sun is 6° below the horizon, closely approximating the 15° angle traditionally used to determine the start of the fast.

[22] The package is available at https://pypi.org/project/hijridate. Cross-checks use Islamic Philosophy Online (https://www.muslimphilosophy.com/ip/hijri.htm), Institute of Oriental Studies, Zürich University.

province level in the Afghan specifications. Because Ramadan falls in the same Hijri month in every country, Ramadan_{mt} varies only over time, and its identifying content comes from where the month lands in the solar calendar in a given year.

The identifying assumption is that, conditional on the two fixed effects, the Gregorian position of Ramadan is uncorrelated with the error term. This is a statement about the assignment mechanism rather than about behavior. The Hijri year is about eleven days shorter than the solar year, so Ramadan begins roughly eleven days earlier each year and cycles through all seasons over thirty-three years. The Gregorian month effects absorb any fixed seasonal pattern in conflict or protest, whether from weather, agricultural cycles, fighting seasons, or school calendars, and they do so even where that pattern differs across countries.[23] The country-by-Hijri-year effects absorb all shocks specific to a country in a given Hijri year, including economic conditions, political transitions, and national holidays. What remains for identification is the year-to-year shift in which Gregorian month coincides with the fast, and that shift is determined by lunar astronomy. It is not set by, nor responsive to, conflict or protest conditions in any country.

Conflict and protest counts have a long right tail, with a few country-months recording hundreds of events. To keep these observations from dominating the estimates, the baseline outcome is $\log(y+1)$, and β_1 approximates the proportional change in $y+1$. Because the outcomes are counts, I also estimate the main specifications with the inverse hyperbolic sine, negative binomial regression, and Poisson pseudo-maximum likelihood (PPML), for conflict in Table A.3 and for protest in Table A.8. The same-date design reports $\log(y+1)$ and PPML estimates side by side (Table 6). The main results generally hold under these alternatives, and I discuss these estimates alongside the corresponding results.

Four classes of confound survive this structure, and each is addressed by a separate specification. The first is any factor tied to a specific Gregorian calendar date rather than to a month on average. Weather varies within a month, fixed-date holidays fall on the same day each year, and the agricultural calendar is day-specific. To absorb these, I implement a same-date design that exploits the precise rate of the lunar drift. Because the Hijri year is about eleven days shorter, the last ten days of Ramadan in year t fall on nearly the same Gregorian dates as the days immediately after Ramadan in year $t+1$, by which point the month has receded. The same calendar date in the following year therefore serves as an untreated control that shares the treated day's solar position. Figure 3 illustrates the construction using the 2017–2018 cycle.

As a robustness check, I restrict the sample to these matched windows, collapse to the country-day level, and estimate a daily analogue of equation (7), replacing country-by-Hijri-year effects with country-by-cycle-year effects that group each treated window with its matched control. The treated window covers the last ten days of Ramadan, the month's most demanding slice: worship peaks as *Laylat al-Qadr* is sought on the odd-numbered nights, and the

[23] The Ramadan indicator is a pure time dummy, so in a balanced panel the country-by-Hijri-year effects residualize it identically in every country, and its mean within any country-by-Gregorian-month cell is the cross-country mean that γ_g removes. Country-specific seasonality therefore enters the omitted-variable term only through its average across countries, and the deviations from that average sum to zero. Replacing γ_g with country-by-Gregorian-month effects leaves the estimates in Table 1 numerically unchanged, as does any country-specific seasonal control of comparable flexibility.

cumulative toll of three weeks of fasting is at its highest. Both the doctrinal and physiological channels predict their largest effects there, making the window the most powerful test of the mechanism rather than a compromise. The cost is sample size. Each country contributes roughly twenty days per cycle and identification relies entirely on within-pair variation across adjacent years sharing the same calendar dates. To my knowledge, this exact-date matching approach has not previously been applied in the Ramadan literature.

The second is that Ramadan changes more than observance. The month compresses working hours, shifts sleep, alters consumption, disrupts local markets, and slows the state, and a month indicator cannot separate any of these from the repricing of devotional time. The ideology gradient addresses this. Every economic and administrative component of the bundle falls on religious and secular-nationalist insurgencies alike: the same shortened working day, the same disrupted markets, the same adversary. Any channel running through the bundle rather than through doctrine predicts escalation in both sets of groups, so the comparison of Ramadan responses across ideological categories in Section 6.1 is a test of the bundle as well as of compatibility. Physiological components are addressed separately below and in the fasting-duration test.

The third is that the response may vary with the size of the observant population. To test whether the effect rises with the Muslim share, I estimate a linear interaction on the full 107-country sample,

$$\log(y_{cmt}+1) = \beta_1 \text{Ramadan}_{mt} + \beta_2 \text{Ramadan}_{mt} \times \%\text{Muslim}_{ct} + \gamma_g + \theta_{ct} + \epsilon_{cmt}, \tag{8}$$

where the level of $\%\text{Muslim}_{ct}$ is absorbed by the country-by-year effects and does not enter separately, and a flexible version that replaces the linear term with discrete population-share categories:

$$\log(y_{cmt}+1) = \beta_1 \text{Ramadan}_{mt} + \sum_{k=1}^{4} \theta_k \cdot (D_{k,ct} \times \text{Ramadan}_{mt}) + \gamma_g + \theta_{ct} + \epsilon_{cmt}. \tag{9}$$

The categories are above 75%, 50–75, 25–50, and 1–25, with below 1 percent as the omitted group.[24] The flexible specification distinguishes an effect concentrated where observance is widespread from one rising smoothly with the Muslim share, and the near-zero categories double as a placebo where Ramadan carries no social salience.

The fourth is that fasting may operate on the timing of violence rather than its level. Under Assumption 1, daytime capacity falls and labor-intensive attacks should shift to the hours after a meal, leaving the monthly total unchanged. I test this within Afghanistan, where declassified military records carry time stamps precise to the minute. I split each day into a daylight fasting window (05:00–17:59) and a post-iftar window (18:00–04:59), cutoffs that approximate the

[24] Sample sizes are 31, 7, 8, 39, and 22 countries respectively in the UCDP panel, and 14, 3, 6, 19, and 6 in the ACLED panel.

sample-averaged sunset (18:05) and civil twilight (05:13) times, and estimate at the province-Hijri-month-shift level:

$$\log(y_{pmst}+1) = \beta_1 \,\text{Ramadan}_{mt} + \beta_2 \,\text{PostIftar}_s + \beta_3 \,(\text{Ramadan}_{mt} \times \text{PostIftar}_s) + \theta_{pt} + \gamma_{g(mt)} + \epsilon_{pmst}. \tag{10}$$

The coefficient of interest is β_3, the difference-in-differences estimate of the within-day reallocation. The total Ramadan effect in the post-iftar window is $\beta_1 + \beta_3$; the corresponding daytime effect is β_1 alone. Because the physiological prediction is specific to the exertion a tactic requires, I estimate equation (10) separately for direct fire, indirect fire, and IED explosions, which sit at the top, middle, and bottom of the exertion gradient. To trace the intraday profile at full resolution, I also estimate an hour-level event study:

$$\log(y_{pdht}+1) = \beta_1 \text{Ramadan}_{dt} + \sum_{j \neq 12} \alpha_j \text{Hour}_j + \sum_{j \neq 12} \delta_j (\text{Hour}_j \times \text{Ramadan}_{dt}) + \gamma_g + \theta_{pt} + \epsilon_{pdht}, \tag{11}$$

with 12:00 as the omitted base hour, a neutral point between post-sunset mobilization and the pre-dawn meal. The coefficients δ_j trace the intraday trajectory of violence during Ramadan relative to that base, providing a direct test of whether labor-intensive attacks concentrate after meals.

Ancillary tests and estimation

Two further specifications close the design. First, I carry the same-date windows into the within-day structure of equation (10), splitting each province-day into a fasting and a non-fasting shift, so that the timing pattern is estimated within pairs of days sharing the same solar position. Second, to distinguish the occurrence of Ramadan from the intensity of the fast, I interact the Ramadan indicator with the logarithm of daily fasting hours at the capital city, following Campante and Yanagizawa-Drott (2015). A national-level test is appropriate here rather than a subnational grid, as Hodler et al. (2024) use, because global subnational conflict panels are sparse, georeferencing in standard datasets is imprecise (Eck, 2012), and local religious-affiliation data are unavailable across the sample.

Two features of the design are worth stating before the results. The rotation of Ramadan through the solar calendar is exogenous by construction, so the design does not rest on a behavioral or institutional argument for its source of variation. And the fixed-effect structure is deliberately more demanding than the standard in this literature, which identifies from month-level or fasting-hour variation. The same-date design and the intraday specifications narrow the identifying variation further, at the cost of precision, in exchange for tests that no seasonal or calendar-day confound can realistically generate.

6 Main Results

This section takes the framework's predictions to the data in the order of the compatibility ranking. I first show that armed conflict rises during Ramadan and that the increase comes from religious insurgencies. Hourly military records from Afghanistan then show how the fast shapes the timing of that escalation. I next turn to civilian protest and the individual survey evidence at the other end of the ranking, and close with an exact-date design that removes calendar-day confounds from both sides of the result.

6.1 Effects on Armed Conflict

The first question is the direction of the conflict response. Most forecasters expected violence to fall during the month, whereas the framework predicts an increase wherever combatants can treat fighting as worship. Table 1 shows that Ramadan raises state-based armed conflict in predominantly Muslim countries: the coefficient of 0.037, significant at the 1% level, corresponds to a 3.8% increase.[25] The estimate is the same with only Gregorian month fixed effects (column 1) and with country-by-Hijri-year effects (column 2).

Column (3) restricts the sample to the 14 countries averaging more than one event per month. The coefficient nearly doubles to 0.068, a 7.0% increase, and a restricted wild cluster bootstrap confirms inference with 14 clusters ($p = 0.006$). These countries average 25.9 events per country-month against 11.8 across all 31 and account for most events in the sample; applied to that base, the coefficient implies roughly 26 additional events in a typical Ramadan.[26]

Column (4) adds Gregorian year-by-quarter fixed effects, which absorb seasonal patterns that change over the sample and any other shock common to all countries in a given quarter. The coefficient then compares Ramadan only with neighboring months in the same quarter and falls to 0.028 (s.e. 0.012). Columns (5) and (6) use all 107 countries to ask whether the response scales with the observant population. The linear interaction with Muslim share is insignificant, but discrete categories locate the effect: countries above 75% Muslim show a 3.6% increase relative to those below 1% ($p < 0.05$), while the intermediate categories are imprecise. The Ramadan main effect in column (6) is the response in those below-1% countries, where the month carries no social salience, and it serves as a placebo: the coefficient is 0.002 (s.e. 0.009), which excludes effects larger than about half the baseline estimate.[27] Because the categories are compared within the same months, this contrast is immune to shocks common to all countries, and it places the effect where observance is widespread rather than rising smoothly with the share.

The following sets of results support this interpretation. First, the estimate does not depend on which months serve as controls, and its variation across control windows is itself informative. Table A.4 restricts controls to the three Hijri months on either side of Ramadan, then drops those months so that controls come only from months four to six removed. The coefficient is 0.030 with near controls, 0.037 in the baseline using all non-Ramadan months,

[25] All percentages are computed as $\exp(\hat{\beta}) - 1$, the proportional change in $y + 1$ (Section 5).
[26] Computed as $14 \times \hat{\beta} \times (\bar{y} + 1)$ with $\hat{\beta} = 0.068$ and $\bar{y} = 25.9$.
[27] A separate regression on the 22 countries whose Muslim population is below 1% gives the same estimate.

and 0.045 with distant controls, and the same ordering holds in the interaction specifications (0.028 against 0.044). Conflict is somewhat elevated in the adjacent months themselves, with a mean of 12.2 in the near window against 11.6 in the distant one. Second, the effect is absent where Ramadan is not observed. In the 22 countries with Muslim populations below 1%, the Ramadan coefficient is 0.002 with a standard error of 0.009. The religious composition of a country, not the position of the month in the solar calendar, governs the response.

Third, the result replicates in independent data. Using ACLED political violence restricted to state-rebel incidents, the Ramadan coefficient is 0.076 and significant at the 1% level, roughly double the UCDP estimate (Table A.5), and the concentration in high-Muslim-share countries recurs there. The estimates in Table 1 are robust to excluding any single country (Figure A.4).

Armed conflict therefore rises during Ramadan, by about 4% in the typical predominantly Muslim country and 7% where fighting is concentrated, and only where the month is widely observed. The direction reverses the forecasters' prior and raises the question the framework is built to answer: who is fighting more?

Group Ideology and Aggression Attribution

The identity of the groups driving the increase is the paper's most direct test of compatibility. If doctrine is the operative parameter, the escalation should concentrate among groups whose objectives can render combat worship and be absent among secular-nationalist organizations facing the same states.

I classify all 139 rebel groups in the sample into two categories from each group's founding charter, stated objectives, and primary recruitment ideology, drawing on the UCDP Actor Dataset (Brosché and Sundberg, 2024) and the Mapping Militants Project (Crenshaw and Robinson, 2025).[28]

Religious groups (n = 55) operate within an Islamic framework that supplies the doctrinal raw material to recast offensive combat during Ramadan as meritorious worship. The most active are the broad Syrian insurgent coalition, the Afghan Taliban, the Islamic State, al-Shabaab, and Tehrik-i-Taliban Pakistan. What unites the category is access to that repertoire, which is available to groups whose identity and objectives are cast in religious terms.

Secular-nationalist groups (n = 84) operate outside any religious framework for armed struggle. Their stated objectives are ethno-nationalist, Marxist-Leninist, separatist, or factional, and religion plays no constitutive role. The most active are the Kurdistan Workers' Party and its affiliates, the Free Aceh Movement, and the secular Sudanese insurgencies. Lacking any doctrinal basis to recast Ramadan combat as worship, they should show no systematic Ramadan effect even where they sustain high baseline violence.

The religious category is dominated by a few large contributors. I re-estimate the main specifications dropping each in turn and as a bloc; the Ramadan coefficient retains its sign,

[28]The classification covers all 139 organized armed groups engaging in state-based conflict against government forces in the 30 Muslim-majority countries that record at least one such group over 1989–2022. State actors appearing as side B in the UCDP data, such as the Government of Pakistan in interstate episodes, are dropped. Appendix C gives the full list and the coding logic.

magnitude, and significance throughout, so the escalation reflects the category as a whole rather than the behavior of any single organization.[29]

Table 2 reports the Ramadan effect by group category. Religious groups show a 3.5% increase in conflict events (column 1) and a 3.9% increase in fatalities (column 2), both significant at the 5% level, against 1.3% and 0.8% for secular-nationalist groups, neither significant. The difference between the two, estimated from a stacked regression that accounts for their covariance, is 0.021 (s.e. 0.013) for events and 0.030 (s.e. 0.019) for fatalities, so total activity points toward the doctrinal contrast. The initiation results below show the gap is concentrated in rebel-initiated events, about a sixth of the total. The fatality response tracks the event response closely enough that the month raises the human cost of conflict at roughly the rate it raises the count of engagements.

Which groups are involved is not the same as who initiates. UCDP GED records both parties to each event but does not code the aggressor. If religious insurgents drive the increase by invoking precedents for offensive warfare, the effect should concentrate in rebel-initiated events. If instead governments intensify preemptive operations during the month, it should appear on the government side.

I classify each event from its own source description, together with the identities of Side A (government) and Side B (rebel group), as government-initiated, rebel-initiated, or neutral, using the GABRIEL framework (Asirvatham et al., 2026). Events whose descriptions do not unambiguously identify an initiator are coded neutral, a conservative default that trades sample size for attribution accuracy. Neutral events constitute 57% of the sample and are dropped; the remaining 43% divide into 26% government-initiated and 17% rebel-initiated (see Table A.6 for examples). Label definitions, the prompt, and the validation exercises are in Appendix B1.

Columns (3) and (4) disaggregate by initiator. Among religious groups, rebel-initiated events rise by 3.3% during Ramadan, significant at the 5% level, while government-initiated events against the same groups rise by 1%, significant only at the 10% level. The escalation is concentrated on the rebel side, at roughly three times the government magnitude. The government estimate is the more precisely estimated of the two, and its small magnitude is consistent with reactive or anticipatory counterinsurgency rather than with state-led escalation.

The secular-nationalist estimates provide the falsification. Rebel-initiated events by secular groups show no response (-0.000, s.e. 0.007), and the difference from religious rebels is 0.032 (s.e. 0.014), significant at the 5% level. This is the discriminating test against a mechanical time-reallocation account: if Ramadan simply rescheduled violence, secular rebels facing the same states under the same calendar would escalate alongside religious ones, and they do not. Government-initiated events, by contrast, rise by about 1% against both kinds of opponent, with no difference between them (0.000, s.e. 0.007), so the state's response to the month does not depend on whom it is fighting. The triple difference, which compares the gap between

[29] These results are available upon request.

religious and secular groups in rebel initiation with the same gap in government initiation, is 0.032, significant at the 5% level.[30]

The escalation is therefore a rebel offensive by religious insurgencies. They initiate more attacks during the month, governments respond modestly and equally to religious and secular opponents, and secular-nationalist rebels facing the same states do not escalate at all. This is the ordering the framework predicts from doctrinal compatibility. Because secular-nationalist groups share the month's economic and administrative changes with their religious counterparts, the contrast is also hard to reconcile with explanations that run through those changes; channels specific to religious groups, such as almsgiving and the publicity value of attacks, are taken up in Section 8.

6.2 Timing Attacks: Evidence from Afghanistan

The conflict panels show that religious insurgencies fight more during Ramadan, but not that the fighters themselves observe the month. Hourly military records from Afghanistan test this directly. If enough fighters keep the fast, the physical cost of dawn-to-sunset abstinence should push high-exertion combat into the hours after a meal, while pre-positioned attacks, which demand no effort at the moment of execution, should not shift. A simple fatigue account predicts something different: if fasting merely sapped fighters, violence would fall.

The main test splits each province-Hijri-month into a fasting window (05:00–17:59) and a post-iftar window (18:00–04:59) and estimates equation (10) separately by tactic (Table 3, Panel B). Direct fire, the most physically demanding tactic, rises 8.7% in the monthly aggregate (Panel A, column 1), and the increase falls after sunset: the daytime differential is a null of -1.8%, while the Ramadan $\times$ Post-Iftar interaction implies a 13.5% larger increase after iftar than during the fasting day, significant at the 1% level.

The gradient across tactics is the discriminating evidence. Indirect fire, which requires only brief positioning effort, shows no level effect and only a weak within-day tilt (column 2). IED explosions, which require no real-time exertion, rise 10.8%, the largest level effect in the table, with no within-day concentration (column 3). The timing response thus orders exactly as the exertion logic predicts, and the least demanding tactic shows the largest monthly rise: the doctrinal channel raises the level of all attack types, while the physiological constraint shapes only the timing of those that demand real-time effort. Fasting reorganizes *when* attacks occur rather than *whether* they occur.

Two further patterns sharpen this reading. Friendly direct fire by ISAF and Afghan government forces rises 7.9% during Ramadan but shows no within-day concentration (column 4), so the insurgent timing does not mirror Coalition patrol schedules, and the state's own aggregate rise rules out the concern that Ramadan degrades conflict measurement.[31] The insurgent shift

[30] As an additional robustness check, I classified the same event descriptions using a standard zero-shot prompting approach through the OpenAI API (`gpt-4o-mini`) rather than GABRIEL's log-probability extraction. The estimates are robust to this alternative classification strategy and are available upon request.

[31] SIGACTS records the target affiliation of each event but not, for Friendly Action events, the initiating actor. Insurgent direct fire can therefore be disaggregated by target (columns 5–6), whereas friendly direct fire (column 4) cannot be split into Coalition-led versus Afghan-led operations.

is also concentrated against Afghan forces, which lack night-fighting capability, and absent against Coalition forces, which have it, so insurgents choose targets to suit the hours the fast leaves them.[32]

Figure 1 shows the shape of the intraday profile relative to noon. Direct fire rises through the hours after the pre-dawn meal, peaking at 08:00, falls to a trough in mid-afternoon, when fasted physical performance is lowest (Chaouachi et al., 2009), and rises again at iftar. Indirect fire and IEDs show no comparable shape. Accounts based on darkness, defender inattention, or publicity cannot explain elevated close combat early in the morning, and a cognitive-depletion account, in which hunger raises impulsive aggression, would place the peak where the profile has its trough.

These results do not hinge on the log transformation. Table A.3 re-estimates the main specifications with the inverse hyperbolic sine, negative binomial regression, and Poisson pseudo-maximum likelihood (PPML). The results remain largely similar. Direct fire shifts toward the post-iftar window under all three, and IED explosions rise in level under all three, although under the count models IEDs also concentrate after iftar. The aggregate conflict response, positive under the inverse hyperbolic sine (0.044) and negative binomial regression (0.069), is zero under PPML (0.006). PPML weights country-months by their event counts, concentrating the estimate on Syria and Afghanistan, where Ramadan fell inside the fighting season throughout their most active years. Under all three, rebel-initiated attacks by religious groups rise, by up to 26% under PPML, while those by secular groups do not. In the exact-date design, which compares Ramadan days with the same calendar dates a year later, the PPML estimate is 0.099 and statistically significant at the 1% level.

The timing evidence adds two things to the conflict panels. Enough of the fighters who escalate are themselves fasting to bend the timing of close combat, even though Islamic jurisprudence generally exempts combatants, which ties the escalation to observance by the fighters rather than to the calendar alone. Therefore, the doctrinal channel sets the level of violence, while physiology sets its timing.

6.3 Effects on Civilian Protests

Civilian protest sits at the other end of the compatibility ranking. For an observant citizen, protest is not worship but time taken away from it, so the framework predicts that Ramadan should reduce protest, and only where the month is widely observed.

Table 4 begins with cross-country evidence from ACLED. Column 1, restricted to the 14 African countries with Muslim population shares above 75%, shows a 6.9% decline in peaceful protests during Ramadan, significant at the 10% level. With 14 clusters the estimate is imprecise, and the pattern emerges more sharply in the full 48-country sample. Column 2 interacts Ramadan with the continuous Muslim population share: the interaction is -0.117 ($p = 0.01$), so the suppression strengthens as religious composition rises. Column 3 replaces the linear

[32] Against Coalition forces (column 5), monthly direct fire rises 7.3% with no within-day concentration. Against Afghan state forces (column 6), the monthly effect is zero, but the interaction is 0.119 (significant at the 1% level): a 4.3% daytime decline offset by a 12.6% post-iftar rise. Coalition night-vision optics, drone surveillance, and rapid air support raise the cost of attacking them after dark, while Afghan units lack equivalent capability.

term with discrete categories. Countries above 75% Muslim show a 12.4% relative decline compared with the group below 1% Muslim, significant at the 5% level, and no other category reaches conventional significance. The Ramadan main effect in that specification, which applies to the negligible-Muslim reference group, is a statistically insignificant $+0.062$ and serves as a built-in placebo: the calendar position of the month does not suppress protest where the month is not observed.[33]

Columns 4 and 5 turn to within-country evidence from the SIGACTS demonstration records for Afghan provinces, which bypass media reporting entirely. Monthly demonstrations fall 4% during Ramadan at the province level (column 4), significant at the 1% level, corroborating the cross-country result in a predominantly Muslim setting from an independent source. Column 5 decomposes the monthly effect into pre-iftar and post-iftar half-day shifts. The suppression is concentrated in the daytime: pre-iftar demonstrations fall 2.5% (significant at the 1% level), while the combined post-iftar effect ($\beta_1 + \beta_3 = -0.006$) is indistinguishable from zero.

The post-iftar null should not be read as a behavioral response. The Post-Iftar main effect of -0.051 confirms that demonstrations cluster in daylight and are rare after sunset, so the post-iftar baseline sits near a floor and leaves little for Ramadan to suppress. What the decomposition does establish is that the decline falls in the window the fast occupies. This is the footprint the time-allocation channel predicts: Ramadan concentrates devotional obligation and compressed work schedules in precisely the daytime hours when protest ordinarily occurs, crowding out the conventional mobilization window.

The decline is not an artifact of ACLED's coding. Because the peaceful-protest sub-event is assigned only when demonstrators face no intervention, a Ramadan increase in policing would reclassify events and lower the count with no change in behavior. Table A.7 rejects this, showing that all three protest sub-events taken together fall by 7.9%, more than the peaceful component alone. Both repression categories fall rather than rise. Conditional on a demonstration occurring, the share coded peaceful rises. Violent demonstrations, recorded under ACLED's separate riot category, show a precise null, so the decline is not displaced across the event-type boundary either.

The protest decline does not hinge on the log transformation (Table A.8). With the baseline fixed effects, the Ramadan coefficient in the 14 predominantly Muslim countries is -0.082 under the inverse hyperbolic sine and -0.181 under negative binomial regression, significant at the 5% and 1% levels respectively, and -0.068 under PPML, close to the log estimate of -0.071 but imprecise. In the full 48-country sample, countries above 75% Muslim show the largest relative decline under all three estimators, significant except under PPML. In Afghanistan, daytime demonstrations fall significantly under all three. The count models also imply a decline after iftar, smaller in proportion than the daytime one, but because demonstrations are

[33]The main effect is positive and significant in the linear specification of column 2 ($+0.052$, $p < 0.01$) and grows under the distant-control window of Table A.9. Because the Ramadan indicator is a pure time dummy, this coefficient is identified from the Hijri-month composition of the comparison period rather than from anything specific to Ramadan, and it is absorbed once the reference group is defined by category as in column 3. The interaction, not the main effect, is the quantity the mechanism predicts.

less than half as common in that window outside Ramadan, most of the reduction in levels still falls in the fasting window.

The estimates do not depend on which months serve as controls, and the variation across control windows is itself informative. Table A.9 restricts the comparison to the three Hijri months on either side of Ramadan, then drops those months so that controls come only from months four to six removed. The suppression is weakest against near controls and strongest against distant ones: the Muslim-sample coefficient moves from -0.056 to -0.071 in the baseline to -0.072, and the above-75% interaction from -0.125 to -0.132 to -0.162. The ordering mirrors the pattern in the conflict results and follows from the same source. Protest is depressed in the months flanking Ramadan as well, so including them among the controls biases the estimate toward zero and the baseline is conservative.[34] The estimates are also robust to excluding any single country (Figure A.5).

Protest therefore moves opposite to armed conflict. It falls where the month is observed, does not fall where it is not, and falls in the daytime hours that fasting and worship occupy. The same month that raises the return to fighting for religious insurgencies raises the cost of protest for observant citizens.

6.4 Individual Survey Evidence

A decline in protest counts could reflect fewer opportunities to protest rather than less desire to. The survey tests the individual footprint the framework predicts: if observance drives the decline, religious practice should rise while political engagement falls, and only among those who observe the month.

The individual-level data show the same divergence and separate the effect from the season. Ramadan raises religious engagement and lowers political engagement among Muslims, and neither movement appears among non-Muslims in the same countries.

The evidence comes from waves 4 through 7 of the WVS, covering 41 737 Muslim respondents in 23 Muslim-majority countries with exact interview dates, fielded between 2001 and 2022. Exact dates allow each respondent to be matched to the Hijri calendar and classified by whether the interview fell during Ramadan. The analytical sample is restricted to respondents who report religion as "very important" or "rather important," a criterion met by approximately 95% of Muslim respondents. The restriction parallels the country-level restriction to Muslim-majority contexts and serves the same purpose: the mechanism operates through observance, so respondents for whom the treatment is inactive would contribute attenuating noise without adding identification leverage.

Interview timing could correlate with respondent characteristics, so I test balance on eight demographic covariates. Five are balanced; three show small imbalances, in age, income, and reported religious importance, all absorbed by the demographic controls. Fieldwork intensity

[34] One result strengthens materially under the distant window. The Ramadan $\times$ Post-Iftar interaction in the Afghan panel is 0.026 ($p = 0.008$) with distant controls, against an insignificant 0.018 with near controls and an insignificant estimate in the baseline. The daytime concentration of the protest decline is therefore detectable once the flanking months are removed from the comparison

and interview days per cell are also balanced across Ramadan and non-Ramadan periods, consistent with schedules set by logistics rather than the religious calendar (Appendix Table A.10).

Table 5 reports the estimates, with each outcome in two specifications: a seven-month Hijri window centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah) and the full twelve-month sample. The narrower window compares Ramadan-period respondents to interviews in Hijri-adjacent months, where seasonality, fieldwork composition, and respondent demographics are closest; the full sample is the broader check. Prayer rises 5.1–5.2 percentage points off a baseline of roughly 76%, significant at the 10% level (columns 1–2). Daily or near-daily prayer is a habitual behavior, and the elevated baseline reflects the religious-importance restriction; that a shift of this size is detectable even in this already-observant population confirms that Ramadan measurably heightens religious engagement. Political interest falls 5.4–5.5 percentage points, about 12% of the baseline mean (columns 3–4), and willingness to demonstrate falls 3.9–4.2 percentage points, about 11% of the baseline (columns 5–6). The coefficients are nearly identical across the seven-month and full-sample specifications, indicating that the results do not depend on the Hijri-month comparison group and that the Gregorian month fixed effects already absorb the seasonality that the narrower window further isolates.

To assess whether these effects reflect individual religious observance rather than country-level fieldwork artifacts or seasonal patterns the fixed effects fail to absorb, I estimate the same specifications on non-Muslim respondents living in the same 23 countries (Appendix Table A.11). Non-Muslims share the same fieldwork timing, country-wave composition, and societal context as Muslim respondents but do not observe Ramadan, so the Ramadan indicator should have no effect on their religiosity or political engagement if the main results operate through observance. None of the six placebo coefficients reaches statistical significance, supporting the interpretation of the Ramadan effects as operating through individual religious observance rather than through country-level confounds.

The survey thus recovers the footprint that separates time reallocation from a supply-side alternative. Prayer rises while political interest and willingness to demonstrate fall, and only among those who observe the month; a state that simply raised the cost of assembly would lower willingness to protest without raising prayer.

6.5 Exact-Date Matching

Seasonality is the leading threat to the baseline estimates, and this design removes it by construction. The Hijri year is roughly eleven days shorter than the solar year, so the last ten days of Ramadan in year t fall on nearly the same Gregorian dates as the early post-Ramadan period in year $t + 1$, by which point the month has receded (see Figure 3). Comparing the treated days to those same calendar dates the following year holds each day's solar position fixed. Country-by-cycle-year fixed effects group each treated window with its matched control, so identification comes entirely from within-pair variation across adjacent years; Gregorian month fixed effects absorb residual cross-cycle seasonality. The design differences out any confound tied to a specific calendar date, such as weather, a fixed-date holiday, or the agricultural calendar, that month-level fixed effects cannot. The treated window is the month's most

demanding slice, when *Laylat al-Qadr* is sought on the odd-numbered nights and the cumulative toll of three weeks of fasting is greatest, so both channels the paper identifies predict their largest effects there.

Table 6 reports the estimates, with OLS on $\log(Y+1)$ in odd columns and PPML on the raw count in even columns. The two estimators differ in magnitude by an order of magnitude or more throughout, and the gap is not a puzzle about which is right. The log specification averages over a daily panel in which most country-days are zero, so its coefficient is dominated by movements between zero and one event; PPML weights by the conditional mean and loads on the high-intensity country-days where most events occur. OLS gives the response of the typical country-day, PPML the response of the event count. The sign and significance agree across both throughout, which is what the design is being asked to establish.

Armed conflict rises. The OLS coefficient is 0.009 ($p < 0.10$), a 0.9% increase per country-day; the PPML coefficient is 0.099 ($p < 0.01$), a 10.4% increase in the event count (columns 1–2).[35]

Within these windows, the within-day concentration of high-exertion combat survives. Using Afghan direct fire, the *Ramadan* $\times$ *Post-iftar* interaction is 0.023 under OLS and 0.129 under PPML, significant at the 5% level in both (columns 3–4), so the post-iftar shift documented in Section 6.2 is not an artifact of seasonal timing. The daytime coefficient is a significant -0.025 under OLS, indicating a genuine reallocation out of fasting hours rather than an addition on top of an unchanged daytime baseline. Under PPML the daytime coefficient is -0.068 with a standard error of 0.094, too imprecise to distinguish a decline from a null, so whether the daytime margin falls or holds flat within these windows is not resolved here.

Protest declines, and by more than in the monthly estimates. The OLS coefficient is -0.020 ($p < 0.01$), a 2.0% daily decline per country, and the PPML coefficient is -0.327 ($p < 0.01$), a 27.9% decline in the event count (columns 5–6). The strengthening is consistent with civilian observance peaking in the final ten days and the crowd-out of secular political activity being sharpest there.

Columns 7–8 extend the within-day decomposition to Afghan demonstrations, where the panel is too sparse to support inference. The outcome averages 0.005 events per province-day-shift, and PPML retains only 1,824 observations. The OLS Ramadan main effect is -0.004 ($p < 0.05$), indicating fewer daytime demonstrations, and the interaction is positive but insignificant. The PPML estimates are not interpretable at this base rate and I do not read magnitudes from them.

Both sides of the Ramadan effect therefore survive a design that differences out the seasonal component of any specific calendar date: insurgent attacks rise, civilian demonstrations fall, and the within-day concentration of high-exertion combat holds.[36]

[35] PPML drops country-cycle groups without events, reducing the Muslim-majority sample from 22,196 to 7,214 and eliminating 9 of 31 countries. These groups carry no information about the coefficient under PPML, so the estimate is the proportional effect among country-cycles with at least one event.

[36] The estimates are robust to excluding Eid al-Fitr from the control window.

7 Mechanisms

The results so far establish the direction of each response and who drives it. This section asks whether the mechanism leaves the further footprints the framework predicts. If the religious return to combat drives the escalation, insurgents' own messaging should turn more devotional during the month. The same text tools test whether the state, rather than the citizen, drives the protest decline. And the escalation should stop at the civilian boundary except where doctrine erases it.

7.1 Religious Messaging

Insurgent Newsletters

If the escalation reflects doctrinal framing, it should appear in the groups' own discourse. Ramadan's sacred salience should sharpen the devotional content of their messaging, and it should do so in the material where religious arguments are made rather than in the reporting of operations, whose register is fixed across the calendar. I test this using the Islamic State's Arabic-language weekly *al-Naba*, a numbered periodical issued almost without interruption for a decade.[37] Appendix B describes the corpus and the segmentation.

I score every section on *devotional and ritual emphasis*, the extent to which the text foregrounds personal worship such as fasting, night prayer, Qur'anic recitation, and supplication rather than combat operations or political argument. The score is assigned by the same GABRIEL framework used for aggressor attribution, on a 0–100 scale, and the attribute definition names no month, date, or historical event, so the rater cannot infer the treatment from the prompt and inflate the score mechanically.[38] The operational news serves as a within-issue placebo. Each item reports a province, a target, a tactic, and a casualty count, a template with no slot for a devotional argument, so its register is fixed across the calendar and any Ramadan movement in the score would indicate the rater keying on surface cues rather than on doctrinal content. Hijri-year fixed effects absorb the secular drift in the newsletter's content, so the comparison is Ramadan against non-Ramadan sections within the same year.

Devotional emphasis in feature essays averages 50.1 outside Ramadan and 78.1 during it, a 28-point shift (Table A.12). Editorials and operational news move far less over the same window, by 5.9 and 7.5 points respectively. The operational base is low, so the proportional movement in that series is not negligible; what the placebo requires is that the absolute reorientation be concentrated where religious arguments are made, and on the common scale the feature essays move roughly three to five times as far as either other series. Because feature-essay frequency is itself flat across the calendar, ranging from 21 to 30 essays per month with 28

[37] A fully parallel test would require comparable, systematically dated messaging across religious and secular-nationalist groups, but no such corpus exists; most armed groups publish no regular archived periodical. *al-Naba* is the closest approximation, and the cost is external validity. I read the text as a single-group probe of the mechanism and leave the cross-group contrast to the conflict panels.

[38] The definition does name fasting among its examples of personal worship, which is innocuous in this corpus but is a treatment cue in the sermon analysis below, where I report a second variant with the enumeration removed.

in Ramadan, the contrast reflects what the group writes about rather than how much it writes. The measure also recovers the devotional calendar in Figure 4.

Reading the high-scoring issues shows what the shift consists of. During Ramadan the feature essays turn to the substance of the month's worship: the virtues of the fast, the night prayer (*qiyam al-layl*), Qur'anic recitation, supplication, and the devotional example of the early Muslims, material of the kind carried by a dedicated piece such as "Our Predecessors in Ramadan: The Night Prayer" that a non-Ramadan issue does not run.[39] This devotional turn is the vehicle of the doctrinal-framing channel, not a retreat from it. It meets a readership at the peak of its religious receptivity, and the group repeatedly yokes the worship to fighting, recasting the combatant's hardship, fasting and sleeplessness under arms, as ascetic worship whose merit the month multiplies.

A second construct points the same way. *Sacralization of the present moment*, the extent to which a section casts the current period as a specially blessed or rewarding time for striving, rises during Ramadan in both editorials and feature essays (Appendix Figure A.6). This is the framing closest to combat-as-worship, since it makes the reward to effort depend on when the effort occurs. I treat it as secondary, since the construct bleeds into the operational reports, but it reinforces the same reorientation.

Together the measures show that the Islamic State reorients its messaging toward the sacred during Ramadan, the discursive counterpart to the behavioral escalation documented above. The reorientation is not generic piety. Devotional messaging peaks in both Ramadan and Dhul-Hijjah, yet the conflict escalation is unique to Ramadan, so heightened devotion alone does not mobilize violence. What singles out Ramadan is the combat-specific precedent and collective ritual that the group invokes to recast the fasting month as a season of war.

State Sermons

The survey footprint in Section 6.4, more prayer and less political interest, is consistent with time reallocation, but a state-messaging channel could produce the same footprint. A regime that preaches political quietism from the Friday pulpit during Ramadan would raise reported devotion and lower political engagement without any change in the value of devotional time. The pulpit is well suited to this purpose: the Friday sermon is delivered weekly in every mosque, attendance is obligatory for adult men, and the sermon is a condition of the prayer's validity rather than an addition to it, so exposure cannot be separated from attendance. Within a country a single state-authored script can therefore reach the entire observant population.[40]

I test this directly using the official sermon corpora of Turkey (*Diyanet*) and Egypt (*Wizārat al-Awqāf*), two systems in which the ministry drafts one script per week and distributes it

[39] *al-Naba* no. 334.

[40] The script must be delivered in congregation and preceded by a two-part sermon (*khutbah*) from the pulpit (*minbar*), so the state's message and the religious obligation are inseparable at the point of delivery.

nationwide.[41] Because the script is written centrally, each observation is the state's chosen message for a given week rather than a sample of what individual preachers said.[42]

I score each script on the same devotional scale applied to *al-Naba*, along with two further constructs. *Political salience* captures regime-relevant content, including national unity, state authority, military sacrifice, loyalty to leadership, and sovereignty, which is what a state loading the channel in its own support would produce. *Strict religious preaching* captures doctrinally authoritative instruction and serves as a baseline placebo analogous to the operational news in the insurgent corpus. All three are measured on a 0–100 scale, with no date passed to the model.

Devotional emphasis surges during Ramadan in both systems (Figure 5): +21.2 points in Turkey on a non-Ramadan mean of 39.0, and +39.0 points in Egypt on a mean of 30.9, both significant at the 1% level.[43] Strict religious preaching rises only marginally in both (+4.8 and +4.7 points on non-Ramadan means of 72.4 and 74.7), and sermon length is unaffected, so the month shifts the composition of the script rather than its volume.

Political salience does not move. It sits at 23 outside Ramadan against 22 within it in Turkey, and 40 against 39 in Egypt, while devotional emphasis rises from 39 to 64 and from 31 to 69 in the same scripts. The estimates bound any increase at 4.5 points, 19% of the Turkish non-Ramadan mean and 11% of the Egyptian. The stability holds across the calendar as well as at the Ramadan margin: the political series sits within a narrow band in every Hijri month, with no month in which the state markedly politicizes the pulpit.

The state therefore does not inject pro-regime political rhetoric into the pulpit when its audience is largest. A residual version of the account holds that the state floods the pulpit with devotion to crowd out politics entirely, leaving congregations absorbed in worship. Text analysis cannot separate a deliberate crowding-out strategy from a ministry observing the holy month, but three patterns limit the reading. Religious content already dominates these scripts year-round, so the devotional shift does not displace a politically loaded baseline. The messaging follows the religious calendar rather than the political one, rising again in Dhul-Hijjah and dropping in Shawwal, which is not the pattern of a tool calibrated to protest risk. And the same surge appears in *al-Naba*, a publication trying to mobilize rather than demobilize its readers.

That parallel is the substantive point. Because both a state ministry and the Islamic State turn sharply toward devotion during Ramadan, heightened devotional intensity alone cannot explain the divergence in behavior. The two actors share a month and a devotional register, but only one possesses a doctrine that maps the shared intensity into violence. The crowding-out

[41] Appendix B documents the corpus construction, the scraping, and the attribute definitions. The Turkish sample runs from March 2014 to February 2026 and the Egyptian from July 2016, when the Awqāf moved to written distribution, to March 2026, giving 610 and 460 weekly scripts of which 50 and 35 fall in Ramadan.

[42] Compliance is institutionally supported and high but incomplete, and the object measured here does not require it. The script is therefore the message the state chose to send, which is what the state-messaging account requires, rather than a transcript of what any particular congregation heard.

[43] Because the devotional definition names fasting among its examples of personal worship, which is innocuous in the insurgent corpus but a treatment cue here, I score a second variant with the enumeration removed. The two variants give Ramadan coefficients within 0.8 points of one another in Turkey and 2.2 in Egypt, so the estimate reflects devotional register rather than a single word.

reading, taken seriously, also relocates the state inside the mechanism rather than outside it: whether the devotional surge reflects the ministry's strategy or the month's demand, its effect on a listener is to raise the value of devotional time, which is the shadow price in Equation (2) rather than a rival to it.

7.2 Civilian Targeting Channel

The paper's central result, that armed conflict rises during Ramadan, runs against a terrorism literature documenting restraint, with insurgents committing fewer attacks on sacred dates to avoid public backlash (Reese et al., 2017; Hodler et al., 2024). The framework reconciles the two by target: the doctrinal reading that sacralizes Ramadan combat licenses carrying the fight to the enemy, while killing civilians during a sacred period carries a reputational cost the same religious environment magnifies. If that holds, the state-directed escalation of Section 6.1 should stop at the civilian boundary for the general run of groups and cross it only where doctrine erases the line between enemy and civilian.

I take the question to two datasets of civilian-directed violence that the state-based panels cannot address: UCDP one-sided events, in which an organized group deliberately kills civilians, over 1989–2022, and attacks on private citizens in the Global Terrorism Database over 1970–2020 (START, 2022). The two differ in inclusion criteria and coverage, which is why agreement between them is informative: GTD records civilian-directed attacks without UCDP's annual battle-death threshold and names a broader set of perpetrators. Perpetrators are classified by the ideology scheme of Section 6.1 and the predominantly Muslim setting is held fixed, under the same country-by-Hijri-year and Gregorian-month fixed effects. Appendix D gives the construction.

According to Table A.13, for the religious category as a whole, civilian targeting rises 2.7% in UCDP and 1.5% in the GTD (columns 1 and 5). Secular-nationalist groups show a null result in UCDP and GTD (columns 2 and 6), extending the ideology gradient to this margin.

The religious category average is a mixture rather than a uniform effect. Dropping the Islamic State collapses the increase to an insignificant 1.7% in UCDP and 0.9% in the GTD (columns 3 and 7), whereas the Islamic State alone raises civilian targeting by 5.2% and 5.1% (columns 4 and 8). The split is therefore escalation concentrated at one organization set against no detectable increase among the rest, and it appears in two differently constructed datasets, so it is unlikely to reflect an artifact of either.

This inverts the state margin. Against the state the escalation is general and does not depend on any single religious rebel group, the Islamic State included (Section 6.1); against civilians it disappears once that group is removed. Two forces produce the divergence. For most religious insurgents the reputational cost of killing civilians is high and the state-directed increase does not carry onto civilians, in keeping with the restraint argument (Reese et al., 2017). The Islamic State is the exception. Its takfiri doctrine designates civilian populations as legitimate targets, dissolving the constraint that binds the others, and its record matches the coeffi-

cient. The group has repeatedly cast Ramadan as a season of intensified attack and mounted mass-casualty operations against civilians during the month.[44]

The finding tightens the doctrinal account rather than qualifying it. The escalation crosses the civilian boundary only for the group whose doctrine sacralizes civilian killing and halts at the state boundary for the rest, so doctrine governs not only whether a group escalates during Ramadan but which targets that escalation reaches. This is the target-side counterpart to the earlier discursive evidence, where the Islamic State's own newsletter turns sharply devotional and martial during the month, and it fixes the paper's relationship to the terrorism literature: restraint is the conduct of the general run of groups, whose civilian targeting does not rise, not that of the Islamic State, whose does.

The civilian-margin null also closes off one account of the protest decline. Under a fear-based reading, rising civilian-directed violence deters public assembly. That would require civilian targeting to climb across the predominantly Muslim countries where protest falls, and outside the Islamic State it does not move, so the increase is far too narrow to demobilize entire societies. A direct test confirms the decoupling: state-based armed conflict and demonstrations are essentially uncorrelated across country-months in the ACLED data, so the months that carry more fighting are not the months that carry less protest.[45] The protest decline therefore rests on time reallocation rather than on displacement by violence.

8 Alternative Accounts

State Capacity and Opportunistic Timing

A strategic alternative holds that the Ramadan increase reflects insurgents exploiting a temporarily weakened adversary rather than any sacralization of combat. State forces in Muslim-majority countries are themselves fasting, operate on compressed schedules, and may reduce patrol tempo, all of which lower the cost of attacking them. Three implications follow. If the state is distracted rather than depleted, inattentive after iftar and during tarawih, attacks should concentrate in those hours across all attack types, since a distracted defender is easier to strike with any weapon. State-initiated activity should fall, since the same constraints that weaken defense weaken offense. And the escalation should be available to any insurgent facing the same state, whatever its ideology.

The first two are unlikely. The post-iftar concentration is specific to high-exertion direct fire and absent for pre-positioned IED explosions (Table 3, columns 1 and 3): against the same fasting Afghan defenders, direct fire shifts sharply into the post-iftar window while IED explosions rise uniformly across the day. A distracted-defender mechanism cannot produce this split, since the defender's inattention would lower the cost of every attack type equally. Friendly direct fire by ISAF and Afghan government forces rises rather than falls during Ramadan (column 4), inconsistent with a state too depleted to fight and more consistent with

[44] The 2016 Ramadan campaign is the clearest instance, a series of mass-casualty bombings against civilians timed to the month (Amarasingam and Winter, 2017).

[45] Available upon request.

reactive engagement. On the third, secular-nationalist groups facing the same fasting states show no detectable escalation.

A relevant account specific to religious groups holds that the sacred month raises the propaganda value of an attack, so groups strike during Ramadan for publicity rather than for religious merit. Messaging evidence cannot separate this account from the doctrinal channel, since devotional framing would appear under either reading. The two differ instead in whose payoff moves. Propaganda accrues to the organization, but the escalation requires individual fighters to accept a greater risk of death, and organizational publicity gives a fighter no reason to do so. What does is the elevated religious reward of combat and martyrdom during Ramadan, which is the doctrinal channel itself. A symbolic return that actually moves fighters is therefore not an alternative to observance repricing but the same channel viewed from the organization's side.

Intertemporal Substitution

Three variants of a displacement concern apply. The Hijri calendar may shift the baseline in ways unrelated to Ramadan; insurgents may suppress operations beforehand to conserve resources for the month; or the month may reschedule a fixed quantity of violence drawn from adjacent months. I test all three with a full Hijri-month event study (Figure 2) and with leads and lags (Table A.14).

Displacement is rejected cleanly. Stockpiling or reallocation would make the adjacent coefficients negative and comparable in magnitude to the Ramadan coefficient. They are positive: the one-month lead is $+0.026$ and the pooled two-month pre-Ramadan window is $+0.026$, both significant at the 5% level (columns 2 and 3). The pooled post-Ramadan coefficient is -0.006 with a standard error of 0.014, so activity does not fall afterward either, and the sum of the four adjacent-month coefficients in the event study is $+0.046$ ($p = 0.52$). Every displacement account predicts the opposite sign.

The event study shows that the conflict effect is the peak of a broader profile rather than an isolated spike. Ramadan is the only month individually distinguishable from the Dhul-Hijjah reference ($\beta = 0.040$, $p = 0.052$), but Jamada al-Thani, Rajab, and Sha'ban sit at 0.025 to 0.033 with intervals that overlap Ramadan's, while the five earliest Hijri months sit below zero. The profile tracks the devotional calendar: Rajab is one of the four sacred months, Sha'ban carries a supererogatory fasting tradition, and observance rises through both before peaking in Ramadan. That is consistent with the mechanism rather than a threat to it, but it does mean the baseline estimate, which uses all eleven non-Ramadan months as controls, partly reflects the elevation of the devotional season against the early-calendar trough. Table A.4 shows the same fact from the other side: restricting controls to the adjacent months lowers the coefficient to 0.030, and restricting them to distant months raises it to 0.045.

The protest profile mirrors the conflict profile with the sign reversed. Ramadan carries the most negative estimate ($\beta = -0.100$), and suppression extends into the flanking months, with Sha'ban at -0.056 in the event study and Shawwal at -0.081 in the leads and lags (column 5), while the earlier Hijri months sit above zero. The same devotional gradient that elevates

conflict from Rajab onward depresses protest over the same window, which is what a common shock to the value of devotional time predicts when it is applied to two activities of opposite doctrinal compatibility. Delayed mobilization is rejected: postponement requires a positive post-Ramadan coefficient, and the pooled window is -0.072 ($p = 0.011$), so the suppression does not unwind after the month ends.

Fasting Intensity

One might ask whether the intensity of the fast, rather than the occurrence of the month, drives the results. Hodler et al. (2024) argue that longer daylight fasts suppress violence in the following year by eroding public support for religious extremism. Table A.15 tests this directly on the main outcomes of this paper, restricting the sample to Ramadan months and regressing outcomes on log fasting hours following the panel structure in Campante and Yanagizawa-Drott (2015) and Hodler et al. (2024).

The estimates are uninformative in this sample. Fasting hours vary across country-years only through the interaction of latitude with Ramadan's seasonal position, and once country and Hijri-year fixed effects absorb both margins separately, little variation remains: the conflict coefficient in Muslim-majority countries is -0.753 with a standard error of 3.224, and the protest placebo in non-Muslim countries is 1.008 with a standard error of 6.820 (columns 1 and 5). These intervals admit effects of any magnitude in either direction. The positive and significant protest coefficient in column 4 should be read the same way: the ACLED Muslim-majority sample has 14 countries concentrated north of the equator, so the estimate is identified off a narrow band of latitudes, and the pooled interaction in column 6 shows no significant effect for the most-Muslim category.

The design does not rest on this table. The within-day evidence carries the physiological argument, and it does so with a sharper prediction. If depletion drove violence, high-exertion attacks would peak in the mid-afternoon trough of fasted performance; instead they reach their minimum there and rise only after each meal. Fasting therefore shapes when violence occurs rather than how much, and the monthly level responds to the Ramadan indicator rather than to fast duration. The same logic disposes of a civilian-exhaustion account of the protest decline: were physical depletion keeping people off the streets, the suppression would deepen with fasting hours, and the protest decline is instead concentrated in the daytime hours the fast occupies rather than scaling with its length.

Ramadan Financing

A financing account holds that Ramadan almsgiving swells militant budgets through charity fronts, and that the additional funds buy more attacks (Limodio, 2022). The composition of the increase is unlikely for a budget shock: capital-intensive attacks stay flat, while the rise is in cheap improvised explosives and labor-intensive direct fire (Table 3). The one causal estimate of this channel is identified from a compulsory bank-deposit Zakat levy specific to Pakistan, whereas the giving that would fund a panel-wide escalation is overwhelmingly informal and in kind. Furthermore, money raised during the month would plausibly fund operations with

a lag, so a budget channel predicts a post-Ramadan increase. Yet the pooled post-Ramadan coefficient shows no such increase and is inconsistent with a delayed financing effect.

News Coverage

Media attention fluctuates around religious holidays (Durante and Zhuravskaya, 2018; Eisensee and Strömberg, 2007), raising the concern that differential reporting during Ramadan mechanically inflates or deflates recorded events. Two features of the design address this. First, both results replicate in declassified military records from Afghanistan, which are logged by on-the-ground personnel under standardized reporting protocols and bypass media entirely: insurgent attacks rise and demonstrations fall at the province level. Second, the within-day patterns are a placebo internal to those records. Direct fire shifts into the post-iftar window while IED explosions rise uniformly across the day, and the two are recorded by the same observers in the same province-months under the same protocol, so no reporting process distinguishes between them by the exertion a tactic requires. The individual survey evidence points the same way from a different measurement technology: prayer frequency, political interest, and protest willingness are elicited from respondents rather than compiled from reports, and are therefore unaffected by news cycles.

The opposite-signed effects in the media-sourced panels are harder to attribute to a single reporting shift. A uniform decline in coverage during Ramadan would suppress both counts rather than raising one and lowering the other. An account in which outlets find Ramadan violence more newsworthy while crowding out routine protest coverage would generate the observed pattern, which is why the non-media evidence above rather than the sign contrast is what closes the concern.

9 Conclusion

A single religious shock moves armed organizations and ordinary citizens in opposite directions. Religious insurgencies escalate during Ramadan and concentrate their most physically demanding operations in the hours after a meal. Civilians pray more, report less interest in politics, and demonstrate less often. Both responses follow from one relative price. Sacred time raises the value of religious output, which subsidizes action that doctrine renders worship and taxes the time of action that competes with it.

The finding relocates where religion's political effect lives. It is not a property of how devout a population is, which is what cross-sectional measures of religiosity capture, nor of whether a conflict is coded as religious, which is what actor-level taxonomies capture. It is a property of a pairing: an act, and the reading an actor brings to it. The same month that makes combat meritorious for an organization founded on doctrinal objectives makes protest costly for a devout individual with no comparable reading available.

The civilian result raises a question the paper does not answer. If Ramadan draws people off the street, the month is a period when the cost of pushing through unpopular policy is temporarily lower. Testing whether regimes act on this would require the timing of policy

announcements and crackdowns rather than protest counts, and I flag it as a conjecture that follows from the mechanism.

The policy implication for international mediation is uncomfortable. While this paper evaluates the intensity of violence rather than the durability of specific negotiated settlements, the mechanism highlights why the standard diplomatic instrument faces a severe headwind. The premise of a Ramadan ceasefire is that heightened devotion opens a window for de-escalation. The evidence suggests the opposite for the actors who matter most: the same doctrine that makes a religious insurgency a formidable adversary makes the month a season of intensified, not reduced, activity. The framework also suggests why material concessions may backfire in this context. Ginges and Atran (2013) show that side-payments offered in exchange for sacralized commitments intensify rather than soften resistance, because the offer signals that the commitment is for sale. A ceasefire package that asks a group to forgo the month's multiplied spiritual merit in exchange for political consideration is that transaction exactly.

Two conditions bound the estimates. The design identifies the effect of the Ramadan period rather than of fasting compliance, since individual-level observance is unobserved across the panel, so the estimates are a lower bound on the response among those who fast. The mechanism operates on the intensive margin of actors already mobilized, and the conflict panel, which conditions on dyads that have already crossed the threshold for an active conflict, cannot speak to whether the month draws new actors in.

Sacred time is a recurring, exogenous, and internally synchronized shock to the value of devotion, and every major tradition has one. What this paper suggests is where to look. The question is not whether observance rises during Lent, the High Holy Days, or Navratri, but whether the political actions available to a given actor can be cast as worship within the readings that actor can credibly hold. Where they can, sacred time is a subsidy; where they cannot, it is a tax. The framework also points to a related question it does not answer. Religious festivals and holidays are the mirror image of Ramadan in one respect: they mark moments of celebration rather than sustained discipline, and their demands on time are shorter and less binding. Whether they produce the same repricing, a weaker version of it, or no effect at all is another open question.

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Figures & Tables

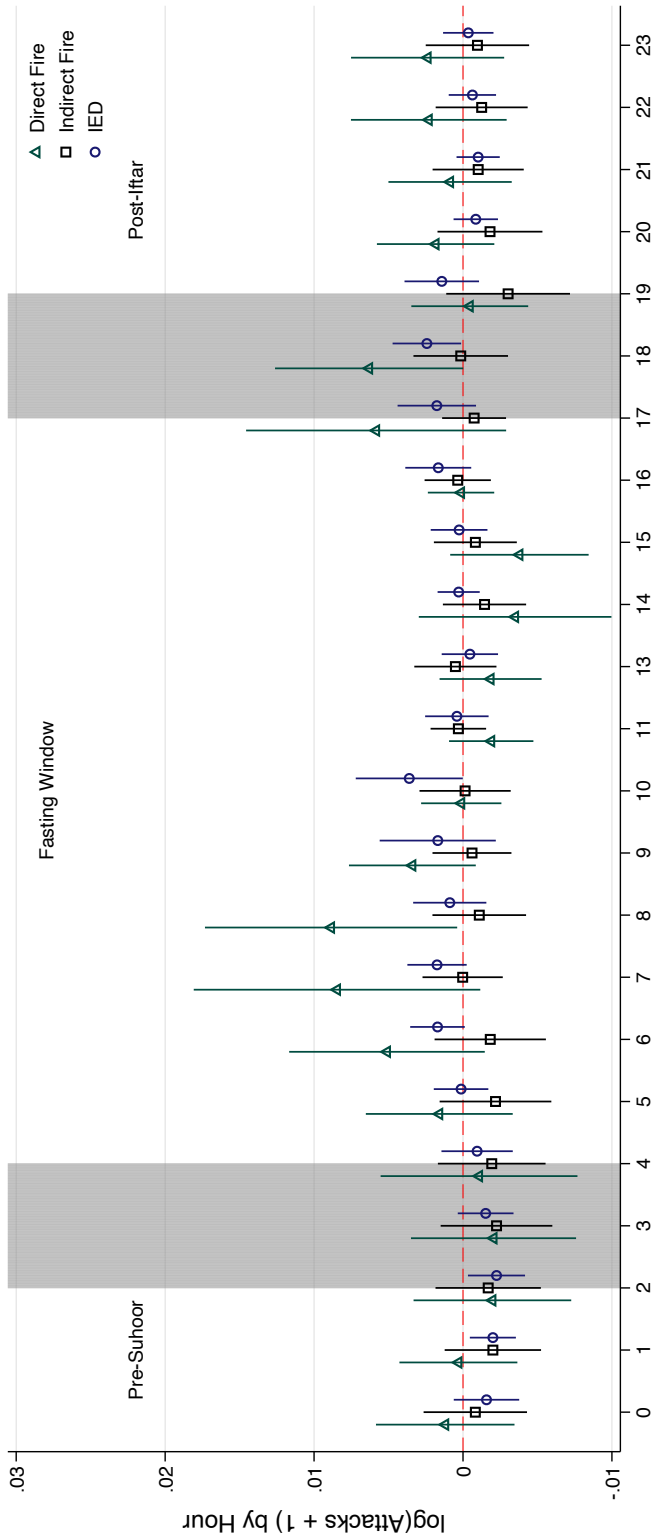


Figure 1: Within-Day Tactical Substitution During Ramadan by Attack Type: Direct Fire, Indirect Fire, and IEDs

This figure plots point estimates and 95% confidence intervals from the interaction between a Ramadan indicator and hour-of-day indicators, estimated separately for three attack-type outcomes at the province-day-hour level. The dependent variable in each model is the log-transformed hourly number of attacks plus one ($\ln(y+1)$). *The omitted base category is 12:00 PM (local time), the middle of the fasting day. Dark teal hollow triangles represent direct fire (DF) attacks involving small arms and rocket-propelled grenades, which require sustained physical exertion in close combat. Black hollow squares represent indirect fire (IDF) attacks (mortars and rockets fired from prepared positions), which involve less continuous exertion. Midnight blue hollow circles represent IED explosions, which are pre-positioned and triggered remotely or by command wire, requiring minimal real-time effort. The middle region denotes the approximate daylight fasting window, bordered on the left by the sample-averaged pre-dawn meal (Suhoor) and on the right by sunset (Iftar). All models include province, Hijri-year, and Gregorian month fixed effects to account for cross-province differences, annual conflict trajectories, and solar seasonality. Robust standard errors are clustered at the province level (34 provinces).*

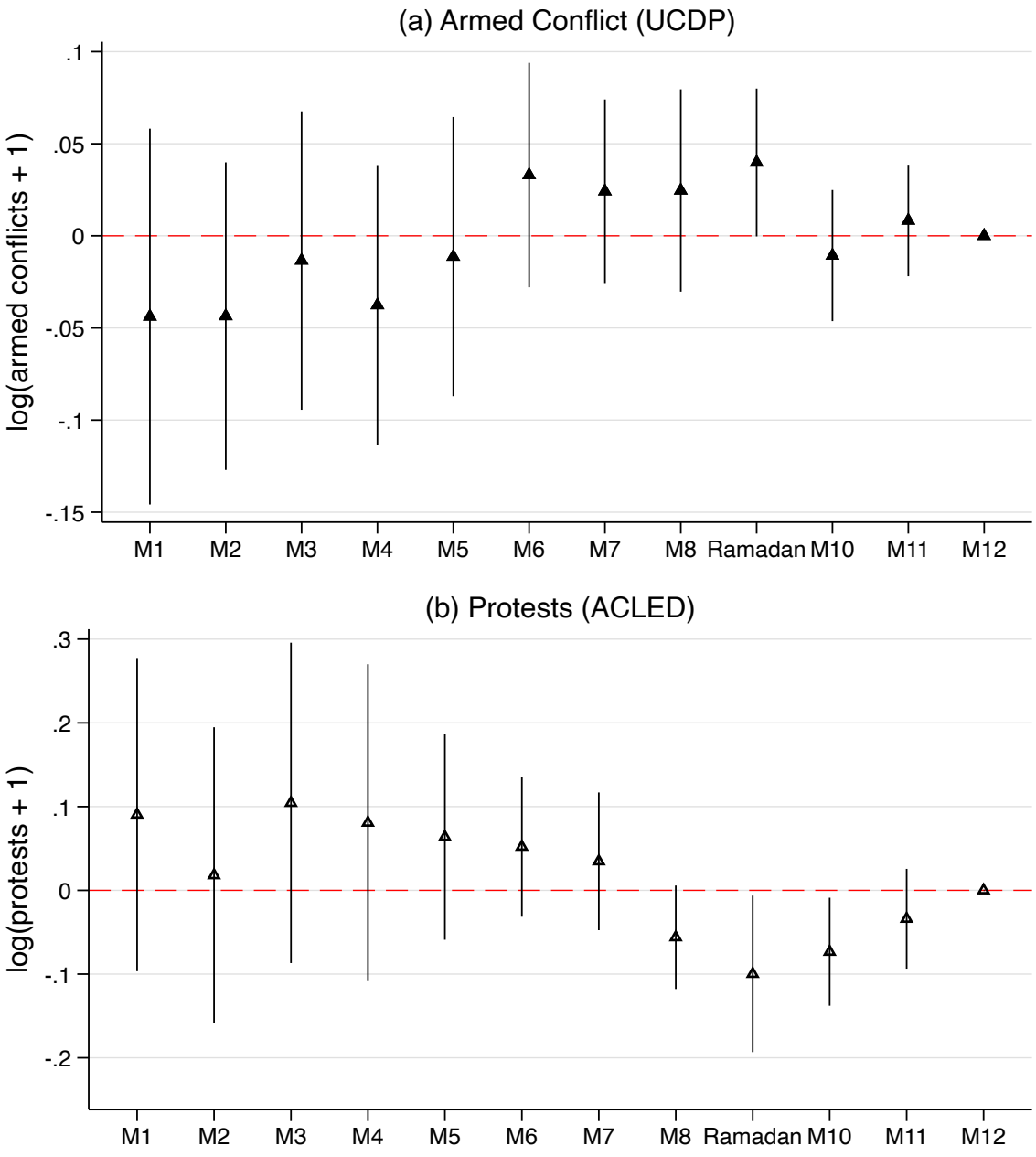


Figure 2: The Ramadan Effect Across All Hijri Months

Notes: This figure plots coefficients on all twelve Hijri month indicators from a single regression for each outcome, with Dhu al-Hijjah (M12) as the omitted base category. Panel (a) estimates the effect on log(state-based armed conflicts + 1) using UCDP data for 31 Muslim-majority countries (1989–2022). Panel (b) estimates the effect on log(demonstrations + 1) using ACLED data for 14 African Muslim-majority countries (1997–2023). Both specifications include country-by-Hijri-year and year-by-quarter fixed effects; the latter absorb year-specific seasonal shocks rather than just average seasonality, providing a more demanding test than the Gregorian month fixed effects used in the baseline regressions. Bars represent 95% confidence intervals based on robust standard errors clustered at the country level (31 clusters in panel a, 14 in panel b). Wider confidence intervals in panel (b) reflect the smaller sample.

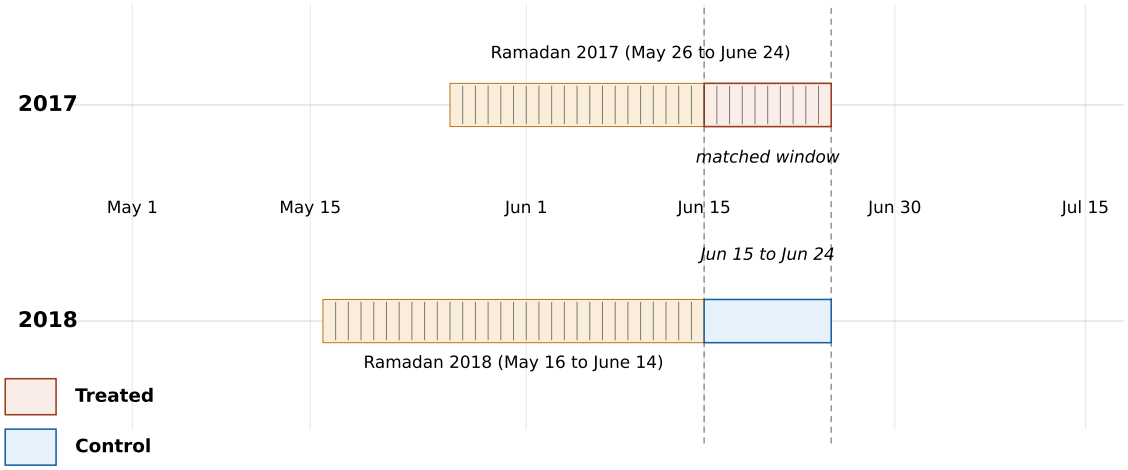


Figure 3: Exact-Date Matching Design

Notes: The Hijri calendar drifts a little more than 10 days per year relative to the Gregorian calendar, so the last 10 days of Ramadan in year t fall on the same Gregorian calendar dates as the days immediately after Ramadan ends in year $t+1$. The figure illustrates the 2017–2018 cycle. In 2017 (top row), Ramadan ran from May 26 to June 24; the last 10 days, June 15 to June 24, constitute the treated window. In 2018 (bottom row), Ramadan ran from May 16 to June 14, and the same Gregorian dates, June 15 to June 24, constitute the control window. The two windows share the same solar position, absorbing any seasonal confound tied to a specific calendar day, such as weather, agricultural cycles, or recurring holidays, that month-level fixed effects cannot capture. The design pools all such consecutive-year pairs across the panel. Daily tick marks within the Ramadan blocks indicate individual days; the matched windows are highlighted with dashed vertical guides.

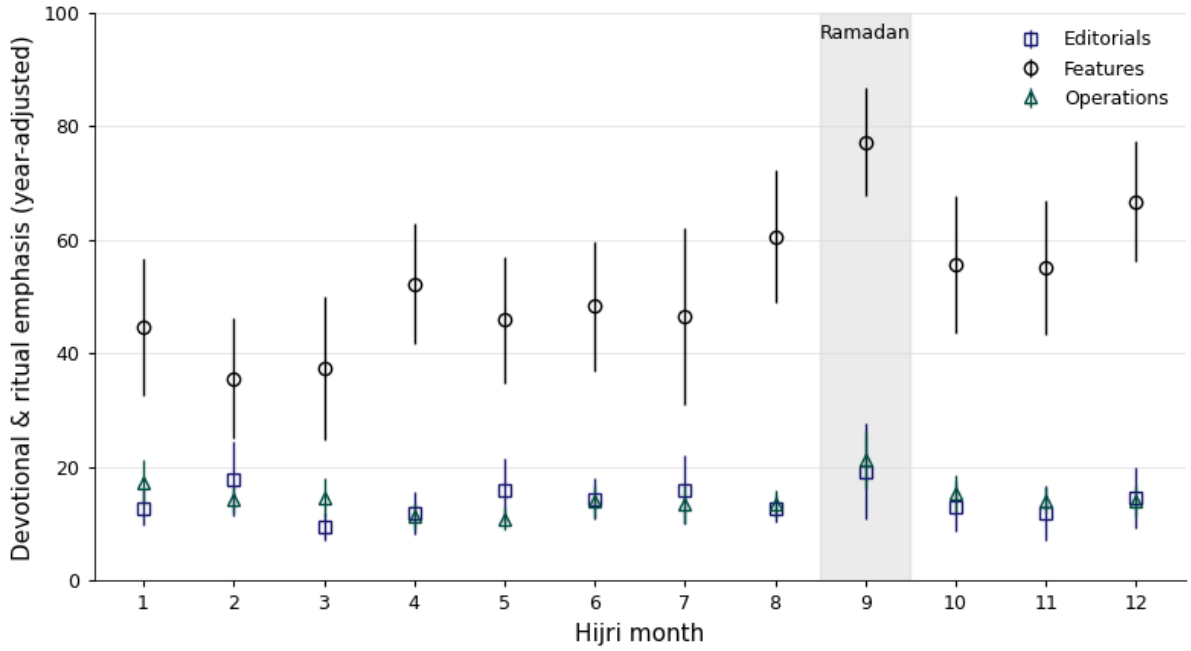


Figure 4: Devotional and Ritual Emphasis in the Islamic State's *al-Naba*, by Hijri Month

Notes: The figure plots devotional and ritual emphasis in the Islamic State's weekly *al-Naba* across 533 issues spanning December 2015 to April 2026, of which 46 fall in Ramadan, separately for editorials, feature essays, and operational news (the *Editorials*, *Features*, and *Operations* series): the signed lead argument, the standalone essays on doctrine, history, or governance, and the weekly attack reports by province, respectively. Devotional and ritual emphasis is a GABRIEL rating on a 0–100 scale of the extent to which a section foregrounds personal worship, such as fasting, night prayer, Qur'anic recitation, and supplication, rather than combat operations or political argument. To net out the pronounced secular drift in the newsletter's content, I residualize each section's score on Hijri-year fixed effects, re-center at the grand mean, and take the mean by Hijri month within each section type. The plotted points are these descriptive year-adjusted monthly averages, not coefficients from a fitted model; spikes span the mean plus or minus 1.96 standard errors across sections within the month and are descriptive, not clustered by Hijri year. Ramadan, the ninth Hijri month, is shaded. Devotional emphasis in the feature essays is far higher in Ramadan, near 78, than in other months, where it averages about 50; editorials remain low throughout, rising only slightly in Ramadan; and operational news, a within-issue placebo, is flat across the calendar. The profile also tracks the Islamic devotional calendar, with an anticipatory rise in Sha'ban (month 8) and a second peak in Dhul-Hijjah (month 12, the month of the Hajj). Appendix B details the corpus, the segmentation, and the prompt.

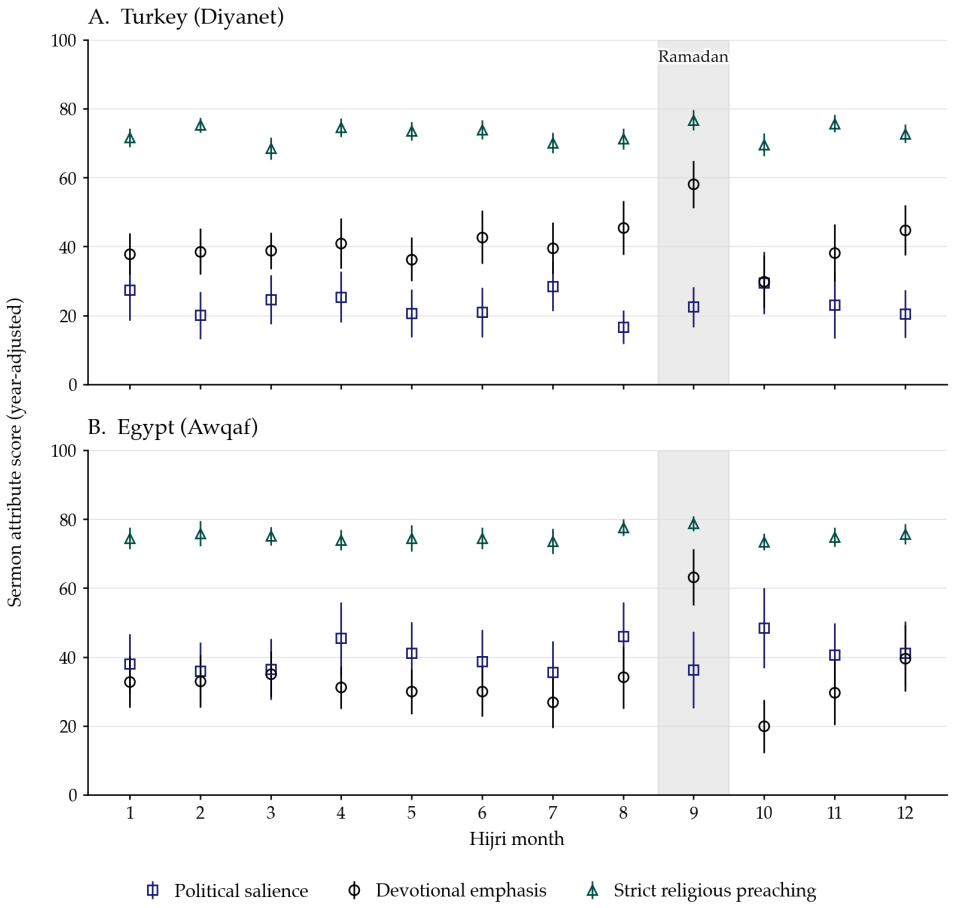


Figure 5: Sermon Content in Two State Systems, by Hijri Month

Notes: The figure plots the content of centrally drafted Friday sermons by Hijri month for the two largest fully standardized systems: Panel A, 610 weekly scripts issued by the Turkish *Diyanet* between March 2014 and February 2026, of which 50 fall in Ramadan; Panel B, 460 weekly scripts issued by the Egyptian *Wizārat al-Awqāf* between July 2016, when the ministry moved to verbatim distribution, and March 2026, of which 35 fall in Ramadan. In both systems the ministry drafts one script and distributes it for verbatim delivery nationwide, so each observation is the state's chosen message for that week. Three GABRIEL attributes are scored on a 0–100 scale. *Devotional emphasis* uses the same attribute definition applied to *al-Naba* in Figure 4, so the two figures are measured on a common scale. *Political salience* is the extent to which the sermon addresses regime-relevant political themes such as national unity, state authority, military sacrifice, and sovereignty, as against religious or moral instruction that makes no reference to the state. *Strict religious preaching* is the intensity of doctrinally authoritative instruction, scriptural citation with attribution, rulings on the permitted and the forbidden, and warnings of divine punishment; it is high and stable year-round and serves the placebo role that operational news serves within an issue of *al-Naba*. I residualize each sermon's score on Hijri-year and Gregorian-month fixed effects, re-center at the attribute mean, and take the mean by Hijri month. The Gregorian-month effects matter because the corpus spans roughly twelve years rather than a full lunar cycle, so Ramadan's solar position is correlated with calendar month over this window. The plotted points are these descriptive year-adjusted monthly averages, not coefficients from a fitted model; spikes span the mean plus or minus 1.96 standard errors within the month and are descriptive, not clustered by Hijri year. Ramadan, the ninth Hijri month, is shaded.

Table 1: The Relative Effects of Ramadan on State-Based Armed Conflict, 1989–2022

	Dependent Variable: log(armed conflict+1)					
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.037***	0.037***	0.068***	0.028**	0.010	0.002
	(0.013)	(0.012)	(0.022)	(0.012)	(0.008)	(0.009)
			[0.006]			
Ramadan × % Muslim					0.022	
					(0.016)	
Ramadan × >75% Muslim						0.035**
						(0.015)
Ramadan × 50–75% Muslim						-0.012
						(0.026)
Ramadan × 25–50% Muslim						-0.001
						(0.033)
Ramadan × 1–25% Muslim						0.020
						(0.015)
Conflict mean	11.84	11.84	25.90	11.84	5.046	5.046
Sample	Muslim	Muslim	Muslim (Intense)	Muslim	All	All
Observations	12648	12648	5712	12648	43656	43656
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-year FE		✓	✓	✓	✓	✓
Year-by-quarter FE				✓		

Notes: Armed conflict is the count of state-based armed conflict events per country-month, from the UCDP Georeferenced Event Dataset (GED), 1989–2022. Ramadan equals one if the calendar month corresponds to Ramadan in the Hijri calendar. Columns (1)–(4) use the 31 countries that are on average at least 75% Muslim (World Religion Project and Pew); column (3) restricts this sample to the 14 of those countries with a mean of more than one conflict event per month ("Intense"). Column (4) adds Gregorian year-by-quarter fixed effects, which absorb seasonal and other shocks common to all countries in each quarter of each year. In columns (5)–(6) the sample consists of 107 countries; in column (6) the reference group is the 22 countries with an average Muslim population of 1% or less. All specifications include Gregorian month fixed effects. Conflict mean is the average of the outcome in levels. Robust standard errors clustered at the country level are in parentheses. Because column (3) has only 14 clusters, the value in square brackets is the p-value from the restricted wild cluster bootstrap (null imposed; Cameron et al. (2008); Roodman et al. (2019)), based on 9,999 replications with Rademacher weights. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: The Ramadan Effect on Armed Conflict and Conflict Initiation by Group Ideology

	All conflict		By initiating side	
	Events	Deaths	Rebel-initiated	Government-initiated
	(1)	(2)	(3)	(4)
Ramadan × Religious	0.034**	0.038**	0.032**	0.010*
	(0.012)	(0.017)	(0.013)	(0.005)
Ramadan × Secular-nationalist	0.013	0.008	-0.000	0.009*
	(0.009)	(0.014)	(0.007)	(0.005)
Difference	0.021	0.030	0.032**	0.000
	(0.013)	(0.019)	(0.014)	(0.007)
Triple difference, (3) minus (4)			0.032**	
			(0.014)	
Outcome mean, religious (levels)	10.88	63.74	1.878	2.730
Outcome mean, secular (levels)	1.210	14.49	0.226	0.302
Observations	24480	24480	24480	24480
Gregorian month FE	✓	✓	✓	✓
Country-by-year FE	✓	✓	✓	✓

Notes: Each column reports a stacked regression of the religious and secular-nationalist series for one outcome, with series-specific country-by-Hijri-year and Gregorian-month fixed effects. The first two rows give the Ramadan effect for each group, identical to estimating each series separately; *Difference* is the religious minus the secular-nationalist coefficient, with a standard error that accounts for the covariance between them. The triple difference is the difference in column (3) minus the difference in column (4), estimated from a stacked regression of all four initiation series. Columns (1)–(2) use all state-based conflict events and fatalities (best estimate); columns (3)–(4) use events by initiating side, classified from UCDP event descriptions with GABRIEL (Appendix B1). *Religious* groups operate within an Islamic framework that supplies the doctrinal basis to cast offensive combat as meritorious worship; *secular-nationalist* groups have ethno-nationalist, Marxist-Leninist, or factional objectives, and religion plays no constitutive role in their armed struggle; Appendix C lists all groups. Dependent variables are the log of the count plus one; outcome means are in levels. Observations count two stacked series of 12,240 country-months each. Robust standard errors clustered at the country level are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effect of Ramadan on Insurgent Violence in Afghanistan: Monthly Aggregates and Within-Day Timing

	Insurgent Attacks			Friendly Attacks	Insurgent DF by Target	
	log(DF+1)	log(IDF+1)	log(IED+1)	log(DF+1)	log(DF+1)	log(DF+1)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Monthly Aggregates						
Ramadan	0.083***	0.010	0.103***	0.076***	0.070***	-0.025
	(0.027)	(0.028)	(0.023)	(0.022)	(0.024)	(0.021)
Outcome mean (levels)	20.37	4.546	6.955	1.551	8.076	10.63
Observations	5,406	5,406	5,406	5,406	5,406	5,406
Panel B: Within-Day Decomposition (pre-iftar vs. post-iftar)						
Ramadan	-0.018	-0.031	0.094***	0.035*	0.028	-0.044*
	(0.024)	(0.028)	(0.021)	(0.018)	(0.021)	(0.023)
Post-Iftar	-0.367***	-0.195***	-0.509***	-0.084***	-0.258***	-0.281***
	(0.055)	(0.055)	(0.057)	(0.017)	(0.049)	(0.040)
Ramadan × Post-Iftar	0.127***	0.047*	0.025	0.015	0.020	0.119***
	(0.025)	(0.027)	(0.025)	(0.020)	(0.017)	(0.025)
Outcome mean (levels)	9.646	2.199	3.325	0.669	3.976	5.012
Observations	10,880	10,880	10,880	10,880	10,880	10,880
Target	State	State	State	Rebels	ISAF	Govt
Gregorian month FE	✓	✓	✓	✓	✓	✓
Province-by-Hijri-year FE	✓	✓	✓	✓	✓	✓

Notes: This table estimates the effect of Ramadan on insurgent violence in Afghanistan using the SIGACTS dataset (2002–2015). Panel A reports the monthly aggregate effect using a province-by-Hijri-month-by-Hijri-year panel. Panel B decomposes the monthly effect using a half-day shift panel that splits each province-Hijri-month into pre-iftar (dawn to sunset, encompassing the daytime fast) and post-iftar (sunset to dawn) windows. Columns 1–3 disaggregate rebel-initiated attacks by exertion type: direct fire (DF, small-arms and rocket-propelled grenade attacks requiring sustained close combat), indirect fire (IDF, mortars and rockets fired from prepared positions with less ongoing exertion), and improvised explosive device explosions (IED, pre-positioned and triggered remotely or by command wire, requiring minimal real-time effort). In columns 1–3, "State" targets comprise all government-side forces, namely Coalition forces (ISAF members and OEF allies) and Afghan State Forces (ANSF, including the Afghan National Army and Afghan National Police). Column 4 reports friendly direct fire (DF initiated by ISAF and Afghan government forces against insurgents). SIGACTS records the target affiliation of each event but not, for Friendly Action events, the initiating actor, so friendly DF in column 4 cannot be decomposed into Coalition-led versus Afghan-led operations. Columns 5 and 6 partition the column 1 DF result by target: column 5 restricts to enemy DF events with at least one Coalition target affiliation, and column 6 restricts to enemy DF events with at least one Afghan State Forces target affiliation. Because a single attack can record multiple target affiliations, the two categories are not mutually exclusive and need not sum exactly to column 1. In Panel B, the Ramadan main effect captures the pre-iftar (fasting-period) differential; the Ramadan × Post-Iftar interaction measures the additional post-iftar concentration during Ramadan relative to non-Ramadan months. The total post-iftar Ramadan effect is the sum of the main effect and the interaction. Post-iftar windows use sample-averaged sunset (18:05) and dawn (05:13) cutoffs. All specifications include province-by-Hijri-year and Gregorian month fixed effects. Outcome mean reports the average level of the dependent variable in the estimation sample. Robust standard errors are in parentheses, clustered at the province level (34 provinces). Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 4: Effect of Ramadan on Civilian Protest Activity

	Cross-Country (ACLED)			Afghanistan (SIGACTS)	
	log(Peaceful Protests + 1)			log(Demonstrations + 1)	
	(1)	(2)	(3)	(4)	(5)
Ramadan	-0.065*	0.052***	0.062	-0.037***	-0.028***
	(0.031)	(0.019)	(0.045)	(0.013)	(0.009)
Ramadan × % Muslim		-0.117***			
		(0.039)			
Ramadan × >75% Muslim			-0.142**		
			(0.053)		
Ramadan × 50–75% Muslim			-0.039		
			(0.072)		
Ramadan × 25–50% Muslim			0.032		
			(0.054)		
Ramadan × 1–25% Muslim			-0.050		
			(0.047)		
Post-Iftar					-0.051***
					(0.015)
Ramadan × Post-Iftar					0.022**
					(0.009)
Outcome mean	7.569	4.215	4.215	0.304	0.136
Sample	Muslim	All	All	Afghanistan	Afghanistan
Observations	4,536	15,552	15,552	5,406	10,880
Gregorian month FE	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓		
Province-by-Hijri-year FE				✓	✓

Notes: This table estimates the effect of Ramadan on civilian protest activity. Columns 1–3 use the ACLED dataset for 48 African countries (1997–2023). The outcome is the logarithm of peaceful protests plus one at the country-Hijri-month level, where ACLED defines peaceful protests as demonstrations in which participants do not engage in violence or property destruction and are not met with force or intervention. Column 1 restricts the sample to 14 countries with a Muslim population share above 75%. Columns 2 and 3 use the full sample, interacting the Ramadan indicator with the continuous Muslim population share (column 2) and with discrete share categories (column 3); the baseline category in column 3 is countries with less than 1% Muslim population. Columns 4 and 5 use the SIGACTS dataset for Afghan provinces (2002–2015). The outcome is the logarithm of demonstration events plus one at the province-Hijri-month level (column 4) and at the province-Hijri-month-shift level (column 5), where each province-month is split into pre-iftar (dawn to sunset) and post-iftar (sunset to dawn) windows. In column 5, the Ramadan main effect captures the pre-iftar (fasting-period) differential and the Ramadan × Post-Iftar interaction measures the differential between the post-iftar and pre-iftar Ramadan effects; the total post-iftar Ramadan effect is the sum of the two coefficients. All specifications include Gregorian month fixed effects. Columns 1–3 include country-by-Hijri-year fixed effects and columns 4–5 include province-by-Hijri-year fixed effects. Outcome mean reports the average level of the dependent variable in the estimation sample. Robust standard errors are in parentheses, clustered at the country level in columns 1–3 and at the province level in columns 4–5 (34 provinces). Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 5: The Ramadan Effect on Spirituality and Political Engagement

	Spirituality/Prayer		Interest in Politics		Attending Demonstrations	
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.052*	0.051*	-0.055**	-0.054*	-0.042**	-0.039**
	(0.025)	(0.026)	(0.026)	(0.026)	(0.018)	(0.017)
Outcome mean	0.770	0.751	0.433	0.480	0.342	0.368
Sample	±3 mo.	All	±3 mo.	All	±3 mo.	All
Observations	16,744	33,149	19,326	41,183	16,718	36,813
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓

Notes: This table examines the effect of Ramadan on individual-level measures of spirituality and political engagement using data from 23 Muslim-majority countries from the World Values Survey, waves 4 through 7 (2001–2022). The analytical sample is restricted to Muslim respondents from Muslim-majority countries who report religion as either "very important" or "rather important" on the WVS importance-of-religion scale; approximately 95% of Muslim respondents in the WVS sample meet this criterion. The restriction is motivated by the mechanism under study: the Ramadan effect operates through religious observance during the holy month and is unlikely to affect respondents whose self-reported religious commitment is low. This individual-level restriction is conceptually analogous to the country-level restriction to Muslim-majority contexts. Observation counts vary across specifications because certain survey questions were not administered to respondents in all countries and waves. Columns (1)–(2) use a binary indicator for high-intensity prayer (praying several times a day or once a day and zero otherwise). Columns (3)–(4) use a binary indicator for political interest (reporting being "very interested" or "somewhat interested" in politics). Columns (5)–(6) use a binary indicator for demonstration willingness (reporting having attended or willingness to attend lawful peaceful demonstrations). Odd-numbered columns restrict the sample to interviews conducted in the seven Hijri months centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah); even-numbered columns use the full twelve-month sample. All specifications include country-by-Hijri-year and Gregorian month fixed effects and the full demographic control set: gender, age, age-squared, marital status, number of children, education, employment status, income, and self-reported importance of religion. Outcome mean reports the baseline average of the dependent variable. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table 6: Last 10 Days of Ramadan vs. Same Dates in Year $t+1$

	Y=Armed Conflict				Y=Demonstrations			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ramadan	0.009*	0.099***	-0.025***	-0.068	-0.020***	-0.327***	-0.004**	-0.938**
	(0.005)	(0.026)	(0.009)	(0.094)	(0.006)	(0.060)	(0.002)	(0.458)
Post-iftar			-0.126***	-1.240***			-0.005**	-1.431***
			(0.033)	(0.104)			(0.002)	(0.397)
Ramadan×Post-iftar			0.023**	0.129**			0.003	0.620
			(0.009)	(0.056)			(0.002)	(0.598)
Outcome mean (levels)	0.423	0.423	0.414	0.414	0.282	0.282	0.005	0.005
Estimator	OLS	PPML	OLS	PPML	OLS	PPML	OLS	PPML
Dep. variable	Log(Y+1)	# of Y	Log(Y+1)	# of Y	Log(Y+1)	# of Y	Log(Y+1)	# of Y
Observations	22,196	7,214	17,408	11,260	8,218	2,924	17,408	1,824
Country/Province × Cycle-year FE	✓	✓	✓	✓	✓	✓	✓	✓
Greg. month FE	✓	✓	✓	✓	✓	✓	✓	✓
Within-day			✓	✓			✓	✓

Notes: This table reports estimates from an exact Gregorian calendar matching design. Because the Islamic lunar year is approximately 10 days shorter than the Gregorian year, the last 10 days of Ramadan in year t fall on nearly the same calendar dates as the early post-Ramadan period in year $t+1$, which encompasses the 10 days of the month after Ramadan. Country (or province) by (cycle-)year fixed effects group each treated window with its matched control, ensuring that identification comes entirely from within-pair variation; Gregorian month fixed effects absorb residual cross-cycle seasonality. Columns (1)–(2) and (5)–(6) report the cross-country day-level results: state-based armed conflicts from UCDP and demonstrations from ACLED, respectively, restricted to countries where the average Muslim population share exceeds 75%. Columns (3)–(4) and (7)–(8) zoom in on Afghanistan using the SIGACTS incident database (direct-fire attacks and demonstrations), with the panel constructed at the province × Gregorian date × shift level. The shift indicator *Post-iftar* equals one for local hours 18:00–04:59 (the non-fasting window during Ramadan) and zero for the 05:00–17:59 fasting window. In within-day columns the *Ramadan* main effect captures the fasting-period differential, the *Post-iftar* coefficient captures the average evening lull, and the interaction measures the differential between the post-iftar and pre-iftar Ramadan effects; the total post-iftar Ramadan effect is the sum of the main effect and the interaction. OLS specifications (odd columns) use $\log(Y+1)$ as the dependent variable; PPML specifications (even columns) model the raw event count directly and are robust to the prevalence of zeros in daily data (Silva and Tenreyro, 2006; Chen and Roth, 2024). PPML drops observations separated by fixed effects, reducing the effective sample. The outcome mean reports the unconditional average of the level variable across the estimation sample (per country-day in columns 1–2 and 5–6; per province-day-shift in columns 3–4 and 7–8). Robust standard errors are in parentheses, clustered at the country (columns 1–2, 5–6) or province (columns 3–4, 7–8) level. Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

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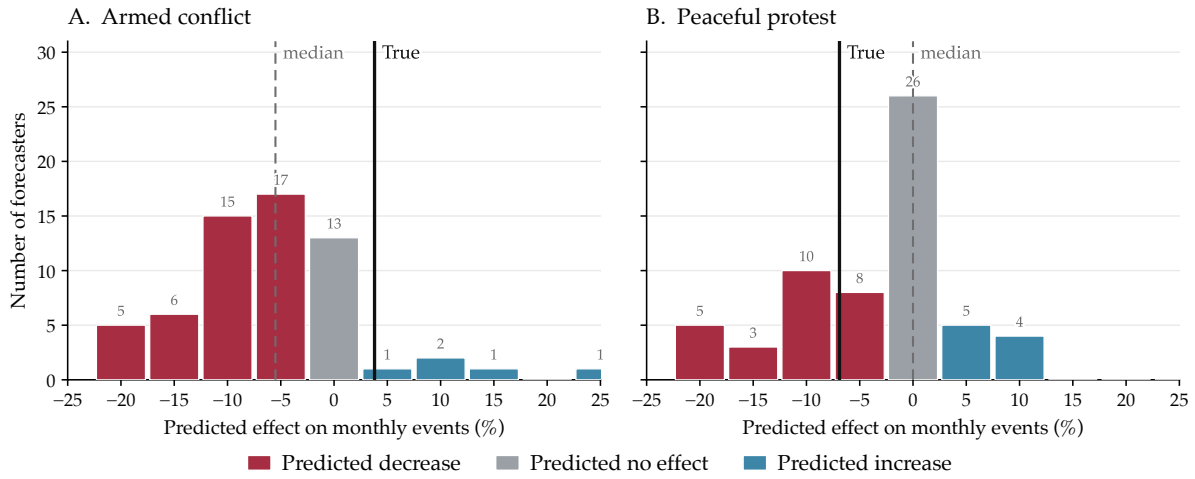


Figure A.1: Expert forecasts versus realized effects of Ramadan

The figure plots expert forecasts of Ramadan's effect on two outcomes in predominantly Muslim countries, elicited through the Social Science Prediction Platform (study ID sspp-2026-0012-v1) and completed by 61 forecasters (in-progress and preview responses excluded). Panel A is the monthly count of state-based armed conflict events between governments and rebel groups; Panel B is the monthly count of peaceful protests in predominantly Muslim countries in Africa, each relative to non-Ramadan months. Each forecaster chose a direction (decrease, no effect, or increase) and entered a magnitude between -25% and +25%; the signed forecast takes its sign from the direction and its size from the absolute value of the entry, with no-effect responses set to zero. Bars group the signed forecasts into 5-point bins centered on multiples of 5 and are colored by the bin's sign, so every forecast within 2.5 points of zero falls in the gray center bin whatever direction the forecaster chose: three decreases in each panel, one increase in Panel A, and four in Panel B. The gray dashed line marks the median forecast and the solid black line the realized estimate from the paper (labeled True): +3.8% for armed conflict and -6.9% for peaceful protest. By direction, three in four forecasters predicted that Ramadan would reduce armed conflict (46 of 61), and a plurality predicted that it would reduce protest (29 of 61). The median forecast was a 5.5% decline in armed conflict and no change in protest, a prior the estimates reverse for armed conflict and exceed in magnitude for protest.

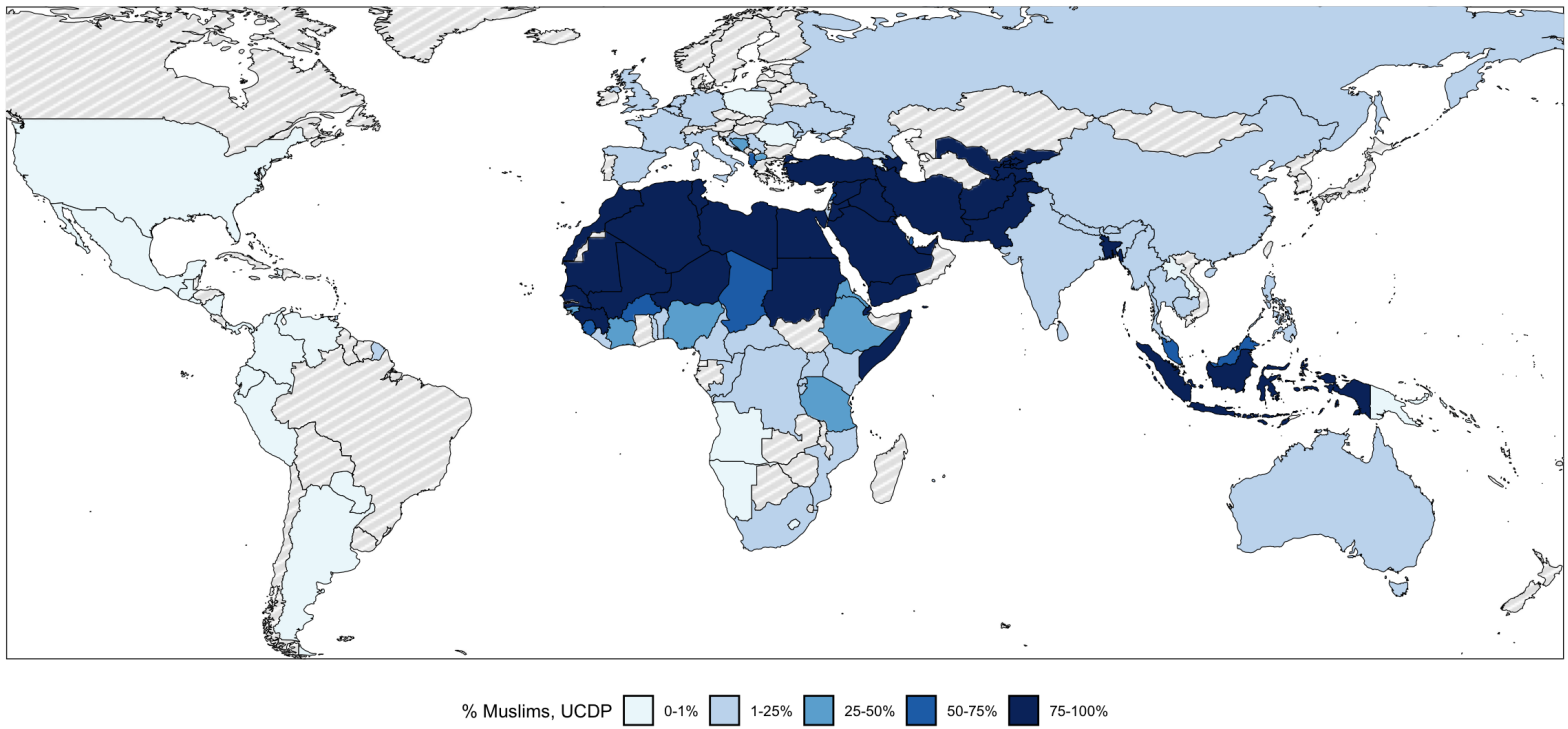


Figure A.2: Countries by Share of Muslim Population

Notes: There are 107 in-sample countries in the UCDP dataset. Hatched countries in gray do not appear in the dataset. Countries are grouped according to their average share of Muslim population from 1989 to 2022.

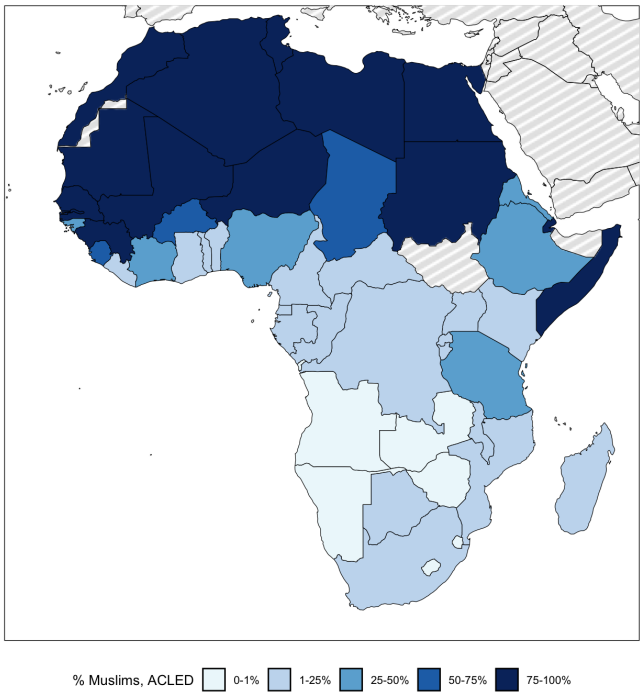


Figure A.3: African Countries by Share of Muslim Population

Notes: There are 48 in-sample countries in the ACLED dataset. Hatched countries in gray do not appear in the dataset. Countries are grouped according to their average share of Muslim population from 1997 to 2023.

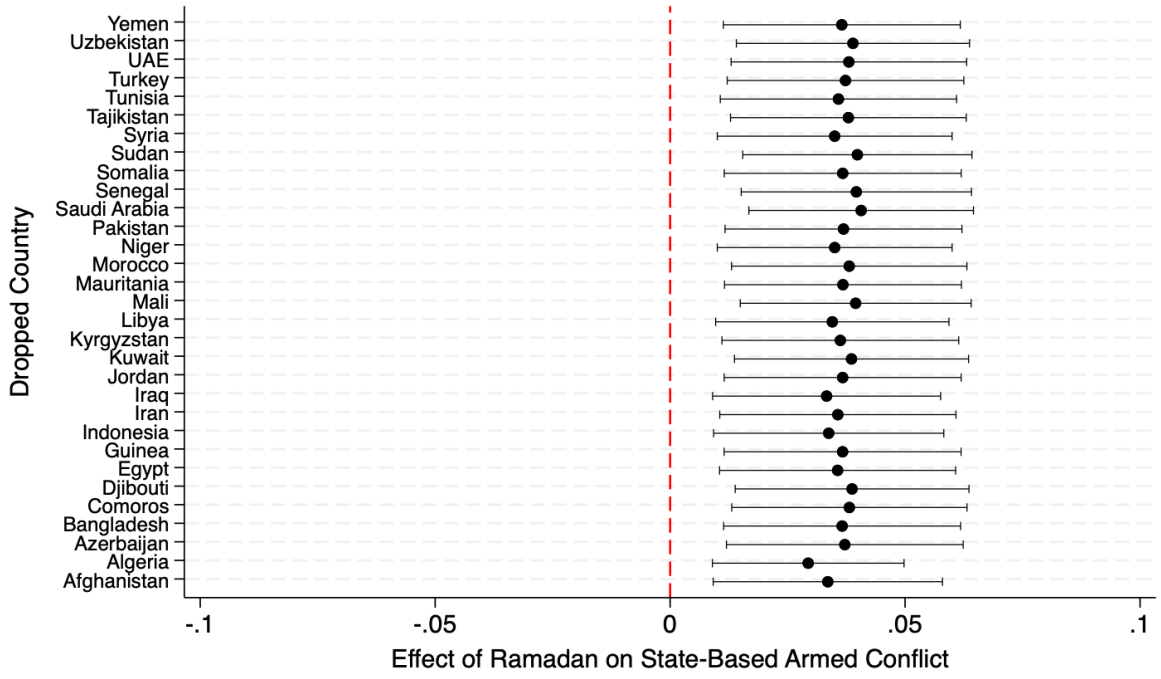


Figure A.4: Relative Effects of Ramadan on Armed Conflict in Muslim Countries Excluding 1 Country

Notes: This figure shows a leave-one-country-out sensitivity analysis of the effect of Ramadan on the intensity of state-based armed conflict, based on UCDP data from 1989–2022. The outcome is the natural logarithm of the number of conflict events plus one, measured monthly. Each point reports the Ramadan coefficient from a separate regression excluding one of the 31 Muslim-majority countries (defined as having a Muslim population share above 75%). All regressions include Gregorian month and country-by-year fixed effects. Standard errors are clustered at the country level. The red vertical dashed line denotes the zero-effect benchmark. This robustness check assesses the influence of individual countries on the overall result.

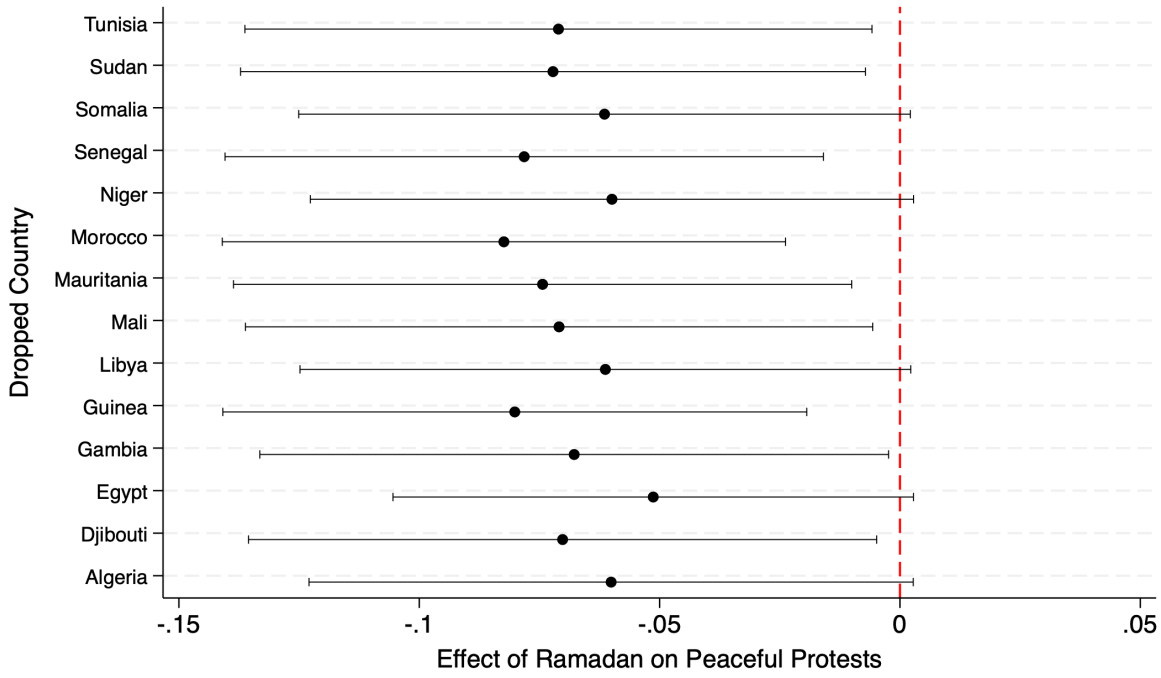


Figure A.5: Relative Effects of Ramadan on Demonstrations in Muslim Countries Excluding 1 Country

Notes: This figure shows a leave-one-country-out sensitivity analysis of the effect of Ramadan on the intensity of peaceful protests, based on ACLED data from 1997–2023. The outcome is the natural logarithm of the number of protests events plus one, measured monthly. Each point reports the Ramadan coefficient from a separate regression excluding one of the 14 Muslim-majority countries (defined as having a Muslim population share above 75%). All regressions include Gregorian month and country-by-year fixed effects. Standard errors are clustered at the country level. The red vertical dashed line denotes the zero-effect benchmark. This robustness check assesses the influence of individual countries on the overall result.

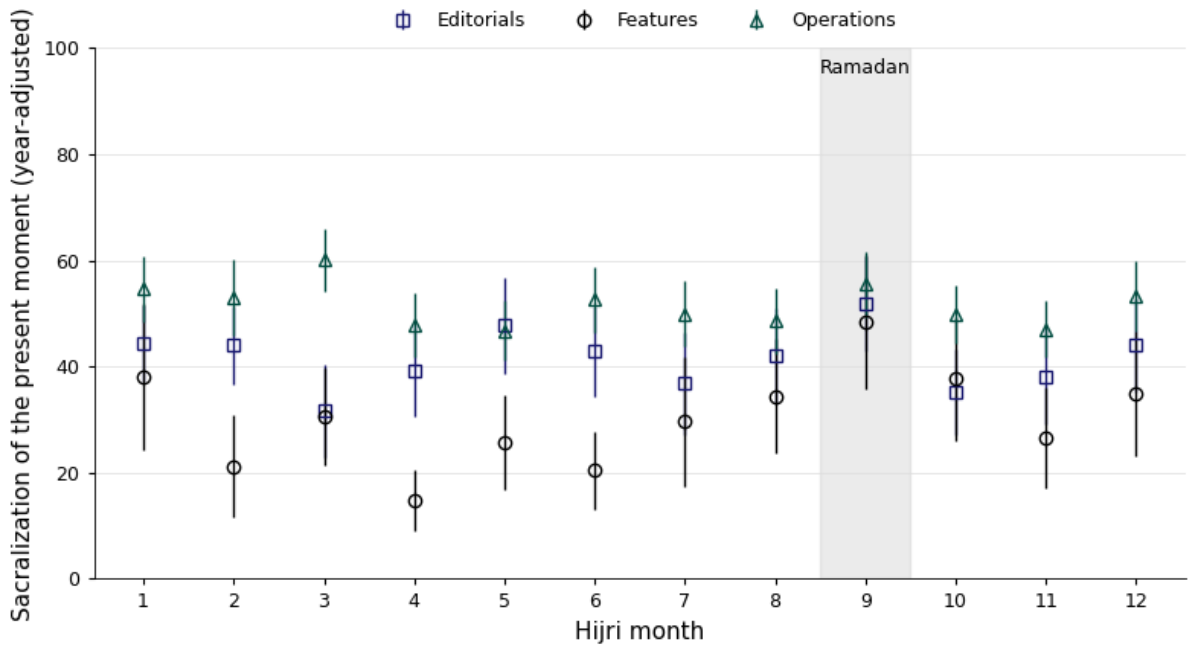


Figure A.6: Sacralization of the Present Moment in the Islamic State's *al-Naba*, by Hijri Month

Notes: The figure plots sacralization of the present moment in the Islamic State's weekly *al-Naba* by Hijri month across 533 issues (46 in Ramadan), separately for editorials, feature essays, and operational news (the *Editorials*, *Features*, and *Operations* series): the signed lead argument, the standalone essays on doctrine, history, or governance, and the weekly attack reports by province, respectively. Sacralization of the present moment is a GABRIEL rating on a 0–100 scale of the extent to which a section presents the current period as a specially blessed or rewarding time for striving, by the text's own argument rather than any date it reports. As in Figure 4, I residualize each section's score on Hijri-year fixed effects, re-center at the grand mean, and take the mean by Hijri month; the plotted points are these descriptive year-adjusted monthly averages, and spikes span the mean plus or minus 1.96 standard errors across sections (descriptive, not clustered by Hijri year). Ramadan is shaded. Editorial and feature-essay sacralization both rise during Ramadan, consistent with the doctrinal-framing channel. The construct is reported as a secondary measure, however, because it bleeds into the operational news, which scores moderately high in every month even though factual battle reports have no doctrinal reason to cast the present as a blessed time.

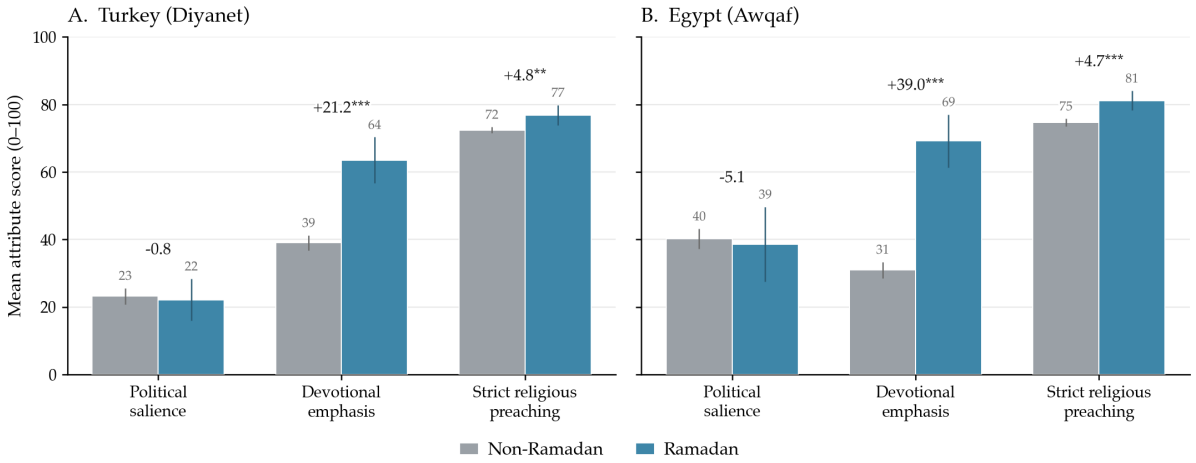


Figure A.7: Ramadan and Non-Ramadan Sermon Content in Two State Systems

Notes: The figure plots mean GABRIEL attribute scores for centrally drafted Friday sermons, separately for Ramadan and non-Ramadan Fridays. Panel A covers 610 weekly scripts issued by the Turkish *Diyanet* between March 2014 and February 2026, of which 50 fall in Ramadan; Panel B covers 460 weekly scripts issued by the Egyptian *Wizārat al-Awqāf* between July 2016, when the ministry moved to verbatim distribution, and March 2026, of which 35 fall in Ramadan. Each attribute is rated on a 0–100 scale: *political salience* captures regime-relevant political themes such as national unity, state authority, military sacrifice, and sovereignty; *devotional emphasis* uses the same attribute definition applied to *al-Naba* in Figure 4, so the two figures are measured on a common scale; and *strict religious preaching* captures the intensity of doctrinally authoritative instruction, which is high and stable year-round and serves the placebo role that operational news serves within an issue of *al-Naba*. Bars are unadjusted group means with spikes spanning the mean plus or minus 1.96 standard errors. Numbers above each pair report the Ramadan coefficient from Equation 24, which conditions on Hijri-year and Gregorian-month fixed effects and therefore differs from the raw difference between the two bars; stars denote significance at the 10, 5, and 1 percent levels using Newey-West standard errors with an eight-week bandwidth. Devotional emphasis rises sharply in Ramadan in both systems and strict religious preaching only slightly, while political salience does not rise in either.

Table A.1: Summary Statistics: Armed Conflict

Variable	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Panel A — UCDP Cross-Country Hijri Month (Muslim Share $>$ 75%, 31 countries)					
Log(State Conflicts+1)$_{cmt}$	12,648	0.729	1.298	0	7.602
Panel B — UCDP by Rebel Ideology (Muslim Share $>$ 75%, 30 countries)					
Log(Religious Events+1)$_{cmt}$	12,240	0.496	1.201	0	7.602
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.260	0.698	0	4.595
Panel C — UCDP by Aggression Attribution (Muslim Share $>$ 75%, 30 countries)					
Rebel-Initiated Events					
Log(Religious Events+1)$_{cmt}$	12,240	0.284	0.796	0	6.250
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.086	0.349	0	3.466
Government-Initiated Events					
Log(Religious Events+1)$_{cmt}$	12,240	0.247	0.798	0	6.479
Log(Secular-Nationalist Events+1)$_{cmt}$	12,240	0.096	0.391	0	3.892
Panel D — SIGACTS Afghanistan					
D.1 Province-Month Panel					
Log(Rebel Attacks+1)$_{pmt}$	5,542	2.252	1.803	0	7.581
Log(Direct Fire+1)$_{pmt}$	5,542	1.833	1.638	0	7.268
Log(IED Attacks+1)$_{pmt}$	5,542	1.608	1.556	0	6.645
D.2 Province-Month-Shift Panel (Pre/Post Iftar)					
Log(Rebel Attacks+1)$_{pmst}$	11,084	1.717	1.583	0	7.401
Log(Direct Fire+1)$_{pmst}$	11,084	1.356	1.416	0	7.090
Log(IED Attacks+1)$_{pmst}$	11,084	1.139	1.313	0	6.506
D.3 Province-Day-Hour Panel					
Log(Rebel Attacks+1)$_{pdht}$	3,804,192	0.038	0.182	0	3.434
Log(Direct Fire+1)$_{pdht}$	3,804,192	0.023	0.138	0	3.367
Log(IED Attacks+1)$_{pdht}$	3,804,192	0.017	0.118	0	2.944

Notes: This table presents summary statistics for all armed conflict variables. Subscripts follow the regression specifications: c denotes country, m Hijri month, t Hijri year, p province, s shift, d Gregorian day, and h hour. Panel A reports descriptives for state-based armed conflict from the UCDP GED (1989–2022) for 31 countries with a Muslim population share above 75%. The full sample covers 107 countries (43,656 observations). Panel B disaggregates log conflict events by rebel group ideology. Panel C further decomposes events by LLM-based aggression attribution using the GABRIEL framework. Panel D reports statistics from the declassified SIGACTS dataset covering insurgent attacks in Afghanistan (2002–2015) at three levels of temporal aggregation. Column (1) reports the number of observations, (2) and (3) the mean and standard deviation, and (4) and (5) the minimum and maximum values.

Table A.2: Summary Statistics: Demonstrations, Survey, Fasting Hours, and Raw Counts

Variable	(1) N	(2) Mean	(3) S.D.	(4) Min	(5) Max
Panel A — ACLED Cross-Country Hijri Month (Muslim Share $>$ 75%, 14 countries)					
Log(Demonstrations+1)$_{cmt}$	4,536	1.082	1.361	0	5.730
Log(Political Violence+1)$_{cmt}$	4,536	0.742	1.166	0	5.652
Panel B — World Values Survey (Individual-Level, Muslim Respondents)					
High-Frequency Prayer (Dummy)$_i$	35,449	0.732	0.443	0	1
Interviewed during Ramadan (Prayer Sample)$_i$	35,449	0.121	0.326	0	1
Participation in Demonstrations (Dummy)$_i$	39,049	0.364	0.481	0	1
Interviewed during Ramadan (Demo Sample)$_i$	39,049	0.106	0.308	0	1
Interest in Politics (Dummy)$_i$	43,582	0.483	0.500	0	1
Interviewed during Ramadan (Politics Sample)$_i$	43,582	0.099	0.299	0	1
Panel C — Fasting Hours (Ramadan-Only Sample)					
C.1 ACLED Sample (Muslim Share $>$ 75%, 14 countries)					
Log(Demonstrations+1)$_{ct}$	1,296	0.855	1.159	0	5.537
Log(Fasting Hours)$_{ct}$	1,296	2.526	0.063	2.318	2.719
C.2 UCDP Sample (Muslim Share $>$ 75%, 31 countries)					
Log(State Conflicts+1)$_{ct}$	3,710	0.494	1.058	0	7.298
Log(Fasting Hours)$_{ct}$	3,710	2.527	0.113	2.062	2.917
Panel D — Raw Event Counts (Levels)					
D.1 UCDP Armed Conflict (Muslim Share $>$ 75%)					
State Conflicts$_{cmt}$	12,648	11.904	80.440	0	2,001
Fatalities$_{cmt}$	12,648	79.768	507.713	0	20,214
Religious Events$_{cmt}$	12,240	10.879	81.562	0	2,001
Secular-Nationalist Events$_{cmt}$	12,240	1.210	5.478	0	98
D.2 SIGACTS Afghanistan					
Rebel Attacks$_{pt}$	5,542	45.931	130.828	0	1,960
Direct Fire$_{pt}$	5,542	26.487	76.880	0	1,433
IED Attacks$_{pt}$	5,542	19.444	59.471	0	768
D.3 ACLED Demonstrations (Muslim Share $>$ 75%)					
Demonstrations$_{cmt}$	4,536	9.789	28.352	0	307

Notes: This table presents summary statistics for civilian demonstrations, individual-level survey outcomes, fasting intensity, and raw count levels for all conflict variables. Subscripts indicate the level of observation: c denotes country, m Hijri month, t Hijri year, p province, and i individual respondent. Panel A reports descriptive statistics from the ACLED dataset. Panel B reports individual-level responses from Muslim participants across four waves of the World Values Survey (1999–2022). Panel C reports cross-sectional fasting hours calculated as the time between civil twilight and sunset at each capital city, restricted strictly to Ramadan months to test for physiological intensity mechanisms. Panel D provides the unlogged raw counts (levels) corresponding to the core armed conflict and demonstration samples. Column (1) reports the number of observations, (2) and (3) the mean and standard deviation, and (4) and (5) the minimum and maximum values.

Table A.3: Robustness of the Conflict Results to Functional Form

	UCDP, >75% Muslim			SIGACTS Afghanistan		
	All events	Rebel-initiated, religious	Rebel-initiated, secular	Direct fire	Indirect fire	IED
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Inverse hyperbolic sine (OLS)						
Ramadan	0.044***	0.037**	0.000	-0.024	-0.047	0.109***
	(0.015)	(0.016)	(0.009)	(0.028)	(0.034)	(0.025)
Post-iftar				-0.414***	-0.230***	-0.624***
				(0.061)	(0.065)	(0.068)
Ramadan × Post-iftar				0.158***	0.070**	0.044
				(0.032)	(0.033)	(0.032)
Observations	12,648	12,240	12,240	10,880	10,880	10,880
Panel B: Negative binomial						
Ramadan	0.069**	0.215***	-0.056	-0.126***	-0.204***	0.076*
	(0.034)	(0.058)	(0.113)	(0.042)	(0.075)	(0.043)
Post-iftar				-0.881***	-0.664***	-1.482***
				(0.078)	(0.138)	(0.048)
Ramadan × Post-iftar				0.283***	0.160**	0.263***
				(0.047)	(0.079)	(0.057)
Observations	6,672	2,892	2,388	9,176	7,880	8,632
Panel C: Poisson pseudo-maximum likelihood						
Ramadan	0.006	0.233***	-0.133*	-0.011	-0.103	0.106***
	(0.038)	(0.034)	(0.074)	(0.062)	(0.083)	(0.040)
Post-iftar				-1.273***	-0.948***	-1.519***
				(0.101)	(0.129)	(0.034)
Ramadan × Post-iftar				0.218***	0.003	0.205***
				(0.051)	(0.069)	(0.050)
Observations	6,672	2,892	2,388	9,176	7,880	8,632
Country-by-Hijri-year FE	✓	✓	✓			
Province-by-Hijri-year FE				✓	✓	✓
Gregorian month FE	✓	✓	✓	✓	✓	✓

Notes: Each panel re-estimates the main specifications with a different functional form. Columns (1) to (3) use UCDP state-based conflict in countries more than 75% Muslim: all events in column (1) (Table 1, column 2), and rebel-initiated events by religious and secular groups in columns (2) and (3) (Table 2, column 3). Columns (4) to (6) use SIGACTS province-month counts in the fasting window (05:00 to 17:59) and the post-iftar window (18:00 to 04:59) (Table 3). The Ramadan row is the effect during the fasting window, the Post-iftar row is the average difference between the post-iftar and fasting windows outside Ramadan, and the interaction is the difference between the post-iftar and fasting-window Ramadan effects. Panel A estimates OLS on the inverse hyperbolic sine of counts. Panel B estimates negative binomial (NB2) models with the fixed effects entered as indicators, and Panel C estimates Poisson pseudo-maximum likelihood; both use the PPML estimation sample, which drops fixed-effect cells without events. All panels include the fixed effects listed at the bottom of the table. Standard errors, clustered by country in columns (1) to (3) and by province in columns (4) to (6), are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: The Relative Effects of Ramadan on State-Based Armed Conflicts, 1989–2022: Near versus Distant Control Months

	Dependent Variable: log(armed conflict+1)						
	Near controls				Distant controls		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.030**	0.030**	0.008	0.002	0.045***	0.013	0.001
	(0.012)	(0.012)	(0.008)	(0.009)	(0.015)	(0.010)	(0.010)
Ramadan × % Muslim			0.019			0.026	
			(0.015)			(0.020)	
Ramadan × >75% Muslim				0.028*			0.044**
				(0.015)			(0.018)
Ramadan × 50–75% Muslim				-0.015			-0.007
				(0.024)			(0.030)
Ramadan × 25–50% Muslim				0.004			-0.006
				(0.032)			(0.035)
Ramadan × 1–25% Muslim				0.015			0.025
				(0.014)			(0.018)
Conflict mean	12.19	12.19	5.192	5.192	11.55	4.921	4.921
Sample	Muslim	Muslim	All	All	Muslim	All	All
Observations	7378	7378	25466	25466	6324	21828	21828
Gregorian month FE	✓	✓	✓	✓	✓	✓	✓
Country FE	✓						
Year FE	✓						
Country-by-year FE		✓	✓	✓	✓	✓	✓

Notes: The table varies the set of control months while holding the treatment fixed. Columns (1)–(4) retain only the three Hijri months on either side of Ramadan as controls (Jamada al-Thani, Rajab, Sha'ban, Shawwal, Dhul-Qa'dah, Dhul-Hijjah). Columns (5)–(7) drop those six months, so that controls are drawn only from months four to six removed from Ramadan (Muharram, Safar, Rabi al-Awwal, Rabi al-Thani, Jamada al-Awwal). The baseline estimate using all eleven non-Ramadan months is 0.037 (Table 1, column 5). Armed conflict is the count of state-based armed conflict events per country-month, drawn from the UCDP Georeferenced Event Dataset (GED), 1989–2022. Ramadan equals one if the Hijri month is the ninth. In columns (1)–(2) and (5), the sample consists of the 31 countries with at least 75% Muslims on average over the period (World Religion Project and Pew). In columns (3)–(4) and (6)–(7), the sample consists of 107 countries, with the reference group in columns (4) and (7) comprising the 22 countries with an average Muslim population of 1% or less. Conflict mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: The Relative Effects of Ramadan on Political Violence, 1997–2023 (ACLED)

	Dependent Variable: log(Political Violence +1)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	0.065***	0.079***	0.079***	0.076***	0.076***	-0.001	0.006
	(0.018)	(0.018)	(0.018)	(0.018)	(0.018)	(0.016)	(0.017)
Ramadan × % Muslim						0.070***	
						(0.025)	
Ramadan× >75% Muslim							0.065**
							(0.024)
Ramadan× 50–75% Muslim							-0.029
							(0.056)
Ramadan× 25–50% Muslim							0.020
							(0.055)
Ramadan× 1–25% Muslim							0.005
							(0.023)
Political Violence mean	5.260	5.260	5.260	5.260	5.260	3.437	3.437
Sample	Muslim	Muslim	Muslim	Muslim	Muslim	All	All
Observations	4536	4536	4536	4536	4536	15552	15552
Gregorian month FE		✓	✓	✓	✓	✓	✓
Country FE			✓	✓			
Year FE				✓			
Country-by-year FE					✓	✓	✓

Notes: Political violence is the count of state-based armed conflict events per country-month, drawn from the ACLED dataset, 1997–20223. In columns (1)–(5), the sample consists of 14 African countries with at least 75% Muslims on average for the given period in the World Religion Project (WRP) and PEW database. In columns (6)–(7), the sample and the control group consist of 48 and 6 African countries, respectively. Political violence mean is the average of the outcome in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Examples of LLM Aggressor Classification from UCDP Source Text

Side A (State)	Side B (Rebel)	UCDP Source Article Text	Class.
Gov. of Somalia	Al-Shabaab	"Xinhua News Agency, 2022-09-23, Somali forces kill 15 al-Shabab militants"	1
Gov. of Pakistan	TTP	"Agence France Presse, 2014-01-22, Bomb blast kills five in northwest Pakistan"	2
Gov. of Syria	Syrian insurgents	"SOHR, 2014-04-19, 203 were killed yesterday"	0
Gov. of Syria	Syrian insurgents	"VDC, 2013-09-20, Unidentified"	0
Gov. of Afghanistan	Hizb-i Islami (Khalis)	"The Globe and Mail, 1989-03-20, Rebel toll 215 in siege at Jalalabad, Kabul says"	1
Gov. of Afghanistan	Taleban	"Reuters, 1995-05-08, Afghan forces claim gains against Taleban militia"	1
Gov. of Syria	Syrian insurgents	"SOHR, 2014-01-05, Nasra Front executes soldiers from regular forces"	2
Gov. of Tunisia	JAK-T	"Agence France Presse, 2016-11-06, IS claims killing of Tunisian soldier"	2
Gov. of Turkey	PKK	"Xinhua News Agency, 2007-06-22, Two PKK members killed in operations in eastern Turkey"	1
Gov. of Uzbekistan	IMU	"AP, 2000-08-07, Militants invade Uzbekistan villages, clash with troops"	2
Gov. of Yemen	AQAP	"Agence France Presse, 2014-08-06, 'Qaeda ambush' kills 5 Yemen soldiers: security"	2

Notes: This table provides illustrative examples of how the GABRIEL LLM framework classified unstructured text from the UCDP GED dataset. Classifications are defined as: 1 = Side A (Government) initiated, 2 = Side B (Rebel) initiated, and 0 = Neutral/Indeterminate. The model conservatively defaults to 0 when the text describes casualties without clear attribution.

Table A.7: Protest Coding, Repression, and the Margin of the Protest Decline

	Counts		Repression		Riots	Share peaceful	
	All	Protests	Interv.	Force	Violent	Unwtd.	Wtd.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ramadan	-0.062	-0.082**	-0.034	-0.021	-0.004	0.033*	0.028
	(0.043)	(0.034)	(0.022)	(0.015)	(0.041)	(0.016)	(0.017)
Outcome mean	9.789	8.593	0.803	0.221	1.416	0.802	0.881
Observations	4,536	4,536	4,536	4,536	4,536	2,149	2,149
Greg. month FE	✓	✓	✓	✓	✓	✓	✓
Country-year FE	✓	✓	✓	✓	✓	✓	✓

Notes: The table tests whether the protest decline reflects fewer demonstrations or the reclassification of demonstrations out of ACLED's peaceful-protest category. The sample is the 14 African countries more than 75% Muslim, 1997 to 2023, at the country-by-Hijri-month level, the same sample as column (1) of Table 4, where the baseline peaceful-protest coefficient is -0.069. ACLED assigns the peaceful-protest sub-event only when demonstrators neither use force nor are met with force or intervention, so a Ramadan increase in policing would move events into the sub-events in columns (3) and (4) and lower the baseline count without any change in civilian behavior. Columns (1) to (5) are OLS on $\log(y+1)$. Column (1), *All*, counts all demonstrations, pooling ACLED's protest and riot categories. Column (2), *Protests*, counts the three protest sub-events, which sum exactly to ACLED's protest event type: peaceful protest, protest with intervention, and excessive force against protesters. Columns (3) and (4), *Interv.* and *Force*, take protests with intervention and protests met with excessive force separately; their low base rates bound the log coefficients near zero mechanically, and under Poisson pseudo-maximum likelihood they fall by 28.5% ($p = 0.027$) and 39.4% ($p = 0.063$) respectively. Column (5), *Violent*, takes violent demonstrations, recorded under ACLED's separate riot category, which tests for displacement across the event-type boundary. Columns (6) and (7) take the share of protests coded peaceful, conditional on at least one protest, unweighted and weighted by the protest count; because the conditioning event is itself treated, these are read alongside column (2) rather than on their own. All specifications include country-by-Hijri-year and Gregorian month fixed effects. Outcome means are in levels over the estimation sample. Standard errors, clustered by country, are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Robustness of the Protest Results to Functional Form

	ACLED peaceful protests			SIGACTS
	>75% Muslim	>75% Muslim	All countries	Demonstrations
	(1)	(2)	(3)	(4)
Panel A: Inverse hyperbolic sine (OLS)				
Ramadan	-0.104**	-0.082**	0.079	-0.035***
	(0.036)	(0.037)	(0.055)	(0.011)
Ramadan × >75% Muslim			-0.174**	
			(0.065)	
Ramadan × 50–75% Muslim			-0.049	
			(0.092)	
Ramadan × 25–50% Muslim			0.041	
			(0.067)	
Ramadan × 1–25% Muslim			-0.064	
			(0.058)	
Post-iftar				-0.066***
				(0.019)
Ramadan × Post-iftar				0.028**
				(0.012)
Observations	4,536	4,536	15,552	10,880
Panel B: Negative binomial				
Ramadan	-0.242***	-0.181***	0.172	-0.354**
	(0.076)	(0.066)	(0.172)	(0.180)
Ramadan × >75% Muslim			-0.404**	
			(0.187)	
Ramadan × 50–75% Muslim			-0.139	
			(0.263)	
Ramadan × 25–50% Muslim			0.002	
			(0.190)	
Ramadan × 1–25% Muslim			-0.155	
			(0.180)	
Post-iftar				-0.806***
				(0.151)
Ramadan × Post-iftar				0.142
				(0.213)
Observations	3,588	3,588	11,988	6,424
Panel C: Poisson pseudo-maximum likelihood				
Ramadan	-0.110	-0.068	0.130	-0.319**
	(0.069)	(0.062)	(0.183)	(0.161)
Ramadan × >75% Muslim			-0.233	
			(0.195)	
Ramadan × 50–75% Muslim			-0.195	
			(0.217)	
Ramadan × 25–50% Muslim			-0.135	
			(0.186)	
Ramadan × 1–25% Muslim			-0.114	
			(0.195)	
Post-iftar				-0.906***
				(0.171)
Ramadan × Post-iftar				0.125
				(0.221)
Observations	3,588	3,588	11,988	6,424
Country-by-Hijri-year FE	✓	✓	✓	
Province-by-Hijri-year FE				✓
Gregorian month FE		✓	✓	✓

Notes: Each panel re-estimates the protest specifications with a different functional form. Columns (1) to (3) use ACLED peaceful protests in African countries: the 14 countries more than 75% Muslim in columns (1) and (2), and all 48 countries in column (3), with Ramadan interacted with indicators for Muslim shares above 75%, 50 to 75%, 25 to 50%, and 1 to 25%, where the Ramadan row is the effect in countries below 1% Muslim. Column (4) uses SIGACTS demonstrations in Afghan provinces in the fasting window (05:00 to 17:59) and the post-iftar window (18:00 to 04:59); the Ramadan row is the effect during the fasting window, the Post-iftar row is the average difference between the post-iftar and fasting windows outside Ramadan, and the interaction is the difference between the post-iftar and fasting-window Ramadan effects (Table 4). Panel A estimates OLS on the inverse hyperbolic sine of counts, Panel B estimates negative binomial (NB2) models with the fixed effects entered as indicators, and Panel C estimates Poisson pseudo-maximum likelihood; both count models use the PPML estimation sample, which drops fixed-effect groups without events. All panels include the fixed effects listed at the bottom of the table. Standard errors, clustered by country in columns (1) to (3) and by province in column (4), are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: The Relative Effects of Ramadan on Civilian Protest Activity: Near versus Distant Control Months

	Cross-Country (ACLED)			Afghanistan (SIGACTS)	
	log(Peaceful Protests + 1)			log(Demonstrations + 1)	
	(1)	(2)	(3)	(4)	(5)
Panel A: Near controls (Ramadan ± 3 Hijri months)					
Ramadan	-0.056**	0.042**	0.060	-0.036***	-0.025**
	(0.024)	(0.019)	(0.042)	(0.013)	(0.009)
Ramadan × % Muslim		-0.094**			
		(0.035)			
Ramadan × >75% Muslim			-0.125**		
			(0.049)		
Ramadan × 50–75% Muslim			-0.049		
			(0.076)		
Ramadan × 25–50% Muslim			0.046		
			(0.053)		
Ramadan × 1–25% Muslim			-0.056		
			(0.045)		
Post-Iftar					-0.047***
					(0.013)
Ramadan × Post-Iftar					0.018
					(0.012)
Outcome mean	7.384	4.185	4.185	0.277	0.123
Observations	2,646	9,072	9,072	3,094	6,188
Panel B: Distant controls (4–6 Hijri months from Ramadan)					
Ramadan	-0.072*	0.063***	0.069	-0.036*	-0.029**
	(0.040)	(0.022)	(0.052)	(0.019)	(0.013)
Ramadan × % Muslim		-0.146***			
		(0.047)			
Ramadan × >75% Muslim			-0.162**		
			(0.063)		
Ramadan × 50–75% Muslim			-0.028		
			(0.072)		
Ramadan × 25–50% Muslim			0.014		
			(0.060)		
Ramadan × 1–25% Muslim			-0.043		
			(0.054)		
Post-Iftar					-0.056***
					(0.018)
Ramadan × Post-Iftar					0.026***
					(0.009)
Outcome mean	7.664	4.216	4.216	0.320	0.143
Observations	2,268	7,776	7,776	2,754	5,576
Sample	Muslim	All	All	Afghanistan	Afghanistan
Gregorian month FE	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓		
Province-by-Hijri-year FE				✓	✓

Notes: The table varies the set of control months while holding the treatment fixed. Panel A retains only the three Hijri months on either side of Ramadan as controls (Jamada al-Thani, Rajab, Sha'ban, Shawwal, Dhul-Qa'dah, Dhul-Hijjah). Panel B drops those six months, so that controls are drawn only from months four to six removed from Ramadan (Muharram, Safar, Rabi al-Awwal, Rabi al-Thani, Jamada al-Awwal). The baseline estimates using all eleven non-Ramadan months appear in Table 4. Columns 1–3 use the ACLED dataset for 48 African countries (1997–2023). Column 1 restricts the sample to the 14 countries with a Muslim population share above 75%. Columns 2 and 3 use the full sample, interacting the Ramadan indicator with the continuous Muslim population share and with discrete share categories respectively; the baseline category in column 3 is countries with less than 1% Muslim population. Columns 4 and 5 use the SIGACTS dataset for Afghan provinces (2002–2015); the outcome is the logarithm of demonstration events plus one at the province-Hijri-month level (column 4) and at the province-Hijri-month-shift level (column 5), where each province-month is split into pre-iftar (dawn to sunset) and post-iftar (sunset to dawn) windows. In column 5 the Ramadan main effect captures the pre-iftar differential and the Ramadan × Post-Iftar interaction measures the additional reallocation toward post-iftar hours during Ramadan; the total post-iftar effect is the sum of the two coefficients. Outcome mean reports the average level of the dependent variable in the estimation sample. Robust standard errors are in parentheses, clustered at the country level in columns 1–3 and at the province level in columns 4–5 (34 provinces). Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.10: Balance Test on Demographic Covariates: Ramadan vs. Non-Ramadan Interviews

	Non-Ramadan	Ramadan	Difference	p-value
Female	0.511	0.517	-0.006	0.513
	(0.500)	(0.500)		
Age	38.135	41.262	-3.127***	0.000
	(13.652)	(14.713)		
Married	0.683	0.678	0.005	0.560
	(0.465)	(0.467)		
Number of children	1.943	1.949	-0.006	0.851
	(1.724)	(1.919)		
Education	1.759	1.770	-0.011	0.448
	(0.821)	(0.807)		
Employed	0.531	0.523	0.008	0.398
	(0.499)	(0.500)		
Income (1–10 scale)	4.756	4.629	0.127***	0.001
	(2.068)	(2.110)		
Religion very important	0.862	0.879	-0.018***	0.003
	(0.345)	(0.326)		
N	15,567	3,759	19,326	

Notes: This table reports balance tests on demographic covariates for the analytical sample used in Table 5. The sample consists of Muslim respondents in 23 Muslim-majority countries from the World Values Survey (1999–2023) who report religion as either "very important" or "rather important" on the WVS importance-of-religion scale (approximately 95% of Muslim respondents in the sample) and who were interviewed in the seven Hijri months centered on Ramadan (Jamada-Al-Thani through Dhul-Hijjah). The Difference column reports the mean difference (Non-Ramadan − Ramadan) from a two-sample t-test with unequal variances; the p-value column reports the corresponding two-sided p-value. Standard deviations are in parentheses below the means. The Employed indicator equals 1 for respondents reporting full-time, part-time, or self-employed work and 0 for retirees, housewives, students, unemployed respondents, and other categories. The Religion very important indicator equals 1 for respondents reporting religion as "very important" and 0 for those reporting "rather important." Ramadan-period interviews account for 19.5% of the seven-month window, above the 14.3% share expected under uniform fieldwork timing within the window. The average number of interviews per country-Hijri-year-Hijri-month cell is statistically indistinguishable across Ramadan and non-Ramadan periods (342 vs. 399, p=0.69), as is the average number of distinct interview days per cell (8.6 vs. 9.3, p=0.76). Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.11: Placebo Test: Non-Muslims in Muslim-Majority Countries

	Spirituality/Prayer		Interest in Politics		Attending Demonstrations	
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	-0.033	-0.039	-0.045	-0.072	-0.154	-0.161
	(0.059)	(0.058)	(0.050)	(0.061)	(0.108)	(0.103)
Outcome mean	0.544	0.595	0.526	0.497	0.370	0.412
Sample	±3 mo.	All	±3 mo.	All	±3 mo.	All
Observations	3,330	6,115	3,460	6,321	3,308	6,109
Adj. R^2	0.219	0.331	0.097	0.084	0.089	0.118
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓
Controls	✓	✓	✓	✓	✓	✓

Notes: This table reports placebo estimates of the Ramadan effect on non-Muslim respondents living in the same 23 Muslim-majority countries used in the main analysis. The sample consists of non-Muslim respondents from the World Values Survey (2001–2022). Because non-Muslims living in Muslim-majority countries do not observe Ramadan but share the same fieldwork timing, country-wave composition, and broader societal context as the Muslim sample, the Ramadan indicator should have no effect on individual-level religiosity or political engagement if the main results documented in Table 5 operate through individual religious observance rather than country-level fieldwork artifacts. Columns (1)–(2) use a binary indicator for high-intensity prayer (praying several times a day or once a day and zero otherwise). Columns (3)–(4) use a binary indicator for political interest (reporting being "very interested" or "somewhat interested"). Columns (5)–(6) use a binary indicator for demonstration willingness (reporting having attended or willingness to attend lawful peaceful demonstrations). Odd-numbered columns restrict the sample to interviews conducted in the seven Hijri months centered on Ramadan; even-numbered columns use the full twelve-month sample. All specifications include country-by-Hijri-year and Gregorian month fixed effects and the full demographic control set: gender, age, age-squared, marital status, number of children, education, employment status, income, and self-reported importance of religion. Outcome mean reports the baseline average of the dependent variable. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.12: The *al-Naba* corpus and section-level content measures

			Mean (SD), 0–100 scale	
Section	Period	*N*	Devotional emphasis	Sacralization
Feature essays	Non-Ramadan	274	50.1 (31.6)	28.0 (26.4)
	Ramadan	28	78.1 (26.8)	49.1 (34.5)
Editorials	Non-Ramadan	365	13.6 (13.8)	40.4 (26.3)
	Ramadan	33	19.5 (27.2)	51.6 (28.1)
Operational news	Non-Ramadan	487	13.9 (10.6)	51.1 (23.4)
	Ramadan	46	21.4 (18.8)	55.3 (22.8)

Notes. The sample comprises 533 issues of the Islamic State's weekly *al-Naba*, spanning 22 December 2015 through 16 April 2026, of which 46 fall during Ramadan. Each issue is segmented into sections; the table reports the three running-prose section types after dropping sections with fewer than 500 extracted characters. *N* is the number of scored sections in each cell. Cells report the mean and, in parentheses, the standard deviation of each GABRIEL construct, rated on a 0–100 scale (Appendix B). Feature-essay frequency is flat across the Hijri calendar, ranging from 21 to 30 essays per month with 28 in Ramadan, so the Ramadan contrast in devotional emphasis reflects content rather than essay supply.

Table A.13: Effect of Ramadan on Civilian Targeting

	UCDP One-Sided Violence				GTD: Attacks on Private Citizens			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ramadan	0.027***	0.000	0.017	0.051***	0.015**	0.007	0.009	0.050**
	(0.009)	(0.008)	(0.011)	(0.016)	(0.006)	(0.004)	(0.006)	(0.020)
Group ideology	Religious	Secular	Religious	Religious	Religious	Secular	Religious	Religious
Group sample	All groups	All groups	Excl. IS	IS only	All groups	All groups	Excl. IS	IS only
Outcome mean (levels)	0.595	0.118	0.351	0.800	0.666	0.096	0.450	1.025
Observations	11,016	11,016	9,384	4,080	18,120	18,120	18,120	3,806
Countries	27	27	23	17	30	30	30	22
Gregorian month FE	✓	✓	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table estimates the effect of Ramadan on civilian-directed violence, in contrast to the state-directed conflict of Table 2. The outcome is the logarithm of civilian-targeting events plus one at the country-Hijri-month level. Columns 1–4 use UCDP one-sided violence, in which an organized non-state group deliberately kills civilians, over 1989–2022; columns 5–8 use attacks on private citizens (GTD target type 14) over 1970–2020. Perpetrating groups are classified as Religious or Secular-nationalist following the scheme in Appendix C. Columns 1, 2, 5, and 6 use the full set of classified groups; columns 3 and 7 exclude the Islamic State; columns 4 and 8 retain the Islamic State alone. Secular-nationalist estimates are reported only for the full sample, as neither the Islamic-State-excluded nor the Islamic-State-only split applies to secular groups. The country counts fall across the religious columns because the samples restrict to Muslim-majority countries in which the relevant groups are active: dropping the Islamic State removes countries where it is the only classified religious perpetrator (27 to 23 in UCDP), and the Islamic-State-only panel retains only the countries in which it operates (17 in UCDP, 22 in GTD). GTD counts exceed UCDP counts because GTD records attacks meeting a civilian-directed terrorism definition without UCDP's 25-death annual inclusion threshold, and because GTD names a broader set of perpetrators; the GTD panel drops the missing 1993 database year and groups responsible for fewer than five incidents, and incidents attributed to groups of unknown ideology are excluded. Appendix D details the construction of both panels. All specifications include Gregorian month and country-by-Hijri-year fixed effects and restrict to Muslim-majority countries. Outcome mean (levels) reports the average level of the dependent variable in the estimation sample. Robust standard errors clustered at the country level are in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.14: Anticipation, Displacement, and the Uniqueness of Ramadan

	Armed Conflict (UCDP)			Demonstrations (ACLED)		
	(1)	(2)	(3)	(4)	(5)	(6)
Ramadan	0.037***	0.038***	0.040***	-0.069**	-0.080**	-0.080*
	(0.012)	(0.013)	(0.014)	(0.031)	(0.034)	(0.038)
Sha'ban (1-month lead)		0.026**			-0.032	
		(0.012)			(0.029)	
Pre-Ramadan (2-month window)			0.026**			0.012
			(0.010)			(0.028)
Shawwal (1-month lag)		-0.012			-0.081**	
		(0.015)			(0.028)	
Post-Ramadan (2-month window)			-0.006			-0.072**
			(0.014)			(0.024)
Specification	Baseline	Triple	Pooled	Baseline	Triple	Pooled
Outcome mean (levels)	5.046	5.046	5.046	4.215	4.215	4.215
Observations	12648	12648	12648	4536	4536	4536
Gregorian month FE	✓	✓	✓	✓	✓	✓
Country-by-Hijri-year FE	✓	✓	✓	✓	✓	✓

Notes: This table reports leads/lags specifications that test whether the Ramadan effect reflects strategic anticipation, intertemporal displacement, or a generalized Hijri-calendar pattern, rather than within-Ramadan religious mobilization. Columns (1)–(3) use the log-transformed number of state-based armed conflict events plus one as the dependent variable, drawing on UCDP-GED data for 31 Muslim-majority countries (1989–2022). Columns (4)–(6) use the log-transformed number of peaceful protests plus one, drawing on ACLED data for 14 African Muslim-majority countries (1997–2023). All columns restrict the sample to countries with a Muslim population share above 75%. Columns (1) and (4) reproduce the baseline Ramadan-only specification from Equation 7. Columns (2) and (5) add separate indicators for the one-month lead (Sha'ban) and one-month lag (Shawwal). Columns (3) and (6) replace these with pooled indicators for the two-month window before Ramadan (Rajab and Sha'ban) and the two-month window after Ramadan (Shawwal and Dhul-Qa'dah). Under strategic anticipation or intertemporal displacement, the lead and lag coefficients should be negative and of comparable magnitude to the Ramadan coefficient; positive or null leads and lags reject these accounts. All specifications include country-by-Hijri-year and Gregorian month fixed effects. Outcome mean reports the average of the dependent variable in levels. Robust standard errors are in parentheses, clustered at the country level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Table A.15: The Effect of Fasting Intensity on Armed Conflicts and Demonstrations

	Dependent Variable: log(armed conflict+1)			Dependent Variable: log(demonstrations+1)		
	(1)	(2)	(3)	(4)	(5)	(6)
RFH	-0.753	0.301	0.262	6.430***	1.008	0.591
	(3.224)	(0.183)	(0.181)	(2.052)	(6.820)	(3.203)
RFH × >75% Muslim			0.477			4.585
			(2.061)			(3.895)
RFH × 50–75% Muslim			-0.682			-13.941***
			(1.309)			(4.791)
RFH × 25–50% Muslim			-1.958*			1.729
			(1.180)			(3.547)
RFH × 1–25% Muslim			-0.077			-2.097
			(0.555)			(4.118)
Outcome Mean (levels)	12.52	0.891	5.636	8.566	2.469	5.730
Sample	Muslim	Non-Muslim	All	Muslim	Non-Muslim	All
Observations	945	770	3,710	378	162	1,296
Country FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Share-by-year FE			✓			✓

Notes: This table tests whether fasting intensity, measured as the logarithm of daily Ramadan fasting hours (RFH), predicts armed conflict or demonstrations. The sample is restricted to the month of Ramadan only, so each country contributes one observation per Hijri year. Fasting hours vary across country-years because Ramadan rotates through Gregorian seasons: the same country experiences longer fasts when Ramadan falls in summer and shorter fasts when it falls in winter. All regressions include controls for annual population levels. Columns (1) and (4) restrict to Muslim-majority countries (Muslim population share above 75%), columns (2) and (5) to non-Muslim countries (below 1%) as a placebo. Columns (3) and (6) pool all countries and interact RFH with indicators for Muslim population share categories; the omitted baseline is countries with less than 1% Muslim population. Country fixed effects absorb time-invariant latitude differences, and Hijri year fixed effects absorb which Gregorian season Ramadan falls in globally. Share-by-year FE controls for the muslim population share bin dummies and their interactions with year dummies. The RFH coefficient is identified from the interaction of latitude and seasonal timing: within a given year, countries at different latitudes experience different fasting durations, and within a given country, fasting duration varies as Ramadan rotates across seasons. Robust standard errors are in parentheses, clustered at the country level (31 countries in column 1, 22 in column 2, 106 in column 3, 14 in column 4, 6 in column 5, and 48 in column 6). Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix A

A Religious Rewards and Time Allocation: Proofs and Extensions

This appendix collects the formal results stated in Section 3. Section A.1 derives the time budget. Section A.2 establishes existence and uniqueness and proves Proposition 1 and the two benchmarks. Section A.3 gives the exact responses behind the individual-level test. Section A.4 gives an organizational implementation. Section A.5 adds target choice. Section A.6 adds within-day timing. Section A.7 states what group comparisons identify once common shocks, coordination, and backlash are allowed. Section A.8 examines the linear religious payoff.

A.1 The time budget and the organizational margin

Let physical time satisfy $a + d + r + \ell = T_{\text{all}}$, where d is time in other duty tasks. Suppose the institutional schedule implies

$$d(a) = D_0 - (1 - \tau)a, \qquad 0 \le \tau \le 1.$$

Substitution gives $\tau a + r + \ell = T_{\text{all}} - D_0 \equiv T$, which is equation (1). Choose $\bar{a}$ so that $d(a) \ge 0$ wherever this schedule applies. At $\tau = 1$, additional political effort comes entirely from discretionary time. At $\tau = 0$, total duty time $a + d = D_0$ is fixed. Intermediate values allow both margins. This is a short-run institutional assumption about how an organization schedules operations, assessed from organizational evidence rather than inferred from ideology. The value of displaced duty tasks and the effort cost of deployment enter payoffs through $B(a)$.

A.2 Existence, the shadow price, and the main comparative static

Lemma 1 (Existence and uniqueness)**.** *For $\beta > 0$, under the stated continuity, strict concavity, and compactness conditions, the allocation problem has a unique optimum. If that optimum is interior, it satisfies equation* (3)*.*

Proof. The feasible set is nonempty and compact when $T > 0$ and $\bar{a} < \infty$, and continuity ensures an optimum. Substituting $\ell = T - \tau a - r$, the objective is

$$B(a) + u(T - \tau a - r) + m\big[f(r) + \theta a\big].$$

Write $E = -B''(a) > 0$, $H = -u''(\ell) > 0$, and $K = -mf''(r) > 0$. The negative Hessian in (a, r) is

$$\begin{pmatrix} E + \tau^2 H & \tau H \\ \tau H & H + K \end{pmatrix},$$

with determinant $E(H + K) + \tau^2 HK > 0$. The objective is therefore strictly concave and the optimum unique, and an interior optimum satisfies the first-order conditions (3). □

Uniqueness does not imply interiority, which is a separate restriction on the solution under examination. The derivatives below apply locally to interior solutions. At a binding constraint the active set must be considered separately.

Define the indirect value of nonpolitical time,

$$W(S, m) = \max_{r+\ell=S} \{u(\ell) + mf(r)\}. \tag{12}$$

The envelope theorem gives $W_S = \lambda$. At an interior solution, differentiating $u'(S-r) = mf'(r)$, first holding S fixed and then holding m fixed, yields

$$r_m\big|_S = \frac{f'}{H+K}, \qquad r_S = \frac{H}{H+K},$$

and therefore

$$\lambda_m\big|_S = \frac{Hf'}{H+K} \equiv \psi > 0, \qquad \lambda_S = -\frac{HK}{H+K} \equiv -h < 0. \tag{13}$$

Because $S = T - \tau a$, holding a fixed is equivalent to holding S fixed, and $\lambda_a = \tau h$. The fraction in ψ lies strictly between zero and one, so letting other activities adjust attenuates the increase in the shadow price relative to a calculation that fixes the worship allocation.

The outer problem is $\max_a B(a) + m\theta a + W(T - \tau a, m)$, with first-order condition (3). Differentiating with respect to m,

$$\frac{\mathrm{d}a^*}{\mathrm{d}m} = \frac{\theta - \tau\psi}{E + \tau^2 h}, \qquad \mathcal{D} \equiv E + \tau^2 h > 0. \tag{14}$$

Multiplying by $m'(R) > 0$ gives (6) and proves the threshold in Proposition 1. Differentiating the worship condition together with the budget gives

$$\frac{\mathrm{d}r^*}{\mathrm{d}m} = \frac{f' - H\tau\,(\mathrm{d}a^*/\mathrm{d}m)}{H+K}. \tag{15}$$

For $\theta \le 0$ and $\tau > 0$, political effort falls and worship rises strictly at an interior optimum, which completes Proposition 1. More generally, worship rises whenever $\theta \le \tau\psi$, since political effort is then nonincreasing.

Corollary 1 (The two benchmarks)**.** *For $\beta > 0$ at an interior solution, (a) at $\tau = 0$, $\mathrm{d}a^*/\mathrm{d}m = \theta/E$, so $\operatorname{sign}(\mathrm{d}a^*/\mathrm{d}R) = \operatorname{sign}(\theta)$, while $\mathrm{d}r^*/\mathrm{d}m = f'/(H+K) > 0$. (b) At $\theta = 0$ and $\tau = 1$, $\mathrm{d}a^*/\mathrm{d}m = -\psi/\mathcal{D} < 0$ and $\mathrm{d}r^*/\mathrm{d}m > 0$.*

Proof. Set $\tau = 0$ in (14), so h drops out and $\mathcal{D} = E$; then $\mathrm{d}a^*/\mathrm{d}m = \theta/E$, and (15) reduces to $f'/(H+K)$ because the second term vanishes. Deployments and worship can both rise, since deployment replaces another duty task rather than devotional time. For (b), set $\theta = 0$ and $\tau = 1$ in (14) and (15). □

Part (a) is the compatibility benchmark: doctrine alone orders responses, and the sign result that a reduced treatment of sacred time would assert for all actors holds for every value of doctrinal credit when political effort draws on no discretionary time. Part (b) is the displacement benchmark. Proposition 1 interpolates and identifies which benchmark governs a given actor-action pair.

These derivatives are local. A finite Ramadan change has a signed effect when the threshold stays on the same side along the relevant solution path. An ordering of θ across heterogeneous actors does not by itself order the magnitudes of their responses, since curvature, β, τ, and the outside allocation also enter.

Positive merit with a negative response. Let $T = 3$, $\tau = 1$, $B(a) = 2.5a - a^2/2$, $u(\ell) = 3\ell - \ell^2/2$, and $f(r) = 2r - r^2/4$ on a neighborhood of the solution. At $m = 1$ and $\theta = 0.3$ the optimum is $(a, r, \ell) = (1.1, 0.6, 1.3)$, with $\lambda = 1.7$, $\psi = 1.1333$, $h = 1/3$, $\mathcal{D} = 4/3$, and $\mathrm{d}a^*/\mathrm{d}m = -0.625$. Political activity earns positive religious merit and still declines, because its opportunity cost rises faster. The numbers illustrate the threshold and are not calibrated estimates.

A.3 The individual-level comparison

Consider an action with $\theta = 0$ and $\tau > 0$, and add $\sigma(R)a$ to (2), where $\sigma'(R) < 0$ represents a higher assembly cost. For $\beta > 0$ at an interior solution,

$$\frac{\mathrm{d}a^*}{\mathrm{d}R} = \frac{\sigma' - m'\tau\psi}{\mathcal{D}}, \qquad \frac{\mathrm{d}r^*}{\mathrm{d}R} = \frac{m'f' - H\tau(\mathrm{d}a^*/\mathrm{d}R)}{H + K}.$$

A religious-reward increase alone reduces political effort and raises worship. A pure assembly-cost shock ($m' = 0$, $\sigma' < 0$) produces the same two signs by releasing time from political participation. Both statements require $\tau > 0$: at $\tau = 0$ the religious-reward channel leaves political effort unchanged, and a cost shock releases no participation time to worship.

Remark 1 (Differential exposure to religious rewards)**.** *For $\beta = 0$, the actor chooses $r = 0$, since $u' > 0$, and maximizes $B(a) + \sigma(R)a + u(T - \tau a)$. At an interior political margin,*

$$\frac{\mathrm{d}a^*}{\mathrm{d}R} = \frac{\sigma'}{E + \tau^2 H}.$$

The religious-reward channel has no direct effect on this actor, whereas an assembly-cost shock lowers participation. The interior worship formulas cannot be evaluated at $m = 0$ to obtain this response: worship is at a corner, and the time opportunity cost remains.

Empirically, the relevant comparison is differential exposure to the Ramadan reward increase, rather than absence of all religious valuation. A person whose own religious returns are constant across Ramadan has $m' = 0$ and can still experience a cost-shock response. Under an interior worship allocation, that response is $\sigma'/\mathcal{D}$. The comparison tests a common assembly-cost explanation under comparable shock exposure and appropriate allowance for differences in curvature. Neither a monotone response gradient in β nor equality of cost-shock responses follows from the baseline model. Surveyed interest and willingness require an additional mapping to political effort.

A.4 An organizational implementation

Suppose a leader obtains political return $G(a)$ from deployments. A representative member bears deployment disutility $e(a)$, receives compensation w, and obtains

$$w - e(a) + m\theta a + W(T - \tau a, m).$$

Normalize the member's outside value to $W(T, m)$. If the participation constraint binds and compensation adjusts at the relevant margin, the required compensation is

$$w_{\min}(a, m) = e(a) + W(T, m) - W(T - \tau a, m) - m\theta a, \tag{16}$$

with marginal cost

$$\partial_a w_{\min} = e'(a) + \tau\lambda - m\theta.$$

Maximizing $G(a) - w_{\min}(a, m)$ reproduces the outer problem of Section A.2 with $B = G - e$, up to a term independent of a. Religious credit lowers the compensation a deployment requires and displaced devotional time raises it, so Proposition 1 applies to the leader's choice with the members' θ and τ entering through (16).

This is an implementation under transferable compensation and a slack compensation floor, not a claim that such contracts are observed. Fixed wages, coercion, recruitment frictions, heterogeneous members, and binding financing constraints change the response. An internal objective that weights the same member costs gives another implementation with the same reduced form.

The implementation also makes one competing channel explicit. A Ramadan-specific organizational return $s_g(R)a$, from publicity or financing, raises the leader's demand for effort even when members' religious credit is unchanged, provided the leader can elicit or compensate that effort. Remark 2 states the restrictions under which common organizational returns cancel in group comparisons.

A.5 Targets under a common resource constraint

Let $j \in \{S, C\}$ index state and civilian targets, and consider

$$\max \; B_S(a_S) + B_C(a_C) + u(\ell) + m\big[f(r) + \theta_S a_S + \theta_C a_C\big], \quad \tau(a_S + a_C) + r + \ell = T,$$

with separable strictly concave political returns, equal time requirements across targets, and $E_j = -B_j'' > 0$. Interior first-order conditions imply

$$\frac{\mathrm{d}a_j}{\mathrm{d}m} = \frac{\theta_j - \tau\lambda_m}{E_j}, \tag{17}$$

$$\lambda_m = \frac{f'/K + \tau\theta_S/E_S + \tau\theta_C/E_C}{1/H + 1/K + \tau^2/E_S + \tau^2/E_C}, \tag{18}$$

where λ_m is the total equilibrium derivative, distinct from the partial derivative ψ in (5). Equation (18) follows by differentiating all first-order conditions and imposing the differentiated time budget.

Corollary 2 (Target allocation)**.** *At an interior solution with $\theta_C \le 0 \le \theta_S$, civilian-directed effort is locally nonincreasing in m, strictly decreasing if $\tau > 0$ or $\theta_C < 0$. State-directed effort rises only if $\theta_S > \tau\lambda_m$. Conditional on a fixed total $A = a_S + a_C$, the allocation shifts toward the target with greater credit:*

$$\left.\frac{\partial a_S}{\partial m}\right|_A = \frac{\theta_S - \theta_C}{E_S + E_C}. \tag{19}$$

Positive credit for a civilian category permits escalation on that margin only if it exceeds $\tau\lambda_m$.

Proof. Multiplying (18) by τ shows that $\tau\lambda_m$ is a weighted average of θ_C, θ_S, $\tau f'$, and zero with nonnegative weights $\{\tau^2/E_C,\ \tau^2/E_S,\ 1/K,\ 1/H\}$ normalized by their sum. With $\theta_C \le 0 \le \theta_S$ this gives $\tau\lambda_m \ge \theta_C$, so $\theta_C - \tau\lambda_m \le 0$ and (17) delivers the nonincrease, strictly when $\tau > 0$ or $\theta_C < 0$. The state-directed condition is (17) at $j = S$. For (19), the target-allocation condition at fixed A is $B_S'(a_S) - B_C'(A - a_S) + m(\theta_S - \theta_C) = 0$; differentiating in m gives the stated derivative. □

The corollary supplies a within-organization composition prediction alongside the level restriction. Unequal time requirements across targets, or complementarities in the two violence technologies, require a modified result. Civilian status in event data and target legitimacy under a group's doctrine are coded separately and are not assumed to coincide.

A.6 Within-day timing and the cost envelope

Assumption 1 (Observance under arms)**.** *Enough participants observe the fast for accumulated fasting to raise the average marginal exertion cost of the relevant activity.*

Assumption 1 is a behavioral premise rather than an implication of the model, and the timing evidence assesses it jointly with the scheduling technology. Jurisprudence permits travelers to break the fast and the canonical precedents arise on campaign, so observance while operating is a choice.

For a fixed tactic and target, allocate total activity a across time bins $x_h \geq 0$ with $\sum_h x_h = a$, and let

$$C(a, R, L) = \min_{\{x_h\}} \sum_h \big\{ [b_h + \kappa F_h(R, L) + \eta_h] x_h + g_h(x_h) \big\}, \tag{20}$$

where g_h is strictly convex, L is fast duration, F_h is a fatigue profile, and η_h captures target-specific costs. Exertion intensity enters through κ, which is smaller for events requiring little effort at the moment of execution. The relevant fatigue profile is low soon after the pre-dawn meal, rises through the day, and falls after iftar, so the model is stated in time since nourishment rather than in fasting hours as such. This is a model of recorded event time, not a claim that preparing such events is effortless.

For two periods at fixed a, suppose Ramadan adds κR to the first period's marginal cost and a fixed night penalty η applies to the second. At an interior optimum,

$$\left.\frac{\partial x_N}{\partial R}\right|_a = \frac{\kappa}{g_D''(x_D) + g_N''(x_N)} > 0. \tag{21}$$

Corollary 3 (Conditional timing response)**.** *At fixed total activity, an increase in the first period's marginal exertion cost shifts activity toward the second period as in* (21)*. A smaller κ gives a smaller shift, holding scheduling curvature fixed. The result does not sign the change in total activity.*

A constant η does not appear in (21). With quadratic g_h, changing η shifts baseline timing but leaves this derivative unchanged, so attenuation of the Ramadan reallocation against a night-capable adversary requires an additional restriction: a binding corner or capacity constraint, sufficiently different curvature, or a relative cost that itself depends on R. Comparing the levels of day and night costs is not enough to deliver a difference-in-differences timing result.

If the timing envelope is an additional resource cost, the outer condition becomes $B'(a) + m\theta = \tau\lambda + C_a$, and under strict concavity

$$\left.\frac{\partial a^*}{\partial L}\right|_m = -\frac{C_{aL}}{E + \tau^2 h + C_{aa}}. \tag{22}$$

The baseline B must then exclude this cost to avoid double counting. Duration invariance requires $C_{aL} = 0$. The availability of an alternative tactic whose own cost is independent of duration is not sufficient: it must leave the optimized marginal cost unchanged, including capacities and substitution costs. Likewise, when timing costs vary with R, the total salience response is proportional to $m'(\theta - \tau\psi) - C_{aR}$. The level and timing margins are therefore connected in general, and a null response of monthly totals to fast duration is an empirical finding about the production technology rather than a prediction of the doctrinal mechanism.

A.7 Common shocks and the interpretation of group comparisons

Remark 2 (What the ideology comparison identifies)**.** *Add a common marginal-return shock $s(R)a$, capturing changes in schedules, markets, state tempo, publicity, or financing. For group g,*

$$\frac{\mathrm{d}a_g}{\mathrm{d}R} = \frac{m_g'(\theta_g - \tau_g\psi_g) + s'(R)}{\mathcal{D}_g}. \tag{23}$$

Differencing a religious group r and a doctrinally neutral group n leaves $s'(1/\mathcal{D}_r - 1/\mathcal{D}_n)$, so common primitives do not imply equal reduced-form effects. Under comparable curvature, $\mathcal{D}_r = \mathcal{D}_n = \mathcal{D}$, the shared term cancels and

$$\frac{\mathrm{d}a_r}{\mathrm{d}R} - \frac{\mathrm{d}a_n}{\mathrm{d}R} = \frac{m_r'(\theta_r - \tau_r\psi_r) + m_n'\tau_n\psi_n}{\mathcal{D}},$$

using $\theta_n = 0$. The contrast combines the religious group's net religious incentive with the neutral group's displacement response. To see what a neutral null restricts, write $x_n = m'_n \tau_n \psi_n \geq 0$. Its population response is $(s' - x_n)/\mathcal{D}_n$. A zero response implies $s' = x_n$, allowing both terms to be large; it does not bound either term separately. Under the additional restriction $s' = 0$, a neutral null gives $x_n = 0$, and a positive religious response implies $\theta_r > \tau_r \psi_r$ when $m'_r > 0$. A near-zero estimate restricts the scaled net response within its sampling uncertainty. Inferring a small time burden also requires restrictions on m'_n, ψ_n, and curvature. These distinctions motivate estimating the neutral response and the group contrast jointly.

Under $s' = 0$, a neutral-credit organization contracts when operations displace valued discretionary time and is unchanged in the fixed-duty benchmark. Institutional evidence is needed to select between these predictions. Allowing an unrestricted s' would accommodate either sign and would remove this sign restriction. Equal curvature cancels a shared shock in the contrast but does not address group-specific publicity, financing, or strategic responses.

Remark 3 (Coordination and backlash)**.** *A coordination benefit $\gamma_g(R)a$ and a reputational cost $\rho_g(R)a$ enter the baseline response as $\gamma'_g(R) - \rho'_g(R)$ (setting $s' = 0$ here):*

$$\frac{\mathrm{d}a_g}{\mathrm{d}R} = \frac{m'_g(\theta_g - \tau_g \psi_g) + \gamma'_g - \rho'_g}{\mathcal{D}_g}.$$

For a doctrinally neutral voluntary action, the predicted decline survives when the net coordination benefit is smaller than the increase in the time opportunity cost, $\gamma' - \rho' < m' \tau \psi$ (Chaney, 2013; Kikuta, 2024). Existing organization does not imply zero coordination benefits for armed groups. A nonnegative marginal backlash effect lowers the response in the single-action problem, while its size and target incidence remain empirical questions (Hassner, 2017; Reese et al., 2017). Where mosque networks organize protest, γ' can be large enough to reverse the civilian prediction. Applying the displacement benchmark therefore requires evidence on the demonstrations' organization and doctrinal status.

A.8 Curvature in the value of religious output

The linear religious-payoff term in (2) is a local assumption about the perceived value of additional merit, and this subsection states what it does. Replace it with $V(mQ)$, where $Q = f(r) + \theta a > 0$, $V' > 0$, and $V'' \leq 0$. The first-order conditions then depend on an effective religious return $t = mV'(mQ)$. Let $Q(t)$ denote the religious output chosen in the corresponding linear-return problem, so concavity gives $Q_t \geq 0$. With $z = mQ$,

$$\frac{\mathrm{d}t}{\mathrm{d}m} = \frac{V'(z) + zV''(z)}{1 - m^2 V''(z) Q_t},$$

where the denominator is positive.

The threshold in Proposition 1 therefore retains its direction whenever relative curvature satisfies $-zV''(z)/V'(z) < 1$, with the shadow-price terms evaluated at t. This is the range in which a proportional rise in the value of merit raises the effective religious return at the margin, and it contains the linear case as well as any V with relative curvature below unity. At the boundary case $V(z) = \log z$, multiplying all religious output by m adds $\log m$ to utility and has no allocation effect, which marks the boundary of this region. Relative curvature below one is a maintained restriction for the mechanism, rather than an estimate from the present data. The comparative statics in the main text should be read as holding on the interior of that region, which is where the linear specification sits.

Appendix B

B1. Aggression Attribution

This appendix documents the construction of the aggressor-attribution variable used in the results section: the UCDP GED sample and the underlying source text, the GABRIEL-based classification of each event's responsible party, the exact label definitions and instructions submitted to the model, the validation exercises, and the descriptive composition of the classified sample. The complete classification pipeline is in the replication package.

Source and sample

The conflict data are the event-level records of the UCDP Georeferenced Event Dataset (GED) [version/citation], restricted to the estimation sample described in the data section. Each event names the two parties to the dyad in the `side_a` and `side_b` fields and summarizes the incident in a free-text `source_article` field. In the UCDP convention Side A is the government or state actor and Side B is the non-state armed group; I preserve this assignment throughout, so that a Side A label denotes a government-initiated event and a Side B label a rebel-initiated one. The sample contains 156,685 events. An event is sent to the classifier only if its `source_article` field is non-empty and not equal to the literal aggregation placeholder (`"AGGREGATION"`); the roughly 1,390 events that fail this test carry no usable narrative and are assigned `Neutral` automatically without reaching the language model. Each remaining event is classified individually.

The classification task

I assign a responsible party to each event with GABRIEL (Asirvatham et al., 2026), the same large-language-model framework used for the devotional-emphasis measure in Appendix B2, here invoking its classification routine (`classify`) rather than its rating routine. For each event the model receives a single text block that combines the two party names with the incident description and returns exactly one of three labels, `Side A`, `Side B`, or `Neutral`, together with a probability for each ([`gpt-5-mini`], a single run; see the replication package for the exact model string and decoding settings). The assigned label is the highest-probability label, and I retain the maximum label probability as a classification-confidence score. Following Asirvatham et al. (2026), the routine reads the model's probabilities over the label tokens rather than a free-form generation, so the assignment is deterministic and reproducible. I map the labels to numeric codes (Side A $= 1$, Side B $= 2$, Neutral $= 0$) and merge them back to the full event panel, preserving the auto-assigned `Neutral` for the events withheld above.

The instructions make `Neutral` a deliberately conservative default. The model is told to label an event `Neutral` whenever no clear aggressor can be determined or both sides engaged in mutual combat, and to default to `Neutral` when in doubt. The label definitions identify the party that initiated or was responsible for the violence rather than strictly the party that struck first, and they instruct the model, where a report records casualties on one side without naming an attacker, to infer the opposing side as the responsible party. After classification, `Neutral` events constitute 57% of the sample and are dropped from the aggressor regressions; the remaining 43% divide into 26% government-initiated (Side A) and 17% rebel-initiated (Side B).

Prompt and label definitions

The label definitions and additional instructions submitted to GABRIEL's `classify` routine are reproduced verbatim:

```
labels:
"Side A":  "Side A initiated or is responsible for the violence described in
the event"
"Side B":  "Side B initiated or is responsible for the violence described in
the event"
"Neutral":  "No clear aggressor can be determined, or both sides engaged in
mutual combat"

additional instructions:
Determine which side initiated the violence based on the event description.
If the description explicitly states or strongly implies one side attacked,
classify that side as aggressor.  If the description lists casualties of one
side without attribution, infer the opposite side as the aggressor.  If both
sides engaged in mutual combat (e.g., 'clashes between'), classify as Neutral.
When in doubt, classify as Neutral.
```

Each event is presented to the model as a single text block combining the two party names with the incident description:

```
Side A: {side_a}
Side B: {side_b}
Event:  {source_article}
```

GABRIEL embeds the label definitions, the additional instructions, and this text in its standard classification template, which instructs the model to assign the passage to exactly one label and to report a probability for each. The complete template, including the decoding settings, is documented in Asirvatham et al. (2026) and reproduced in the replication package. Table A.5 reproduces representative event descriptions alongside their assigned labels for each of the three categories.

Validation

I benchmark the classifier against human coding. I hand-code a random sample of 1000 events for the responsible party, reading only the source description and blind to the model's label, and compare these codes against the GABRIEL assignments. Restricting to events a human coder can adjudicate, GABRIEL and the human agree on the aggressor in nearly 100% of cases, and the model's `Neutral` rate tracks the human "cannot determine" rate. The aggressor regressions are robust to restricting the sample to high-confidence classifications and to re-classifying the same descriptions with a standard zero-shot prompt through the OpenAI API ([`gpt-4o-mini`]) in place of GABRIEL's probability-based routine. I do not benchmark against the Global Terrorism Database cross-reference of **?**, which recovers only the subset of events meeting the GTD's narrower terrorism definition and miscodes conventional rebel-initiated engagements.

Limitations

Three caveats bound the interpretation. First, the source descriptions are drawn from media and agency reporting and inherit any reporting bias in who is named as the attacker; unlike the within-issue placebo in Appendix B2, no such design is available here, so I read the aggressor split as evidence on the locus of escalation rather than as a bias-free measure of initiation. Second, the classifier identifies the party responsible for the violence rather than strictly the party that moved first, and where a report gives casualties without naming an attacker it infers the opposing side as responsible; this inference is reasonable for most incident reports but can misattribute events in which the initiating side itself sustained the reported casualties, and such cases bias against detecting rebel-initiated escalation, so I read the headline split as conservative. Third, the language-model label is measured with error; I mitigate this with the

conservative `Neutral` default and the human validation above, and I confirm that the headline Ramadan estimates are unchanged under the alternative classifier and the high-confidence subsample.

B.2 The *al-Naba* Corpus

This appendix documents the construction of the *al-Naba* corpus, the segmentation of each issue into sections, the GABRIEL-based devotional-emphasis measure, the estimating equation, the validation exercises, and the descriptive statistics. The complete extraction and scoring pipeline is in the replication package.

Source and corpus

al-Naba ("the News") is the Islamic State's flagship Arabic-language weekly, issued by its central media office since late 2015. I obtain every issue from the primary-source archive maintained by Zelin (2026), yielding 533 issues (numbers 10 through 543) released between December 2015 and April 2026, of which 46 fall during Ramadan. Issues are distributed as Arabic-language PDFs of roughly eight to twelve pages. I extract the text layer of each PDF with the `PyMuPDF` library and apply standard Arabic orthographic normalization, unifying alif and ya variants and stripping elongation and diacritics for matching while retaining the raw text for scoring. Each issue's masthead carries its Hijri release date, which I convert to and from the Gregorian calendar with the `hijridate` package and cross-check against an independent converter; the Ramadan indicator equals one for issues dated to the ninth Hijri month. One issue (no. 346) is distributed as a scanned image with no text layer and is dropped; throughout I retain only sections with at least 500 extracted characters, which removes this issue and a handful of near-empty fragments. To confirm the text layer is faithful, I compare the extracted characters against a raw re-extraction for every page: the median capture rate is 1.00 and never falls below 0.975, the only systematic gap being text rendered inside raster infographics, which the section analysis does not use.

Issue structure and section segmentation

A typical issue follows a stable template: a front-page lead, a signed editorial, one or more feature essays (*maqalat*) on doctrine, history, or governance, several pages of operational news reporting the week's attacks by province, a weekly statistical infographic (*hasad*) tallying claimed operations, and a back-page devotional panel. I segment each issue into these components with a content-based classifier that partitions the extracted text into blocks and assigns each block to a section type using competing lexical signals: the editorial is identified by its masthead heading or, failing that, by position and register; the remaining prose is sorted into feature essays and operational news. The analysis uses the three running-prose sections, editorials, feature essays, and operational news, and discards the infographic and back panel, which are not continuous text. After the 500-character filter the corpus contains 1232 sections: 398 editorials, 302 feature essays, and 532 operational sections. Editorials are present in 74.5% of issues and feature essays in 56.6%; operational news appears in nearly all. The segmentation is heuristic. I verify it by manual inspection of a random sample of issues, and the identifying contrast below is within-issue, so any classifier error that is symmetric across the Hijri calendar cannot generate the Ramadan pattern.

The devotional-emphasis measure

I score section content with GABRIEL (Asirvatham et al., 2026), the same large-language-model rating framework used for aggressor attribution, here invoking its rating routine rather than its classification routine. For each section, GABRIEL submits the full text together with

the attribute definitions and a fixed rubric to the language model ([`gpt-5-mini`]; see the replication package for the exact model string and decoding settings), which returns an integer from 0 to 100 for each attribute. All attributes are scored in a single call per section, and I take one rating per section. The framework preserves each section's covariates, issue number, Hijri date, and section type, alongside the scores.

My primary construct is *devotional and ritual emphasis*, the extent to which a section foregrounds personal worship rather than combat or political argument. I define the construct in deliberately general terms and include no cue tied to the treatment: the definition names no month, no date, and no historical battle, so the rater cannot infer Ramadan from the prompt and inflate the score mechanically. I additionally score another construct that I report for transparency here but do not use as the headline measure, *sacralization of the present moment*; the full definitions appear below.

A note on the unit of analysis motivates the section-level design. Scoring whole issues fails by construction. *al-Naba* is uniformly militant prose, so an attribute such as religious justification of violence saturates near the ceiling and carries no usable variance, and the Ramadan offensive mechanically loads each issue with additional operational reporting, which dilutes any issue-level average of religious content and pushes it in the wrong direction. Measuring at the section level isolates the discursive content, where doctrine is articulated, from the operational reporting that serves as a within-issue placebo: a genuine doctrinal reorientation should surface in the feature essays, not in factual battle reports, whose register is fixed across the calendar. I report the whole-issue null and the ceiling-bound constructs in the descriptive table so that the section-level design is transparent rather than chosen on the result.

Prompt and attribute definitions

The three attribute definitions submitted to GABRIEL are reproduced verbatim:

```
"devotional and ritual emphasis": "How strongly the text foregrounds personal
acts of worship such as fasting, night prayer, Quran recitation, supplication,
or spiritual self-purification, as opposed to combat operations or political
argument."

"sacralization of the present moment": "How strongly the text presents the
current period as a specially blessed or rewarding time for striving, based
on the text's own argument rather than any date it reports."
```

GABRIEL embeds these definitions in its standard rating template, which instructs the model to read the full passage start to finish, rate each attribute independently on a 0–100 integer scale using the full range, draw no cross-attribute inference, and return the ratings as JSON. The complete template is reproduced in the replication package and documented in Asirvatham et al. (2026).

B.3 The State Sermon Corpus

This appendix documents the construction of the state Friday sermon corpus, the GABRIEL-based attribute measures, and the estimating equation. The complete scraping and scoring pipeline is in the replication package.

Institutional setting and corpus

The Friday congregational prayer (*jumu'ah*) is preceded by a sermon (*khutbah*) delivered from a raised pulpit, and attendance is religiously obligatory for adult men. The two states have built the administrative capacity to write that sermon centrally and distribute it nationwide. Turkey's Directorate of Religious Affairs publishes one weekly script and distributes it to approximately 90,000 mosques. Egypt's Ministry of Religious Endowments began issuing standardized weekly scripts in mid-2016. Because the script is written centrally, the text is a direct observation of the state's chosen message in a given week, without the aggregation problem that arises where sermon content varies across mosques.

I scrape Egyptian scripts from `doaah.com`, the publishing portal of the Awqāf, and Turkish scripts from the Diyanet's official portal. Egyptian scripts published before 2017 were distributed as image-based PDFs with corrupt font encoding and were re-extracted by OCR using Tesseract's Arabic engine; all later files are machine-readable. All texts remain in the original Turkish or Arabic, since the measurement model operates directly on source-language input. For the main estimates I restrict the Egyptian sample to Fridays from July 2016 onward, the point from which the Awqāf distributes a standardized weekly script. I drop duplicate dates, retaining the first record for each Friday, and exclude the Friday of Eid al-Fitr, which carries a fixed festival script in both systems. The final Friday of Ramadan (*jumu'at al-widā'*) is coded as treated, since it falls within the month. This leaves 610 Turkish and 460 Egyptian scripts, of which 50 and 35 fall in Ramadan. Gregorian dates are converted to the Hijri calendar with the `hijridate` package and cross-checked against an independent converter. The Ramadan indicator equals one for Fridays dated to the ninth Hijri month.

The measurement instrument

I score each script with GABRIEL (Asirvatham et al., 2026), the same large-language-model rating framework used for aggressor attribution and for the *al-Naba* sections, invoking its rating routine. For each sermon, GABRIEL submits the full text together with the attribute definitions and a fixed rubric to the language model, which returns an integer from 0 to 100 for each attribute. All attributes are scored in a single call per sermon at temperature zero with structured output, so the ratings are deterministic and reproducible.

Three constructs enter the analysis. *Devotional emphasis* reproduces the *al-Naba* definition verbatim, which is what allows the two corpora to be compared on one scale. *Political salience* captures regime-relevant political themes, and *strict religious preaching* the intensity of doctrinally authoritative instruction. The latter two are language-specific in their anchors, since the state-discourse vocabulary of Turkish and Egyptian official religious speech differs, but identical in structure.

The three attribute definitions submitted to GABRIEL are reproduced verbatim below; the complete system message and the language-specific anchor lists are in the replication package.

```
"devotional emphasis":  "How strongly the text foregrounds personal acts of
worship such as fasting, night prayer, Quran recitation, supplication, or spiritual
self-purification, as opposed to combat operations or political argument."

"political salience":  "The degree to which the sermon addresses regime-relevant
political themes such as national unity, state authority, military sacrifice,
loyalty to leadership, sovereignty, territorial integrity, stability, or security
threats, as opposed to religious or moral instruction that makes no reference
```

```
to the state, the nation, or public authority."

"strict religious preaching":  "The degree to which the sermon employs doctrinally
strict, orthodox Islamic instruction:  citing Quran and hadith with explicit
source attribution, issuing clear commands about what is permitted and forbidden,
emphasizing obligatory duties, warning of divine punishment for sin, and insisting
on correct ritual practice.  Do not consider which particular ritual is discussed."
```

One design point deserves emphasis, because it is the analogue of the no-treatment-cue discipline applied to *al-Naba* in Section B.2. The devotional definition names fasting among its examples of personal worship, which is innocuous in the insurgent corpus but is a treatment cue here, since the treatment is the fasting month. I therefore score a second variant of the construct with the enumeration of specific acts of worship removed altogether. The two variants give Ramadan coefficients within 0.8 points of one another in Turkey and 2.2 in Egypt, so the rating reflects devotional register rather than the appearance of a single word. No reference to fasting appears in the political attributes, and I instruct the model explicitly to disregard the date of the sermon and its position in the religious calendar; no date is passed to the model at any point.

Estimating equation and results

Let s_{it} denote the score on a given attribute for the sermon issued in country i in Gregorian week t. I estimate, separately by country,

$$s_{it} = \beta \, \text{Ramadan}_t + \theta_{h(t)} + \gamma_{g(t)} + \varepsilon_{it}, \tag{24}$$

where Ramadan_t indicates that the Friday falls within Ramadan, $\theta_{h(t)}$ are Hijri-year fixed effects, and $\gamma_{g(t)}$ are Gregorian month fixed effects. The structure mirrors the conflict panels of Section 5, with Hijri-year effects replacing country-by-Hijri-year effects because estimation is by country, and identification again comes from the rotation of Ramadan through the solar calendar. That rotation matters more here, because the corpus spans roughly twelve years rather than a full lunar cycle: over this window Ramadan moves from June and July at the start of the sample to March and April at the end, and in Egypt those later months coincide with the 2018 presidential election and the 2019 constitutional referendum. The Gregorian month fixed effects break that correlation, and they are the reason the conditional estimates differ from raw Ramadan minus non-Ramadan differences. The series is weekly and serially correlated, since both authorities run multi-week thematic blocks, so I report Newey-West standard errors with an eight-week bandwidth and confirm each estimate with a wild cluster bootstrap on Hijri year using Rademacher weights and 999 replications.

Limitations

Three caveats bound the interpretation. First, the corpus covers two countries, so the exercise speaks to the plausibility of the state-messaging mechanism rather than decomposing the protest estimate across the panel. Egypt is in the ACLED sample and Turkey is not. Second, the script bounds what the state says from the pulpit, not what it does elsewhere. A regime could pair devotional sermons with heavier policing of assembly, and the text is silent on that margin; what the evidence closes off is persuasion through the religious channel itself, the one alternative the survey footprint in Section 6.4 cannot distinguish from time reallocation. Third, the scripts record what imams are instructed to read, not what congregations hear or retain. Delivery fidelity is unobserved, though a centrally drafted script is a closer proxy for delivered content than a sample of locally written sermons would be.

Appendix C

This appendix lists the 139 organized rebel groups in the sample with their assigned ideological category and a brief justification for each, and documents the procedure used to assign them to one of two categories: religious ideological or secular-nationalist.

Decision criteria. The dividing line between the two categories is religious identity. *Religious identity* is present when the founding charter, recruitment base, or organizational rhetoric explicitly treats Islamic identity as constitutive of group membership, beyond the incidental fact that the population from which fighters are drawn is Muslim. Groups satisfying this criterion are coded as religious; groups failing it, whose objectives are ethno-nationalist, separatist, Marxist-Leninist, or factional and for which religion plays no constitutive role, are coded as secular-nationalist. The classification rests on stated objectives and foundational documents rather than tactical behavior alone, since organizationally similar operational profiles can arise from very different ideological bases.

The religious category is internally heterogeneous, and I record a second dimension to characterize that variation. *Doctrinal mandate for offensive warfare* is present when a group's principal stated objective is the establishment of religious governance, whether sharia rule, an Islamic state, or a caliphate, and the group frames armed struggle itself as religiously obligatory rather than as an instrumental means to a political end. Groups with a strong religious identity but more territorially bounded aims, for which religion supplies cohesion, recruitment, and external solidarity rather than the doctrinal justification for combat, satisfy religious identity without this mandate. The relevant theoretical condition for the Ramadan mechanism is access to the doctrinal repertoire that recasts combat as worship, which the religious category as a whole possesses; the doctrinal-mandate dimension orders groups within that category from its most doctrinally committed members to its more territorial ones, and motivates the strict-sample robustness check reported in the main text.

The boundary with secular-nationalist groups is rarely ambiguous. The internal variation within the religious category is, by contrast, occasionally contested for individual groups. Where a group's placement along the doctrinal-mandate dimension is genuinely unclear, two pieces of behavioral evidence are informative: known operational alliances with secular factions, and formal participation in established state governance structures. Both sit uneasily with a strict doctrinal mandate for offensive religious warfare, since either would require subordinating religious-governance objectives to political pragmatism, and both therefore place a group toward the more territorial end of the religious category. Neither, however, removes a group from the religious category, since religious identity can remain constitutive even where the doctrinal mandate is weak or contested.

Sources. Classification draws on three layers of evidence. The UCDP Actor Dataset (Brosché and Sundberg, 2024) provides the canonical organizational identifier used throughout the conflict data. The Mapping Militants Project at Stanford's Center for International Security and Cooperation (Crenshaw and Robinson, 2025) provides standardized profiles of major armed organizations, including founding ideology, organizational lineage, and primary recruitment networks. Where these sources are silent, ambiguous, or contested, I consult the group's own founding charter or manifesto and contemporaneous reporting.

Coding procedure. Each group was evaluated against the religious-identity criterion: a group failing it is classified as secular-nationalist, and a group satisfying it is classified as religious. For religious groups, I additionally record whether the doctrinal mandate for offensive warfare is present, a designation that does not affect the two-category assignment used in the main analysis but identifies the doctrinally committed subset used in the strict-sample robustness check.

Religious-Ideological

1. **Afghan Taliban (Taleban)** — Deobandi Islamist movement founded in 1994 in Kandahar; established the Islamic Emirate of Afghanistan in 1996 and again in 2021. Primary stated objective is establishing governance based on sharia under the authority of an Amir al-Mu'minin (Commander of the Faithful). The group's founding charter declares that sovereignty belongs only to God and designates sharia as the sole source of legislation. Ideology rooted in Hanafi jurisprudence and the Deobandi tradition; armed jihad is invoked as an obligatory religious duty. The ideology of the Taliban is considered a shift from traditional Islamist views held by anti-Soviet mujahedeen fighters in the 1980s and early 1990s. It can be characterized as a combination of strict anti-modern Pashtun tribal ideology mixed with certain radicalized Deobandi interpretations of Islam. The UN Security Council grouped the Taliban with Al-Qaeda.

2. **Tehrik-i-Taliban Pakistan (TTP)** — Deobandi jihadist umbrella organization formed in December 2007 from approximately 40 militant factions in Pakistan's Federally Administered Tribal Areas (FATA). Stated objectives include enforcing a strict Deobandi interpretation of sharia throughout Pakistan and ultimately establishing an Islamic caliphate. The group explicitly frames its armed struggle as defensive jihad and seeks to replace Pakistan's secular constitution with religious law. The TTP claims that its jihad against the Pakistani government is an act of self-defense against an apostate and puppet regime of the United States.

3. **TTP – Tariq Afridi faction (TTP - TA)** — Splinter faction of the TTP operating in Pakistan's tribal regions, retaining the parent organization's ideological orientation. Same doctrinal basis as the TTP: enforcement of sharia and armed jihad against the Pakistani state on explicitly religious grounds.

4. **Islamic Movement of Uzbekistan (IMU)** — Islamist militant organization co-founded in 1998 by Tahir Yuldashev and Juma Namangani in Afghanistan. Declared jihad to overthrow the secular government of Uzbekistan and establish an Islamic state governed by sharia. Later expanded its goal to the creation of a caliphate across all of "Turkestan" (Central Asia from the Caspian Sea to western China). Closely affiliated with al-Qaeda; received substantial financial backing from Osama bin Laden and fought alongside the Taliban.

5. **Jamaat-ul-Ahrar (JuA)** — Splinter faction that broke from the TTP in August 2014, merging elements of TTP's Mohmand Agency branch with Ahrar-ul-Hind. Retained the TTP's core objective of imposing sharia in Pakistan. Its founding emir, Omar Khalid Khorasani, explicitly called for overthrowing the Pakistani government, seizing its nuclear weapons, and waging jihad until a global caliphate is established. The group pledged to continue attacks until sharia law was implemented nationwide.

6. **Lashkar-e-Islam (LeI)** — Deobandi militant group founded in 2004 in Pakistan's Khyber Agency (now Khyber District) by Mufti Munir Shakir. Established with the explicit purpose of enforcing strict Deobandi sharia in Khyber Agency. Under the subsequent leadership of Mangal Bagh Afridi, the group set up parallel Islamic courts, mandated mosque attendance, imposed Islamic dress codes, and banned activities it deemed un-Islamic. LeI only accepted sharia as legitimate law and refused to recognize elected Pakistani officials. Before 2008, LeI was primarily a religious reformist organization. After Mangal Bagh assumed leadership, LeI sought to gain extensive control over the Khyber District and the Khyber Pass. During this period, LeI began conducting more suicide and violent attacks against members of the Pakistani state. Since 2014, this goal has largely been com-

promised because of LeI's relocation to Afghanistan. However, LeI still launches attacks and raises funds from Afghanistan with the hope of returning to the Khyber District.

7. **Ansar al-Islam (Jama'at Ansar al-Islam)** — Kurdish Salafist-jihadist group formed in December 2001 in Iraqi Kurdistan from the merger of Jund al-Islam and a splinter faction of the Islamic Movement of Kurdistan. Founded with the stated objective of expelling non-Muslim influences from Kurdistan and governing all territory under strict sharia. Established an "Islamic Emirate of Kurdistan" where it implemented Taliban-style governance: barring women from education and employment, banning music and television, and threatening amputation, flogging, and stoning for religious offenses. Funded by al-Qaeda and closely affiliated with its network.

8. **Ittihad-i Islami (High Council of Afghanistan Islamic Emirate)** — Islamic Union for the Liberation of Afghanistan, a Wahhabist mujahideen faction founded in 1981 by Abdul Rasul Sayyaf. The only major Afghan mujahideen party to embrace Wahhabi theology explicitly and in its entirety. Sayyaf, a Saudi-trained theologian with an MA from Al-Azhar University, maintained close ties to the Saudi religious establishment and to Osama bin Laden. The organization was strongly committed to sharia governance and, in the post-Taliban era, secured control over parts of Afghanistan's judiciary to advance that agenda.

9. **Forces of Mullo Abdullo** — Armed faction led by Abdullo Rahimov (known as Mullo Abdullo), a veteran Islamist field commander of the UTO during Tajikistan's 1992–1997 civil war. Mullo Abdullo rejected the 1997 peace agreement and fled to Afghanistan, where he sheltered with the Taliban and maintained links to the Islamic Movement of Uzbekistan. He returned to Tajikistan in 2009 with a multinational band of insurgents and proclaimed jihad against the Rahmon government. His forces carried out a major ambush in the Rasht Valley in September 2010 that killed at least 28 Tajik military personnel. Tajik authorities identified him as an al-Qaeda-linked commander.

10. **East Turkestan Islamic Movement (ETIM)** — Uyghur jihadist organization founded in 1997 by Hasan Mahsum in Xinjiang, China; also known as the Turkistan Islamic Party (TIP). The group's stated objective is the creation of an independent Islamic state of "East Turkestan" governed by sharia. Since its establishment, ETIM has maintained close organizational ties with al-Qaeda and the Taliban, with its leader Abdul Haq reportedly serving on al-Qaeda's shura council from 2005. The group relocated its headquarters to Taliban-controlled Kabul in 1998, trained fighters in al-Qaeda camps in Afghanistan, and later deployed several thousand fighters to Syria's civil war under the Turkistan Islamic Party banner. Its propaganda calls explicitly for armed jihad against the Chinese state, framing the struggle in religious terms against "atheist" occupation.

11. **Forces of the Caucasus Emirate (Imirat Kavkaz)** — North Caucasus jihadist umbrella organization founded in October 2007 by Chechen militant leader Doku Umarov. Umarov's declaration of the Caucasus Emirate marked a decisive shift from Chechen separatist nationalism toward a transnational jihadist agenda. The group's stated objective was the establishment of an Islamic state governed by sharia across the entire North Caucasus, encompassing Chechnya, Dagestan, Ingushetia, Kabardino-Balkaria, and other republics. Umarov explicitly rejected the secular nationalist goals of the earlier Chechen independence movement and declared armed jihad against Russia as a religious obligation. The organization conducted suicide bombings against civilian targets in Moscow and elsewhere, and was designated a terrorist organization by the United States, the United Nations, and Russia.

12. **Islamic State (IS)** — Salafi-jihadist organization that emerged from al-Qaeda in Iraq (founded 2004 by Abu Musab al-Zarqawi) and declared a global caliphate in June 2014 under Abu Bakr al-Baghdadi. The group's central objective is the establishment and expansion of a caliphate governed by a literalist interpretation of sharia. The declaration of the caliphate explicitly invoked religious authority, claiming sovereignty over all Muslims worldwide and demanding their allegiance. IS implemented comprehensive sharia governance in territories it controlled across Iraq and Syria, including religious courts, enforcement of hudud punishments, and systematic persecution of religious minorities. The group frames its armed struggle as an obligatory religious duty incumbent upon all Muslims.

13. **Al-Qaeda** — Transnational Salafi-jihadist organization founded in 1988 by Osama bin Laden and Abdullah Azzam in Peshawar, Pakistan. Al-Qaeda's stated objectives center on a global armed jihad to expel Western influence from Muslim lands and ultimately establish Islamic governance under sharia across the Muslim world. The group's founding ideology frames armed struggle as an individual religious obligation (fard ayn) for all Muslims. Al-Qaeda's 1998 fatwa explicitly invoked Quranic authority to call for attacks on Americans and their allies worldwide. The organization has served as the ideological and operational nucleus for a network of regional affiliates (AQAP, AQIM, al-Shabaab, JNIM) that share its religious-ideological framework.

14. **Al-Qaeda in the Arabian Peninsula (AQAP)** — Regional al-Qaeda affiliate formed in January 2009 from the merger of the Saudi and Yemeni branches of al-Qaeda, based in Yemen. AQAP's stated objective is the establishment of Islamic governance under sharia across the Arabian Peninsula and the expulsion of Western influence from Muslim lands. The group operates within al-Qaeda's global jihadist framework, framing armed struggle as a religious obligation. AQAP has pursued both local insurgency against the Yemeni government and transnational attacks, including the 2009 attempted bombing of a U.S. airliner and the 2015 attack on the offices of Charlie Hebdo in Paris.

15. **Al-Qaeda in the Islamic Maghreb (AQIM)** — Regional al-Qaeda affiliate that formally adopted its current name in 2007, evolving from Algeria's Salafist Group for Preaching and Combat (GSPC, founded 1998). AQIM's objectives include the establishment of an Islamic caliphate across North and West Africa and the expulsion of Western presence from Muslim lands. The group pledged allegiance to al-Qaeda's central leadership and adopted its transnational jihadist agenda. AQIM has conducted kidnappings, bombings, and armed attacks across Algeria, Mali, Mauritania, Niger, and Burkina Faso, and its Saharan branch later merged into JNIM.

16. **Armed Islamic Group of Algeria (GIA)** — Salafi-jihadist organization founded in 1992 during the Algerian Civil War following the military's cancellation of the 1991 elections that the Islamic Salvation Front (FIS) was poised to win. The GIA's stated objective was the violent overthrow of the Algerian secular state and the establishment of an Islamic state governed by sharia. The group rejected any political compromise and pursued an uncompromising takfiri ideology, declaring all who did not support its armed struggle to be apostates. The GIA conducted mass civilian massacres, assassinations, and bombings throughout the 1990s.

17. **Takfir wa'l Hijra** — Egyptian jihadist group (formally Jama'at al-Muslimin) founded in the 1960s–1970s by Shukri Mustafa, a former Muslim Brotherhood member radicalized during imprisonment. The group's ideology centers on the concept of takfir (excommunication of other Muslims deemed insufficiently pious) and hijra (withdrawal from corrupt society). It declared Egyptian society to be in a state of pre-Islamic ignorance

(jahiliyya) and sought to establish a purified Islamic community that would eventually overthrow the secular state. The group carried out the 1977 kidnapping and murder of a former Egyptian minister of religious endowments.

18. **Al-Gama'a al-Islamiyya (Egyptian Islamic Group)** — Egyptian Islamist militant organization active primarily from the 1970s through the late 1990s. The group's stated objective was the overthrow of the Egyptian secular state and the establishment of an Islamic state governed by sharia, drawing ideological inspiration from Sayyid Qutb's writings. Al-Gama'a al-Islamiyya carried out attacks on government officials, Coptic Christians, secular intellectuals, and foreign tourists, most notably the 1997 Luxor massacre that killed 62 people. Its spiritual leader, Omar Abdel-Rahman, was convicted for his role in the 1993 World Trade Center bombing conspiracy.

19. **Ansar Bayt al-Maqdis** — Sinai-based jihadist group that emerged around 2011 in the security vacuum following the Egyptian revolution. Initially focused on attacks against the Israel-Egypt gas pipeline and Israeli targets, the group expanded to attack Egyptian military and police forces in the Sinai Peninsula. In November 2014, Ansar Bayt al-Maqdis pledged allegiance to the Islamic State and rebranded as "Wilayat Sinai" (Sinai Province). The group seeks to establish Islamic governance in the Sinai through armed jihad against the Egyptian state, which it considers apostate.

20. **Signed-in-Blood Battalion (al-Muwaqi'un bi-l-Dima)** — AQIM splinter group formed in 2012 by veteran jihadist Mokhtar Belmokhtar after his expulsion from AQIM's leadership. The group operated within a Salafi-jihadist framework and is most known for the January 2013 attack on the In Amenas gas facility in Algeria, which killed 39 foreign hostages. Belmokhtar framed the attack as retaliation for France's military intervention in Mali. The group subsequently merged with MUJAO to form al-Murabitun in August 2013.

21. **Movement for Unity and Jihad in West Africa (MUJAO)** — West African jihadist organization that split from AQIM in 2011. MUJAO's stated objective was to spread jihad across West Africa, particularly targeting what it considered apostate governments and Western interests. The group occupied the city of Gao in northern Mali in 2012, where it imposed strict sharia governance including amputations and public floggings. MUJAO merged with Belmokhtar's Signed-in-Blood Battalion to form al-Murabitun in August 2013.

22. **Al-Murabitun** — Jihadist organization formed in August 2013 through the merger of MUJAO and the Signed-in-Blood Battalion, led by Mokhtar Belmokhtar. Al-Murabitun's stated goal was to spread jihad across North and West Africa. The group reaffirmed its allegiance to al-Qaeda in 2015 and conducted joint operations with AQIM in Mali, Burkina Faso, and Ivory Coast, including the November 2015 Radisson Blu hotel attack in Bamako. In March 2017, al-Murabitun merged with AQIM's Saharan branch and other groups to form Jama'at Nusrat al-Islam wal-Muslimin (JNIM).

23. **Al-Shabaab** — Somali Salafi-jihadist organization that emerged from the militant wing of the Islamic Courts Union following the 2006 Ethiopian intervention, formally pledging allegiance to al-Qaeda in 2012. Al-Shabaab's stated objective is the establishment of an Islamic state governed by sharia across Somalia and the broader Horn of Africa. The group has implemented strict sharia governance in territories under its control, including religious courts, hudud punishments, and bans on music, cinema, and secular education. Al-Shabaab frames its armed struggle as jihad against the "apostate" Somali government and its foreign backers, particularly the African Union Mission in Somalia (AMISOM).

24. **Hizbul Islam** — Somali Islamist alliance formed in 2009 from the merger of several factions, led by Sheikh Hassan Dahir Aweys. Hizbul Islam sought to establish Islamic governance in Somalia and opposed the Transitional Federal Government and the Ethiopian military presence. The group imposed sharia in areas under its control and rejected any political accommodation with the Western-backed government. In December 2010, Hizbul Islam formally merged with al-Shabaab after losing most of its territory to the larger organization.

25. **Ansar Dine** — Malian Salafist Islamist group founded in late 2011 by Iyad ag Ghaly, a veteran Tuareg political and military leader. Ansar Dine's stated objective is the implementation of sharia law across all of Mali, distinguishing it from the secular Tuareg separatist movement (MNLA) with which it briefly cooperated. During the 2012 occupation of northern Mali, the group imposed strict sharia governance in Timbuktu and other cities, destroying Sufi shrines designated as UNESCO World Heritage Sites. In March 2017, Ansar Dine merged into JNIM under ag Ghaly's overall leadership.

26. **Jama'at Nusrat al-Islam wal-Muslimin (JNIM)** — Al-Qaeda-affiliated jihadist umbrella organization formed in March 2017 from the merger of Ansar Dine, AQIM's Saharan branch, al-Murabitun, and the Macina Liberation Front, led by Iyad ag Ghaly. JNIM's stated purpose is to unite jihadist efforts across the Sahel under al-Qaeda's global framework. The group pursues the establishment of sharia governance, the expulsion of French and Western military forces, and the overthrow of what it considers apostate governments in Mali, Burkina Faso, and Niger. JNIM has become the dominant jihadist force in the Sahel, conducting attacks across a vast operational area.

27. **Macina Liberation Front (FLM / Katiba Macina)** — Fulani-dominated jihadist organization founded in 2015 by the radical preacher Amadou Koufa, operating primarily in central Mali. The FLM seeks to impose sharia governance in the Macina region (the historic Fulani heartland of the inner Niger Delta) and draws on the legacy of the 19th-century Macina Empire, a Fulani theocratic state. While the group recruits heavily along Fulani ethnic lines and taps into longstanding grievances over land rights and intercommunal marginalization, it operates under JNIM's al-Qaeda-affiliated umbrella and explicitly frames its struggle in religious terms. Koufa's preaching calls for armed jihad and the rejection of secular state authority.

28. **Jama'atu Ahlis Sunna Lidda'awati wal-Jihad (Boko Haram)** — Nigerian Salafi-jihadist organization founded around 2002 by Mohammed Yusuf in Maiduguri, Borno State. The group's name translates roughly to "People Committed to the Propagation of the Prophet's Teachings and Jihad," and it rejects Western secular education and governance as incompatible with Islam. Following Yusuf's extrajudicial killing in 2009, the group launched a full-scale insurgency under Abubakar Shekau, seeking to establish an Islamic state governed by sharia in northeastern Nigeria and the broader Lake Chad Basin.

29. **Al-Itihaad al-Islamiya (AIAI)** — Somali Salafi-jihadist organization active primarily in the 1990s, seeking to establish an Islamic state in Somalia and unite ethnic Somali territories in the Horn of Africa. Founded in the early 1980s and influenced by Wahhabi ideology, AIAI drew support from Saudi and Gulf donors and operated training camps in Somalia. The group imposed sharia governance in areas under its control, notably in the Gedo and Luuq regions. AIAI was linked to al-Qaeda and was accused of involvement in the 1998 U.S. embassy bombings in East Africa. Following Ethiopian military operations and internal fragmentation, many former AIAI members later joined the Islamic Courts Union and subsequently al-Shabaab.

30. **Islamic Front for Armed Jihad (FIAA / FIDA)** — Islamist militant group that split from Algeria's GIA in 1993 during the Algerian Civil War. The group pursued the establishment of an Islamic state in Algeria through armed jihad, sharing the broader Islamist insurgency's goal of replacing the secular government with sharia-based governance. It operated during the most violent phase of Algeria's civil conflict. The Islamic Front for Armed Jihad (FIDA) was notorious during the Algerian Civil War for a very specific tactical choice: they primarily assassinated secular intellectuals, journalists, feminists, and secular politicians. Their violence was not aimed at capturing territory or redressing an ethnic grievance; it was explicitly designed to purge Algerian society of secular and Western ideological influence to pave the way for a strict Sharia state.

31. **Jama'at al-Islamiyya al-Gama'a (JIG)** — Algerian Islamist militant faction active during the Algerian Civil War. Operating as part of the broader constellation of armed jihadist groups that emerged in the 1990s, the group rejected secular state authority and sought to establish strict Islamic governance in Algeria through armed struggle, framing its violent campaign against the state as a religious mandate rather than a conventional political dispute.

32. **Palestinian Islamic Jihad (PIJ)** — Palestinian Islamist militant organization founded in 1981 as an explicit break from the secular nationalist factions of the PLO. PIJ was established precisely to place Islamic religious mandate at the center of the Palestinian armed struggle, and its founding ideology invokes jihad as a religious obligation rather than a strategic instrument. Unlike Hamas, PIJ maintains no formal political wing or social services apparatus, reflecting an organizational structure oriented almost entirely around religiously framed armed resistance. While its operational theater is territorially bounded to the Palestinian context, its primary justification for violence is doctrinal rather than nationalist, placing it in the religious-ideological category.

33. **Abu Sayyaf Group (ASL)** — A violent jihadist militant network founded in 1991 in the southern Philippines. While operating within the ethno-religious geography of Mindanao, the group split from the broader Moro separatist movement specifically to pursue a hardline transnational jihadist agenda. The group rejected political autonomy in favor of establishing an independent Islamic state governed by strict sharia, maintained deep historical ties to al-Qaeda, and later officially pledged allegiance to the Islamic State.

34. **Jama'atu Ansaril Muslimina Fi Biladis Sudan (Ansaru)** — A Salafi-jihadist militant organization based in northeastern Nigeria that splintered from Boko Haram in 2012. Translating to the "Vanguard for the Protection of Muslims in Black Africa," the group is formally aligned with Al-Qaeda in the Islamic Maghreb (AQIM). Its explicitly stated objectives are the restoration of the historical Islamic Sokoto Caliphate across West Africa and the violently enforced defense of Muslims against Western and secular influence.

35. **Alliance for the Re-liberation of Somalia / Union of Islamic Courts (ARS/UIC)** — Broad Somali Islamist coalition. The Islamic Courts Union rose to power in Mogadishu in 2006 by providing governance and security in the absence of a functioning state, before being ousted by the Ethiopian military intervention in December 2006. The ARS was subsequently formed in Asmara in 2007 as an opposition umbrella. The ICU imposed sharia in areas under its control, but its primary appeal was the restoration of order and Somali self-governance against foreign intervention. The coalition included moderates alongside hardliners, and the ARS later split, with the Djibouti faction under Sheikh Sharif Sheikh Ahmed joining the transitional government and eventually assuming the presidency. Islam provided the organizational framework and legitimacy, but the movement's objectives centered on Somali sovereignty and state-building rather than transnational jihad.

36. **Islamic Salvation Army (AIS)** — Armed wing of Algeria's Islamic Salvation Front (FIS), active during the Algerian Civil War (1992–2002). The AIS was formed after the military annulled the 1991 elections that the FIS was poised to win. Its armed struggle sought to restore the political rights of Algeria's Islamist constituency rather than to impose religious governance through jihad. The AIS targeted primarily military and state installations, maintained a conventional insurgent posture, declared a unilateral ceasefire in 1997, and formally disbanded in 2000 under a government amnesty. Islam defined the movement's identity and constituency, but the armed campaign was driven by political exclusion.

37. **Harakat Sawa'id Misr (HASM)** — Militant group formed in Egypt in 2015 following the military ouster of President Mohamed Morsi. HASM has been linked by Egyptian authorities to the Muslim Brotherhood, though the Brotherhood denies the connection. The group targets Egyptian security officials and government installations, framing its struggle as nationalist resistance against an illegitimate regime. HASM's slogans and communiqués employ nationalist rather than jihadist language, and the group has not articulated goals of establishing sharia governance or a caliphate. Islam serves as an organizational identity rooted in the Brotherhood tradition, but the armed campaign is oriented toward political regime change.

38. **Jondullah (Jundallah)** — Sunni Baloch militant organization founded in 2003 by Abdolmalek Rigi in Iran's Sistan-Baluchestan province. The group's stated objectives center on securing recognition of Baloch cultural, economic, and political rights within Iran. Rigi framed the struggle as a fight for equal citizenship for Sunni Baloch, explicitly denying separatist or radical sectarian aims. However, the group recruited through Sunni religious seminaries and increasingly adopted sectarian rhetoric. Sunni Islam serves as a key component of group identity and a basis for external solidarity, but the underlying grievances are ethnic marginalization and economic underdevelopment of the Baloch population. Not to be confused with the Pakistani organization of the same name, a TTP-linked sectarian group that pledged allegiance to IS in 2014.

39. **Jaish al-Adl** — Successor organization to Jondullah, emerging in 2012 after the execution of Abdolmalek Rigi. Inherited Jondullah's organizational base and membership while adopting a more explicitly Salafi posture. Frames its insurgency as defending Sunni Baloch from Shia state oppression, blending Baloch nationalist objectives with Salafi ideology. Primarily targets Iranian Revolutionary Guards and security personnel through ambushes, assassinations, and kidnappings.

40. **Rohingya Solidarity Organisation (RSO)** — Rohingya insurgent group founded in 1982 following the Tatmadaw's mass displacement operations against the Rohingya in Arakan (Rakhine) State, Myanmar. The RSO's stated objectives center on Rohingya self-determination, restoration of political rights, and return to ancestral lands in Arakan. The group split from the more moderate Rohingya Patriotic Front as its religiously oriented faction and cultivated ties with Islamist networks in Pakistan and Bangladesh, recruiting fighters through religious channels. However, its manifesto frames the armed struggle around ethnic persecution and racial discrimination rather than religious governance, with no stated goal of establishing sharia or an Islamic state. Islam serves as a component of group identity and a basis for external solidarity rather than the primary ideological framework for armed struggle.

41. **United Tajik Opposition (UTO)** — Coalition of Islamist, democratic, nationalist, and regionalist factions formed in 1993 during the Tajikistani Civil War, led by Sayid Abdulloh Nuri. The dominant component was the Islamic Renaissance Party of Tajikistan (IRPT),

which gave the coalition its Islamic identity and facilitated support from Afghan mujahideen. However, the UTO's objectives centered on political power-sharing, regional representation for marginalized Gharmi and Pamiri populations, and opposition to the post-Soviet Kulobi-Khujandi ruling elite. The coalition included secular democrats and Pamiri autonomists alongside Islamists. The IRPT never demanded an Islamic state and transitioned to legal party politics after the 1997 peace agreement, underscoring that religion served primarily as a component of group cohesion rather than the central justification for armed struggle.

42. **Hamas** — Sunni Islamist militant organization and Palestinian nationalist movement founded in 1987. While its founding charter is steeped in Islamic rhetoric and frames the liberation of Palestine as a religious duty, its primary operational objective is territorial and nationalist: the establishment of an independent Palestinian state. Religion serves as a powerful mobilizing identity and ideological framework, but the conflict is fundamentally rooted in territorial disputes rather than a transnational jihadist mandate.

43. **Hezbollah** — Lebanese Shia militant organization and political party founded in 1982 with Iranian Revolutionary Guard support in response to the Israeli invasion of Lebanon. Hezbollah's founding 1985 Open Letter declared allegiance to Ayatollah Khomeini's wilayat al-faqih and called for an Islamic republic in Lebanon, placing its origins squarely in religious-ideological territory. However, the organization underwent substantial ideological evolution, and its 2009 manifesto dropped the Islamic state objective entirely in favor of Lebanese sovereignty, resistance to Israeli occupation, and confessional power-sharing within the existing political system. Hezbollah operates an extensive social services network, holds parliamentary seats, and functions as the primary political representative of Lebanon's Shia community. Its armed struggle is framed around national resistance and defense of the Shia community rather than the imposition of religious governance, and its operational alliances with secular actors such as the Syrian Social Nationalist Party and the Free Patriotic Movement reflect a pragmatic political orientation. Shia religious identity remains central to recruitment, internal cohesion, and external patronage networks, but it operates as a marker of communal solidarity rather than the doctrinal basis for violence.

44. **Hizb-i Islami-yi Afghanistan (Hekmatyar faction)** — Afghan Islamist party founded by Gulbuddin Hekmatyar in 1977 as an offshoot of the Muslim Youth movement at Kabul University. Hizb-i Islami's founding charter explicitly calls for an Islamic state in Afghanistan, and the party received substantial Pakistani ISI and Saudi funding during the anti-Soviet jihad on the basis of its Islamist credentials. However, Hekmatyar's operational behavior consistently prioritized Pashtun factional interests over ideological objectives. During the 1992–1996 civil war, Hizb-i Islami fought other Islamist mujahideen factions with equal or greater intensity than it had fought the Soviets, allied with the secularist Dostum militia against the Islamist Rabbani government, and shelled Kabul in campaigns that followed ethnic-factional fault lines rather than ideological ones. Hekmatyar's eventual reconciliation with the Afghan government in 2016 and entry into electoral politics further suggest that Islamic governance was instrumentally deployed rather than constitutively central. The party's recruitment base drew overwhelmingly from eastern Pashtun (particularly Ghilzai) networks, and its conflicts tracked tribal and regional rivalries as much as ideological commitments.

45. **Hizb-i Islami-yi Afghanistan (Khalis faction)** — Splinter faction established by Yunus Khalis in 1979 following a break with Hekmatyar over leadership and organizational style. Khalis, a religious scholar from Nangarhar, oriented his faction more explicitly around traditional religious authority than Hekmatyar's university-educated cadre, and

the faction maintained closer ties to local ulema networks in eastern Afghanistan. Nevertheless, its operational footprint was heavily concentrated among Pashtun communities in Nangarhar and Paktia provinces, and its armed struggle during the anti-Soviet jihad and subsequent civil war followed regional and tribal patterns consistent with localized power competition rather than a unified project of religious state-building. Several prominent Khalis commanders, including Jalaluddin Haqqani, later defected to the Taliban, suggesting that the faction functioned more as a vehicle for eastern Pashtun mobilization than as a cohesive ideological movement.

46. **Jam'iyyat-i Islami-yi Afghanistan** — Afghan Islamist party founded by Burhanuddin Rabbani in 1971, with Ahmad Shah Massoud as its most prominent military commander. Jam'iyyat's founding ideology drew on the Muslim Brotherhood tradition and called for Islamic governance, and Rabbani served as president of the Islamic State of Afghanistan from 1992 to 1996. However, the party's recruitment and operational base was overwhelmingly Tajik, centered in the Panjshir Valley and northeastern Afghanistan. During the civil war, Jam'iyyat's alliances and enmities tracked ethnic geography with striking consistency: cooperation with Hazara and Uzbek factions against Pashtun-dominated groups, regardless of the latter's Islamic credentials. Massoud's resistance to the Taliban was framed as much around Tajik survival and anti-Pashtun domination as around competing visions of Islamic governance. The party's ethnic character was sufficiently pronounced that its post-2001 political successor, the National Coalition of Afghanistan, operated as a de facto vehicle for Tajik political representation.

47. **Hizb-i Wahdat** — Afghan political-military organization founded in 1989 through the unification of several Hazara Shia factions, led by Abdul Ali Mazari. The party's name translates to "Party of Unity," reflecting its primary objective of consolidating Hazara political representation in a context of severe ethnic marginalization. Hazara communities had faced systematic discrimination under successive Pashtun-dominated governments, and Hizb-i Wahdat's armed struggle centered on securing Hazara political rights, territorial autonomy in the Hazarajat, and protection against ethnic violence. Iran provided patronage on the basis of Shia solidarity, and the party's religious identity was inseparable from its ethnic one, as the Hazara are Afghanistan's largest Shia community. However, Hizb-i Wahdat's conflicts were driven by ethnic survival and power-sharing demands rather than any project of imposing Shia governance, and the party fought Sunni Islamist groups (the Taliban, Hizb-i Islami) and allied with secular or Sunni actors (Dostum's Junbish-i Milli) on the basis of ethnic-strategic calculus.

48. **United Islamic Front for the Salvation of Afghanistan (UIFSA / Northern Alliance)** — Military-political coalition formed in 1996 to resist the Taliban, bringing together Jam'iyyat-i Islami, Hizb-i Wahdat, and Junbish-i Milli-yi Islami under the nominal leadership of Rabbani's government. The coalition's name invoked Islamic unity, and its component factions each carried Islamic credentials of varying degrees. However, the UIFSA's formation was driven entirely by the shared existential threat posed by the Taliban's military advance against non-Pashtun regions, and its internal cohesion derived from ethnic complementarity (Tajik, Hazara, Uzbek) rather than ideological agreement. The coalition included Dostum's Junbish-i Milli, a largely secular Uzbek militia, alongside Shia and Sunni Islamist factions whose theological positions were mutually incompatible. The UIFSA's armed struggle was framed around national sovereignty and resistance to Taliban domination rather than any shared vision of Islamic governance.

49. **Mahaz-i Milli-yi Islami-yi Afghanistan (National Islamic Front)** — Afghan mujahideen party founded by Pir Sayyid Ahmad Gailani in 1978. Gailani, a hereditary Sufi pir of the Qadiriyya order, derived his authority from traditional religious lineage rather than

Islamist ideology, and his political orientation was notably moderate and pro-Western within the mujahideen spectrum. The party advocated for the restoration of the Zahir Shah monarchy and a constitutional system, positioning it closer to nationalist traditionalism than to the Islamist state-building projects of Hizb-i Islami or the Taliban. Its recruitment base drew on Gailani's Sufi network and southern Pashtun tribal affiliations. Islam functioned as a source of traditional authority and communal identity rather than as a programmatic ideology for governance, and Gailani consistently favored negotiated political settlements over maximalist religious objectives.

50. **National Resistance Front (NRF)** — An Afghan military-political alliance formed in 2021 as the de facto successor to the Northern Alliance. Operating primarily out of the Panjshir Valley and led by Ahmad Massoud, the group is structurally an ethno-nationalist resistance movement dominated by ethnic Tajiks. While entirely composed of devout Muslims, its armed struggle is fundamentally oriented toward territorial defense, anti-Taliban resistance, and the establishment of a decentralized, pluralistic Afghan republic, sharply contrasting with the Taliban's theocratic project.

51. **Supreme Council for Islamic Revolution in Iraq (SCIRI)** — An Iraqi Shia political party and militia founded in Iran in 1982. Despite its name, following the 2003 US invasion of Iraq, SCIRI integrated heavily into the newly established Iraqi political system. Its primary objectives shifted from a strict ideological revolution to securing and consolidating the political, territorial, and institutional dominance of Iraq's Shia majority against Sunni and Baathist rivals

52. **al-Mahdi Army** — An Iraqi Shia militia formed in 2003 by the populist cleric Muqtada al-Sadr. While intensely religious in its rhetoric and drawing legitimacy from the Sadr clerical lineage, the militia functioned primarily as a militant Iraqi nationalist and communal defense organization. Its armed struggle was driven by opposition to the US occupation and the defense of impoverished Shia neighborhoods in Baghdad and southern Iraq against Sunni militant groups.

53. **Syrian Insurgents** — A broad, fragmented coalition of armed factions fighting against the Assad regime during the Syrian Civil War. Similar to the United Tajik Opposition (UTO), while the most militarily dominant elements of this umbrella (such as Ahrar al-Sham or HTS) maintain strict Islamist ideologies, the overarching category includes a spectrum of actors—from secular Free Syrian Army defectors to localized defense militias. Across the coalition, Sunni Islamic identity serves as a unifying mechanism against the Alawite-dominated government, but the primary shared objective is contesting political and territorial control of the Syrian state.

54. **Kashmir Insurgents** — A broad umbrella classification encompassing the constellation of armed groups fighting the Indian state in Jammu and Kashmir since 1989. While the coalition includes hardline jihadist factions (like Lashkar-e-Taiba) alongside more secular-leaning liberation fronts (like the JKLF), the conflict is fundamentally driven by the territorial dispute over the Kashmir region. Islamic identity serves as a powerful mobilizing force and a boundary marker against the Hindu-majority Indian state, utilized to pursue either an independent Kashmiri state or integration with Pakistan.

55. **Ansarallah (Houthi movement)** — A Zaydi Shia political and militant movement operating in Yemen. Emerging in the 1990s as a theological revivalist movement to protect Zaydi cultural identity against perceived Sunni/Wahhabi encroachment, it evolved into a powerful insurgent force. While adopting radical religious rhetoric and receiving backing from Iran, its armed struggle is deeply rooted in Yemeni ethno-political dynamics,

primarily seeking territorial control, economic resources, and political dominance within the Yemeni state.

Secular Nationalist

1. **Mujahedin-e Khalq (MEK)** — An Iranian dissident militant organization founded in 1965. Originally rooted in a unique syncretism of Marxist-Leninist ideology and Islamic traditionalism, the group participated in the 1979 revolution but quickly took up arms against the newly established Islamic Republic. Its armed struggle against Tehran is fundamentally political and regime-change oriented, and the organization officially campaigns for a secular, democratic Iranian state.

2. **Popular Front for the Liberation of Palestine - General Command (PFLP-GC)** — A Palestinian nationalist militant organization founded in 1968 by Ahmed Jibril following a split from the main PFLP. The group maintains a strict secular, Marxist-Leninist ideological orientation. While operating within the heavily religiously contested geography of the Levant and receiving state backing from Syria and Iran, its foundational ideology and stated objectives are centered entirely on secular Palestinian nationalism and armed resistance against Israel, devoid of any Islamist state-building mandate.

3. **Kurdistan Workers' Party (PKK)** — A Kurdish militant and political organization founded in 1978 by Abdullah Öcalan. Originating as a strict Marxist-Leninist revolutionary movement, its initial objective was the establishment of an independent, socialist Kurdish state in southeastern Turkey. While its ideology later evolved toward "democratic confederalism," the organization remains staunchly secular. Its armed struggle is driven purely by Kurdish ethno-nationalism and the demand for political autonomy, consistently rejecting religious framing.

4. **Kurdistan Freedom Hawks (TAK)** — A radical Kurdish militant faction that emerged as a splinter group from the PKK. Sharing the secular-nationalist roots of its parent organization, TAK conducts urban terrorism and bombings across Turkey in pursuit of Kurdish ethno-nationalist retaliation against the Turkish state. It operates outside of any religious-ideological framework.

5. **Democratic Union Party (PYD)** — A Kurdish political party and militant organization (via its armed wing, the YPG) operating primarily in northern Syria. As the Syrian affiliate of the PKK's broader transnational network, it adheres to Abdullah Öcalan's secular ideology of democratic confederalism. Its armed conflict against the Syrian government and various jihadist groups (including the Islamic State) is driven by the defense of Kurdish territorial autonomy and the establishment of a secular, decentralized administrative region (Rojava).

6. **Party of Free Life of Kurdistan (PJAK)** — An Iranian Kurdish militant group founded in 2004 as part of the PKK's broader umbrella network. Following the same secular-nationalist and democratic confederalist ideology as the PKK and PYD, the group conducts an armed insurgency against the Iranian government to secure self determination and cultural rights for the Kurdish minority in northwestern Iran, explicitly rejecting the theocratic framework of the Islamic Republic.

7. **Democratic Party of Iranian Kurdistan (KDPI)** — A Kurdish nationalist party and militant group operating in Iran, founded in 1945. As one of the oldest Kurdish political organizations, it maintains a strictly secular and left-leaning orientation. Its armed struggle against both the Pahlavi monarchy and the subsequent Islamic Republic has been driven

entirely by the goal of securing Kurdish autonomy and federalism within a democratic, secular Iran.

8. **Kurdistan Democratic Party (KDP)** — A prominent Iraqi Kurdish political party and paramilitary organization (Peshmerga) founded in 1946 by Mustafa Barzani. Functioning as a secular, traditionalist ethno-nationalist movement, its historical armed struggle against the Iraqi state was waged to achieve Kurdish self-determination, territorial autonomy, and the protection of Kurdish cultural identity, culminating in the establishment of the Kurdistan Region of Iraq.

9. **Patriotic Union of Kurdistan (PUK)** — An Iraqi Kurdish political party and paramilitary organization founded in 1975 by Jalal Talabani following a split from the KDP. The group adopted a more secular, socialist, and progressive ideological platform compared to the traditionalist KDP. Its armed insurgency against the Ba'athist Iraqi regime was fundamentally rooted in Kurdish ethno-nationalism and the pursuit of political autonomy.

10. **Communist Party of Turkey/Marxist-Leninist (TKP-ML)** — A radical Turkish insurgent organization founded in 1972. Adhering strictly to Maoist and Marxist-Leninist ideology, the group's armed struggle is aimed at the violent overthrow of the Turkish state to establish a communist regime. Its motivations and operational framework are entirely secular and rooted in revolutionary class struggle.

11. **Maoist Communist Party (MKP)** — A Turkish revolutionary insurgent group that emerged as a splinter faction from the TKP-ML. The group pursues a hardline Maoist ideological agenda, conducting armed attacks against Turkish state targets with the objective of sparking a communist revolution. It operates completely outside of any religious or ethno-religious parameters.

12. **Peace at Home Council (Yurtta Sulh Konseyi)** — A clandestine faction within the Turkish Armed Forces that orchestrated the violent, failed coup d'état attempt on July 15, 2016. In their official broadcast declaration, the plotters framed their armed takeover as a necessary intervention to restore the secular, democratic, and constitutional order of the Turkish Republic, making their stated ideological justification explicitly secular-nationalist.

13. **Revolutionary People's Liberation Party/Front (DHKP-C)** — A radical Marxist-Leninist militant organization operating in Turkey, founded in 1978 (originally as Revolutionary Left). The group pursues the violent overthrow of the Turkish government to establish a socialist state. It conducts assassinations and bombings motivated purely by secular, anti-imperialist, and communist ideology.

14. **Popular Front for the Liberation of Palestine (PFLP)** — A Palestinian Marxist-Leninist and revolutionary socialist organization founded in 1967 by George Habash. The group's ideology is fundamentally secular, viewing the Israeli-Palestinian conflict through the lens of anti-imperialism and class struggle rather than religious duty. It pursues the establishment of a secular, democratic Palestinian state.

15. **Balochistan Liberation Army (BLA)** — A secular ethno-nationalist militant organization based in the Balochistan region of Pakistan. The group wages an armed insurgency against the Pakistani state to achieve independence for the Baloch people. Its grievances are rooted in the economic exploitation of regional resources and political marginalization, and its ideology explicitly rejects religious framing in favor of secular Baloch nationalism.

16. **Balochistan Liberation Front (BLF)** — A Baloch ethno-nationalist militant group founded in 1964, operating primarily in Pakistan's Balochistan province. With historical roots in Marxist ideology and left-wing nationalism, the organization's armed struggle against the Pakistani state is strictly secular. It seeks an independent Balochistan based on ethnic identity rather than religious affiliation.

17. **Baloch Republican Army (BRA)** — A secular, ethno-nationalist insurgent group in Pakistan's Balochistan province, largely serving as the militant wing of the Baloch Republican Party. Like other Baloch separatist organizations, its armed struggle is driven by demands for ethnic sovereignty, political rights, and control over natural resources, entirely separate from the religious militant networks operating in the same region.

18. **Baloch Raaji Aajoi Sangar (BRAS)** — An operational umbrella organization formed in 2018 to unify the efforts of several secular Baloch ethno-nationalist insurgent groups, including the BLA, BLF, and BRA. The coalition coordinates armed attacks against Pakistani security forces and Chinese infrastructure projects. Its foundational mandate is strictly secular and separatist, consolidating the armed struggle for an independent Baloch state.

19. **Muttahida Qaumi Movement (MQM)** — A secular political party in Pakistan representing the Muhajir community (Urdu-speaking migrants from India). Its armed factions engaged in extensive urban conflict in Karachi to secure political and economic power for Muhajirs, operating strictly along ethnic and secular lines

20. **Purbo Banglar Communist Party (PBCP)** — A clandestine Marxist-Leninist and Maoist insurgent group operating in Bangladesh. The group's armed struggle is driven by revolutionary communist ideology, seeking to overthrow the state through agrarian revolution and class warfare, firmly placing it in the secular-ideological category.

21. **Purbo Banglar Communist Party - Janajuddha (PBCP-J)** — A hardline splinter faction of the PBCP in Bangladesh. It maintains the same secular, Maoist revolutionary ideology as its parent organization, conducting assassinations and extortion to fund its localized class-based armed struggle.

22. **Parbatya Chattagram Jana Samhati Samiti (PCJSS)** — A secular political organization representing the indigenous Jumma people of the Chittagong Hill Tracts in Bangladesh. Its armed wing, the Shanti Bahini, waged a decades-long ethno-nationalist insurgency to secure autonomy, land rights, and cultural preservation against Bengali settlement and state marginalization.

23. **Sudan People's Liberation Movement/Army (SPLM/A)** — The primary secular insurgent movement that fought the Sudanese government during the Second Sudanese Civil War. Founded by John Garang, the group initially pursued a Marxist-oriented vision for a unified, secular, and democratic "New Sudan," vigorously opposing the Khartoum government's imposition of Islamic law (sharia) on the diverse and largely non-Muslim southern population.

24. **Sudan People's Liberation Movement-North (SPLM/A-North)** — The northern branch of the SPLM/A that continued fighting the Sudanese state after the secession of South Sudan in 2011. Operating primarily in the Nuba Mountains and Blue Nile states, it fights against regional marginalization and advocates for a secular, democratic, and decentralized Sudan.

25. **Sudan Liberation Movement/Army (SLM/A)** — A secular, ethno-nationalist rebel group founded in 2002 to fight for the rights of the marginalized non-Arab communities in the

Darfur region of Sudan. It explicitly rejects the Islamist and Arab-nationalist policies of the central government, demanding political representation and economic development.

26. **Sudan Liberation Movement/Army - Minni Minnawi (SLM/A - MM)** — A major splinter faction of the SLM/A, drawing its support predominantly from the Zaghawa ethnic group. While it has occasionally signed peace agreements with the government, its ideological foundation remains rooted in secular Darfuri ethno-nationalism and regional autonomy.

27. **Justice and Equality Movement (JEM)** — A powerful Darfuri rebel group. While its early leadership had historical ties to the Islamist politician Hassan al-Turabi, the organization's armed struggle is overwhelmingly centered on regional equality, Darfuri ethno-nationalist rights, and the overthrow of the Bashir regime. It allied with strictly secular rebel groups, placing its operational focus squarely on regional political objectives rather than a religious mandate.

28. **Sudan Revolutionary Front (SRF)** — A broad alliance formed in 2011 comprising major Sudanese rebel groups, including the SPLM-N, JEM, and various SLM factions. The coalition's unified mandate is entirely secular and political, aiming to overthrow the Islamist regime in Khartoum and establish a democratic, federal state that respects the rights of Sudan's marginalized peripheries.

29. **Darfur Joint Resistance Forces** — An umbrella coalition of Darfuri rebel factions formed to coordinate military operations against Sudanese government forces and allied militias. Its objectives are firmly rooted in the defense of Darfuri communities and secular regional autonomy.

30. **National Democratic Alliance (NDA)** — An umbrella coalition of Sudanese opposition groups formed in 1989 to oppose Omar al-Bashir's Islamist military regime. It brought together the secular SPLM/A with northern political parties, unified by a secular, democratic, and anti-regime political platform.

31. **South Sudan Democratic Movement/Army (SSDM/A)** — A militant group formed in South Sudan shortly before its independence. Driven by ethnic grievances (particularly among the Murle and Shilluk) and opposition to the dominance of the SPLM government, its insurgency was entirely secular and based on localized political and ethnic power struggles.

32. **South Sudan Liberation Movement/Army (SSLM/A)** — A predominantly Nuer rebel militia in South Sudan. Formed in opposition to the Dinka-dominated SPLM/A state apparatus, its armed campaign was driven by secular ethno-nationalist grievances, seeking greater political power and resource distribution for its constituency.

33. **Free Aceh Movement (GAM)** — An ethno-nationalist separatist organization that waged an insurgency against the Indonesian government from 1976 to 2005. Despite Aceh's strong Islamic identity and the eventual implementation of regional sharia law, GAM's foundational ideology under founder Hasan di Tiro was explicitly secular, anti-colonial, and ethno-nationalist. The group's stated objective was independence from perceived Javanese neo-colonialism and the restoration of the historic Acehnese state, consciously rejecting the religious-state objectives of earlier rebellions like Darul Islam.

34. **Free Papua Movement (OPM)** — An umbrella term for the decentralized ethno-nationalist separatist movement in West Papua, Indonesia. Founded in 1965, the movement's armed

factions wage a low-level guerrilla war against the Indonesian state. Their armed struggle is strictly secular and driven by demands for indigenous self-determination, independence, and the cessation of economic exploitation and demographic marginalization by the Indonesian government.

35. **National Transitional Council (NTC)** — The de facto political body and armed coalition formed by anti-Gaddafi forces during the 2011 Libyan Civil War. While functioning as a broad umbrella encompassing various local and Islamist militias, the central leadership of the NTC officially pursued the overthrow of the Gaddafi regime to establish a pluralistic, democratic, and secular civil state, prioritizing national political transition over ideological mandates.

36. **Forces of the House of Representatives (LNA)** — The military coalition, predominantly known as the Libyan National Army (LNA) led by Khalifa Haftar, aligned with the Tobruk-based parliament during the Second Libyan Civil War. The faction's stated objective was to defeat rival Islamist and UN-backed coalitions in Tripoli to unify Libya under a centralized military-backed government. Haftar explicitly framed his military campaigns, such as Operation Dignity, as a secular-nationalist war to eradicate Islamist militias and political Islam from Libya.

37. **Movement of Democratic Forces of Casamance (MFDC)** — A separatist movement operating in the Casamance region of Senegal since 1982. The group's armed insurgency is driven by the ethno-nationalist grievances of the predominantly Diola population, citing economic neglect and cultural marginalization by the northern Wolof-dominated government in Dakar. Its demands for independence are secular and rooted entirely in regional and ethnic identity.

38. **Polisario Front** — A Sahrawi nationalist liberation movement fighting for the independence of Western Sahara from Moroccan control. Originating with strong Marxist-Leninist and anti-colonial influences, the organization is strictly secular. Its armed struggle and diplomatic efforts are dedicated entirely to securing territorial sovereignty and self-determination through the establishment of the Sahrawi Arab Democratic Republic (SADR).

39. **Lord's Resistance Army (LRA)** — A radical syncretic militant group operating across Uganda, South Sudan, the Central African Republic, and the DRC. Founded by Joseph Kony in 1987, the group's foundational objective was the violent overthrow of the Ugandan government to establish a state governed by the biblical Ten Commandments and Acholi mysticism. While lacking a cohesive political manifesto and driven by a rigid, prophetic religious-ideological mandate rather than secular political grievances, the group operates entirely outside the Islamic ideological frameworks that characterize other religiously motivated insurgencies in this dataset.

40. **Junbish-i Milli-yi Islami** — Afghan political and paramilitary organization founded by Abdul Rashid Dostum in 1992. Despite the word "Islami" in its name, it is a staunchly secular, predominantly ethnic Uzbek militia. It fought for secular governance and Uzbek factional power during the Afghan civil wars, allying at different times with both Soviet-backed and Western forces against Islamist factions.

41. **Afghanistan Freedom Front (AFF)** — An anti-Taliban militant group formed in 2022 following the Taliban takeover. Composed largely of former Afghan National Defense and Security Forces (ANDSF) personnel, the group's objectives are secular and nationalist. It aims to overthrow the Taliban's theocratic regime and restore a democratic, constitutional republic.

42. **Republic of Artsakh** — The unrecognized breakaway state (formerly the Nagorno-Karabakh Republic) that fought for independence and integration with Armenia against Azerbaijan. Its armed forces functioned as a conventional military fighting a strictly ethno-nationalist and territorial war, entirely devoid of religious-ideological goals, despite the broader religious differences between the warring populations.

43. **Somali National Movement (SNM)** — A Somali insurgent group founded in 1981, drawing its support almost exclusively from the Isaaq clan. The group waged a secular, ethno-nationalist guerrilla war against the Siad Barre regime, ultimately declaring the independent (though internationally unrecognized) state of Somaliland in 1991.

44. **United Somali Congress / Somali Salvation Alliance (USC/SSA)** — A major military faction in the Somali Civil War, originally founded in 1989 and rooted primarily in the Hawiye clan. Its armed struggle to overthrow Siad Barre and its subsequent factional warfare in Mogadishu were driven strictly by clan-based political and territorial competition.

45. **Somali Reconciliation and Restoration Council (SRRC)** — A coalition of secular, clan-based Somali warlords formed in 2001 (with significant backing from Ethiopia) to oppose the Transitional National Government and emerging Islamist groups. Its objectives were entirely political and factional, seeking power-sharing and territorial control within a decentralized Somali state.

46. **Forces of Khudoberdiyev** — A rebel military faction led by mutinous Tajik warlord Colonel Mahmud Khudoberdiyev. He led secular military uprisings in 1997 and 1998 against the Tajik government, driven by regional factionalism and strict opposition to the integration of Islamist forces (from the United Tajik Opposition) into the national army following the civil war peace accords.

47. **Mouvement Populaire de l'Azawad (MPA)** — A Tuareg rebel organization founded in 1990 in Mali. While its founder, Iyad Ag Ghaly, much later embraced jihadism, the MPA itself operated during the 1990s Tuareg rebellion as a strictly secular, ethno-nationalist movement seeking autonomy and political representation for the marginalized Tuareg people.

48. **Anjouan People's Movement (MPA / Republic of Anjouan)** — A separatist movement that declared the independence of the island of Anjouan from the Comoros in 1997. The armed struggle was driven entirely by localized political and economic grievances against the central government in Moroni, operating without any religious-ideological framework.

49. **Niger Movement for Justice (MNJ)** — A predominantly Tuareg rebel group that waged an insurgency against the government of Niger from 2007 to 2009. The group's objectives were secular and ethno-nationalist, demanding a greater share of the region's uranium wealth, regional development, and political integration for northern nomadic populations.

50. **Coordination of Azawad Movements (CMA)** — A coalition of Tuareg independence and Arab nationalist movements in northern Mali formed in 2014. Serving as the primary secular and ethno-nationalist counterweight to jihadist coalitions like JNIM, the CMA seeks political autonomy and self-determination for the Azawad region, consistently rejecting the imposition of transnational sharia governance.

51. **National Council of Timorese Resistance (CNRT)** — An umbrella organization formed in 1998 representing the East Timorese struggle for independence from Indonesia. Integrating the armed wing FALINTIL with diplomatic and political fronts, the movement was strictly secular, nationalistic, and focused on achieving sovereign statehood and self-determination for East Timor.

52. **Front for the Restoration of Unity and Democracy (FRUD)** — A rebel group in Djibouti that launched an insurgency in 1991. Drawing its support primarily from the Afar ethnic group, the organization fought against the perceived political and economic marginalization by the Issa-dominated central government. Its objectives were strictly secular and focused on ethno-political power-sharing.

53. **Rally of Democratic Forces of Guinea (RFDG)** — A Guinean rebel coalition that launched cross-border attacks from Liberia and Sierra Leone into Guinea in 2000 and 2001. Backed by Liberian President Charles Taylor, the group's armed campaign was a secular, political proxy conflict aimed at destabilizing the government of Lansana Conté, entirely devoid of religious framing.

54. **Syrian Democratic Forces (SDF)** — A multi-ethnic and multi-religious military alliance formed in 2015, dominated by the Kurdish YPG but including Arab, Assyrian, and Turkmen militias. The SDF was the primary ground force fighting the Islamic State in Syria. Its ideology is rooted in secular democratic confederalism, aiming to establish a decentralized, secular, and pluralistic governance model in northeastern Syria.

55. **Patriotic Salvation Movement (MPS)** — A Chadian rebel group formed in 1990 by Idriss Déby. Operating from bases in Sudan, the group launched a successful insurgency to overthrow the dictatorial regime of Hissène Habré. Its foundational objectives and subsequent governance as Chad's ruling party were rooted in secular political control and regional factionalism rather than any religious mandate.

56. **Popular Front for the Liberation of Azawad (FPLA)** — A Tuareg rebel faction active during the 1990s Tuareg rebellion in Mali. Splintering from other groups over strategic differences, the FPLA pursued a strictly secular, ethno-nationalist agenda, fighting for the political and economic rights of the Tuareg population in the northern Azawad region.

57. **Sudan Liberation Forces Alliance (SLFA)** — A rebel coalition operating in the Darfur region of Sudan. Formed by merging several secular Darfuri factions, its armed struggle is driven by ethno-nationalist grievances, seeking regional autonomy, political representation, and an end to the marginalization of Darfur's non-Arab populations by the central government.

58. **Rwandan Secular Militant Factions (RJF)** — An armed faction operating within the context of the Rwandan conflict system. Its armed struggle is driven by secular political objectives, regional grievances, and factional competition for state resources, operating entirely outside of a religious-ideological mandate.

59. **Popular Resistance Committees (PRC)** — A Palestinian militant coalition operating primarily in the Gaza Strip. Formed in 2000 during the Second Intifada, it functions as a cross-ideological umbrella group. While operating alongside Islamist groups, its foundational mandate is rooted in secular Palestinian nationalism and armed resistance against Israel, lacking a specific Islamist state-building agenda.

60. **Popular Front of Tajikistan (PFT)** — A secular, pro-government paramilitary organization active during the Tajikistani Civil War. Composed primarily of Kulyabi and Uzbek

militias, the group fought on behalf of the post-Soviet secular government against the Islamist and regionalist forces of the United Tajik Opposition (UTO), motivated by regional factionalism and the preservation of the secular state apparatus.

61. **United Baloch Army (UBA)** — A secular ethno-nationalist militant group in Pakistan's Balochistan province, formed as a splinter faction from the Balochistan Liberation Army (BLA). The group wages a separatist armed struggle against the Pakistani state, driven entirely by demands for Baloch sovereignty and control over regional resources, distinct from religious militant networks.

62. **Alliance of Tuareg of Niger and Mali for Change (ATNMC)** — A cross-border Tuareg rebel alliance formed to coordinate insurgent activities in both Mali and Niger. The group's objectives were strictly secular and ethno-nationalist, demanding economic development, political integration, and cultural rights for the marginalized nomadic Tuareg populations across the Sahel.

63. **Tigray People's Liberation Front (TPLF)** — An ethno-nationalist paramilitary group and political party in Ethiopia. Rooted in Marxist-Leninist ideology at its founding, the organization pursues secular, ethno-nationalist objectives. Following its ouster from the federal government, its recent armed struggle was driven by demands for Tigrayan regional autonomy and resistance against the central government, entirely outside any religious framework.

64. **OPON Forces** — The Special Purpose Police Unit (OPON) of Azerbaijan. In 1995, this heavily armed state security faction, led by Colonel Rovshan Javadov, engaged in a mutinous armed rebellion against the government of Heydar Aliyev. The conflict was a strictly secular, factional power struggle for control of the Azerbaijani state.

65. **Military Faction (Red Berets)** — An elite airborne commando unit of the Malian army. Following the 2012 military coup by the rival "Green Berets," this faction (loyal to ousted President Amadou Toumani Touré) launched an armed counter-coup attempt in Bamako. The fighting was a secular, internal military factional dispute over state control.

66. **Military Faction (Forces of Shah Nawaz Tanai)** — A mutinous faction of the Afghan communist military led by Defense Minister General Shah Nawaz Tanai, who launched an attempted coup against President Mohammad Najibullah in 1990. The armed uprising was a secular, factional power struggle within the Soviet-backed regime.

67. **Military Faction (Forces of Suret Huseynov)** — A rogue military faction in Azerbaijan led by warlord and army commander Suret Huseynov, who launched a military rebellion in 1993 that ousted President Abulfaz Elchibey. The uprising was an entirely secular, politically motivated military coup.

68. **Presidential Guard (Comoros)** — A mercenary-led military unit, commanded by Bob Denard, that launched an armed takeover of the Comorian state in 1989. Despite its origins as a state security force, it operated during the conflict as an independent, secular military faction driven by political control rather than ideological motives.

69. **Sudan Allied Revolutionary Forces (SARC)** — A rebel coalition operating in the Darfur region. Like other Darfuri groups in this category, its armed struggle against the Sudanese government is driven by secular, ethno-nationalist demands for regional autonomy and an end to systemic marginalization.

70. **Baloch Ittehad** — An ethnic separatist alliance operating in Pakistan's Balochistan province. Its armed struggle is firmly secular and ethno-nationalist, coordinating efforts to achieve an independent Baloch state and control over regional resources.

71. **Council for the Restoration of Arakan (CRA)** — An ethnic Rakhine insurgent group in Myanmar. Operating in Rakhine State, its militant campaign against the central government is driven by secular ethno-nationalist objectives, seeking self-determination and political rights for the Rakhine people.

72. **Front pour la Défense de la République (FDR)** — An armed rebel faction in Burundi. Operating within the context of the Burundian Civil War, the group's mobilization and armed struggle were driven by secular ethno-political grievances and factional competition for state power.

73. **Front de Libération de l'Aïr et de l'Azawad (FLAA)** — A prominent Tuareg rebel organization active during the early 1990s insurgency in Niger. Its armed struggle was strictly secular and ethno-nationalist, demanding political integration, economic development, and autonomy for the marginalized Tuareg populations of the Aïr region.

74. **Front Patriotique de Résistance du Niger (FPRN)** — A rebel faction active in Niger. The group's insurgency was driven by secular political opposition and regional grievances against the central government, entirely devoid of religious-ideological framing.

75. **FRUD-Combattant (FRUD-C)** — A hardline splinter faction of the Front for the Restoration of Unity and Democracy (FRUD) in Djibouti. After the mainstream FRUD signed a peace agreement with the government, this faction continued a secular, ethno-nationalist armed struggle to secure greater political representation for the Afar people.

76. **Forces of Khalifa al-Ghawil** — The armed factions and militias loyal to Khalifa al-Ghawil's National Salvation Government in Tripoli during the Libyan civil war. The conflict with rival governments (like the UN-backed GNA or Haftar's LNA) was a secular political struggle for institutional legitimacy and territorial control over the Libyan state.

77. **Ogaden National Liberation Front (ONLF)** — An ethno-nationalist rebel group operating in the Somali Region of Ethiopia. The group's armed struggle against the Ethiopian state is strictly secular, driven by demands for self-determination, regional autonomy, and the protection of ethnic Somali rights against perceived marginalization by the central government.

78. **Popular Front for the Liberation of Libya (PFLL)** — A Libyan political and militant organization comprised of loyalists to the ousted Muammar Gaddafi regime. Formed in 2016, the group seeks to restore the secular, nationalist Jamahiriya system, operating outside of the religious and tribal factionalism that characterizes much of the ongoing Libyan conflict.

79. **Revolutionary Forces of 1 April** — A Chadian rebel faction that opposed the dictatorial regime of Hissène Habré. Formed in the late 1980s, the group later merged with other secular rebel factions to form the Patriotic Salvation Movement (MPS), which successfully overthrew Habré in 1990. The group's armed struggle was driven by secular political opposition and ethnic factionalism rather than a religious ideology.

80. **Somali Patriotic Movement (SPM)** — A Somali rebel organization founded in 1989, drawing its support predominantly from the Ogaden clan. The group waged an armed insurgency during the Somali Civil War, driven strictly by clan-based factionalism, territorial control, and political power-sharing following the collapse of the Siad Barre regime.

81. **Union des Forces de la Résistance de l'Azawad (UFRA)** — A Tuareg rebel group operating in Mali. Active during the ethno-nationalist insurgencies in the Sahel, the group's

armed struggle is strictly secular, seeking regional autonomy, political representation, and the defense of the marginalized Tuareg populations in the Azawad region.

82. **United Liberation Army (ULA)** — An ethno-nationalist militant group operating in Pakistan's Balochistan province. Like other secular Baloch separatist organizations, its armed struggle against the Pakistani state is driven by demands for ethnic sovereignty, political rights, and control over natural resources, explicitly rejecting the religious militant networks operating in the same region.

83. **United Somali Congress / Somali National Alliance (USC/SNA)** — A powerful Somali political and military alliance led by General Mohamed Farah Aidid. Drawing its primary strength from the Hawiye clan, the faction waged a secular, clan-based armed struggle for control of Mogadishu and the broader Somali state during the civil war, notably clashing with rival warlords and UN peacekeeping forces.

84. **Zintan Military Council** — A powerful coalition of militias based in the Libyan city of Zintan. Emerging during the 2011 revolution, the group later aligned with secular and nationalist forces (such as Khalifa Haftar's LNA) against Islamist and UN-backed coalitions in Tripoli. Its operations are driven by local tribal interests, anti-Islamist sentiment, and the pursuit of political power within the fragmented Libyan state.

Appendix D

The civilian-targeting analysis draws on two sources that record violence directed at civilians rather than at the state. This appendix describes each source and the construction of the panels behind Table A.13.

D.1 UCDP One-Sided Violence

UCDP records one-sided violence as a category distinct from the state-based conflict used in the main analysis. A one-sided event is the deliberate use of armed force by a government or a formally organized group against civilians, subject to the same 25-death annual inclusion threshold that governs the state-based series (Sundberg and Melander, 2013). Side A of each event is the perpetrating actor and Side B is the civilian population. I begin from the same UCDP Georeferenced Event Dataset (1989–2022) and the same 31 Muslim-majority countries as the state-based analysis, keep only one-sided events, and drop events whose perpetrator is a government, retaining organized non-state actors. This mirrors the rebel-initiated focus of the aggressor analysis in 6.1: the outcome is civilian killing carried out by armed groups rather than by states.

I classify each perpetrating group into the Religious or Secular-nationalist category using the founding charter, stated objectives, and recruitment ideology criteria of Appendix C. Events whose perpetrator falls outside the classification are excluded. I count classified one-sided events by ideology at the country-day level, collapse to a balanced country-by-Hijri-month-by-Hijri-year panel, and merge the interpolated Muslim-share series. The outcome is $\log(\text{events} + 1)$, and all specifications restrict to countries above the 75% Muslim-share threshold, absorb country-by-Hijri-year and Gregorian-month fixed effects, and cluster by country, exactly as in the baseline conflict regressions.

To locate the source of the religious-group effect I build three versions of the panel: the full set of classified groups, the same set with the Islamic State removed, and the Islamic State alone. The Islamic State is identified from the UCDP actor label. Columns 1 through 4 of Table A.13 report, in order, the religious and secular coefficients on the full panel, the religious coefficient with the Islamic State excluded, and the religious coefficient on the Islamic State alone.

D.2 Global Terrorism Database

The Global Terrorism Database (GTD), maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism at the University of Maryland, records terrorist incidents worldwide from 1970 onward (**?**). The 1993 year is missing database-wide owing to a loss of source records. Because the database is organized around a civilian-directed definition of terrorism, it under-records the rebel-state combat that the state-based analysis studies, the property that led the main text to prefer UCDP for the conflict outcome. That same property makes GTD the natural complement here: it captures the civilian-targeting margin that UCDP state-based events exclude by construction, from an entirely separate compilation.

I harmonize country names to the religion-share file, restrict to the Muslim-majority countries in the sample, and drop groups responsible for fewer than five incidents. This threshold removes a long tail of groups tied to isolated attacks, many of them named too vaguely to classify reliably, and follows standard practice in applied work on the GTD. The outcome is attacks on private citizens, GTD target type 14 (private citizens and property), the largest and least ambiguous civilian category. I classify each named perpetrator into the Religious or Secular-nationalist category using the criteria of Appendix C and drop incidents attributed to groups of unknown ideology, which the database frequently leaves unnamed. I collapse to a country-by-Hijri-month-by-Hijri-year panel, form $\log(\text{attacks} + 1)$, and estimate with country-

by-Hijri-year and Gregorian-month fixed effects and country-clustered standard errors. Two Hijri years affected by data gaps are dropped: the years spanning the missing 1993 data and the truncated final year. As with UCDP, I build full, Islamic-State-excluded, and Islamic-State-only versions, the last dropping a small number of singleton country cells.